# INSLIB

Software Documentation and Mathematical Models




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**Abstract**

INSLIB is a portable library for the estimation of the 3D navigation state, written in C11 only. The core is a 15-state (optionally 18-state) error-state Kalman filter. It fuses inertial measurements with (delayed) GNSS position and velocity, barometer, magnetometer, local position references, absolute yaw references, scalar ground speed, ranges to known anchors as well as zero-velocity and zero-rotation information. Furthermore the library provides two standalone attitude filters (ARS/AHRS), a baro/accelerometer vertical channel filter (`baro_alt`), a World Magnetic Model (WMM) lookup and a wrapper (`nav_suite`), which runs all filters in parallel and degrades gracefully. All covariances are maintained in the UDU ("square-root") factorisation after Bierman and Thornton. The code needs no heap and no OS services, and the execution time per epoch is bounded. This document describes the software architecture, the design decisions and the implemented mathematical models.

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# 1 Introduction

## 1.1 Scope and goals

INSLIB is meant for embedded navigation applications (vehicles, aircraft, robots, indoor platforms) with engineering constraints taken from the aerospace domain:

- Portable C11, no OS dependencies and no compiler extensions
- No heap or dynamic memory. All state lives in structs provided by the caller, the library never calls `malloc`/`free`
- Bounded execution time (WCET[1]): no unbounded loops, no recursion and no input dependent iteration counts in the per-epoch path
- Single-precision hot path: strapdown, prediction and fusion run in 32-bit IEEE 754 `float`. The 64-bit `double` is only used for coordinate calculations, where a higher precision is required
- Fail-operational robustness: non-finite inputs (NaN, Inf) are dropped at the API boundary, implausible measurements are tracked statistically and rejections are counted in the diagnostics

The development follows a requirements-first process: the behaviour is captured in a machine-checked requirements database (`requirements/`, checked by `make reqs`) and is verified with unit and integration tests (`make test`) as well as replays of real datasets (`make datasets`). This document does not repeat the requirements. It explains the architecture, the mathematics and the tuning parameters.

[1] Worst-case execution time.

### 1.2 Module overview

| Module | Files | Purpose |
|---|---|---|
| INS | `ins.[ch]` | 15/18-state Error-State Kalman Filter: position, velocity, attitude, IMU biases, optional magnetometer hard-iron bias. |
| ahrs | `ahrs.[ch]` | Standalone 5/6-state attitude filter (ARS: roll/pitch, AHRS: roll/pitch/yaw with magnetometer heading). Compiling `ahrs.c` with `-DAHRS_NO_MAG` removes the magnetometer code, so a roll/pitch-only build needs neither `magnetic_model` nor `wmm_lut.h`. |
| baro_alt | `baro_alt.[ch]` | 3-state baro/accelerometer vertical channel filter, plus `local_gnss_alt_t`, a 1-state local-height-to-ellipsoid offset filter. |
| nav_suite | `nav_suite.[ch]` | Wrapper: runs INS + two AHRS instances + vertical filters in parallel, mode arbitration, vertical datum alignment, best-source accessors. |
| geodetic_toolbox | `geodetic_toolbox.[ch]` | Quaternion algebra, WGS84 geodesy, gravity model, Earth/transport rate. Reusable outside INSLIB. |
| magnetic_model | `magnetic_model.[ch]`, `wmm_lut.h` | World Magnetic Model lookup (declination, inclination, field strength, NED reference vector) from a generated grid. Reusable. |
| sensor_defaults | `sensor_defaults.h` | Shared consumer-MEMS gyro noise defaults for INS and AHRS (see §3.6). |
| KFCore | submodule | `linalg` (column-major matrix functions), `kalman_udu` (Bierman/Thornton UDU routines). |

## 2 Conventions and notation

- **Body frame ($b$):** FRD, i. e. $x$ forward, $y$ right, $z$ down (right-handed).
- **Navigation frame ($n$):** NED (north, east, down), local-level frame.
- **Earth frame ($e$):** ECEF[2] (WGS84).
- **Attitude:** The attitude is represented by a Hamilton quaternion $\boldsymbol{q}_b^n$ with the scalar part first ($q_0 = w$), which describes the body-to-NED rotation $\boldsymbol{R}_b^n$. Euler angles are Tait–Bryan angles in ZYX order (roll $\phi$, pitch $\theta$, yaw $\psi$).
- **Matrices** are stored column-major (BLAS/LAPACK convention).
- **Timestamps** are monotonic `int64` microseconds.
- $[\boldsymbol{a}]_\times$ is the skew-symmetric cross-product matrix with $[\boldsymbol{a}]_\times \boldsymbol{b} = \boldsymbol{a} \times \boldsymbol{b}$.
- $\boldsymbol{f}^b$ is the specific force measured by the accelerometer, $\boldsymbol{\omega}_{ib}^b$ the angular rate measured by the gyroscope, $\boldsymbol{g}^n$ the local gravity (including the centrifugal term) and $\boldsymbol{m}^n$ the magnetic reference field.

Throughout the document SI units are used: m, $\mathrm{m\,s^{-1}}$, rad, $\mathrm{rad\,s^{-1}}$, $\mathrm{m\,s^{-2}}$ and Pa. Only the magnetometer quantities are given in µT, to match the World Magnetic Model.

[2] Earth-centered, Earth-fixed.

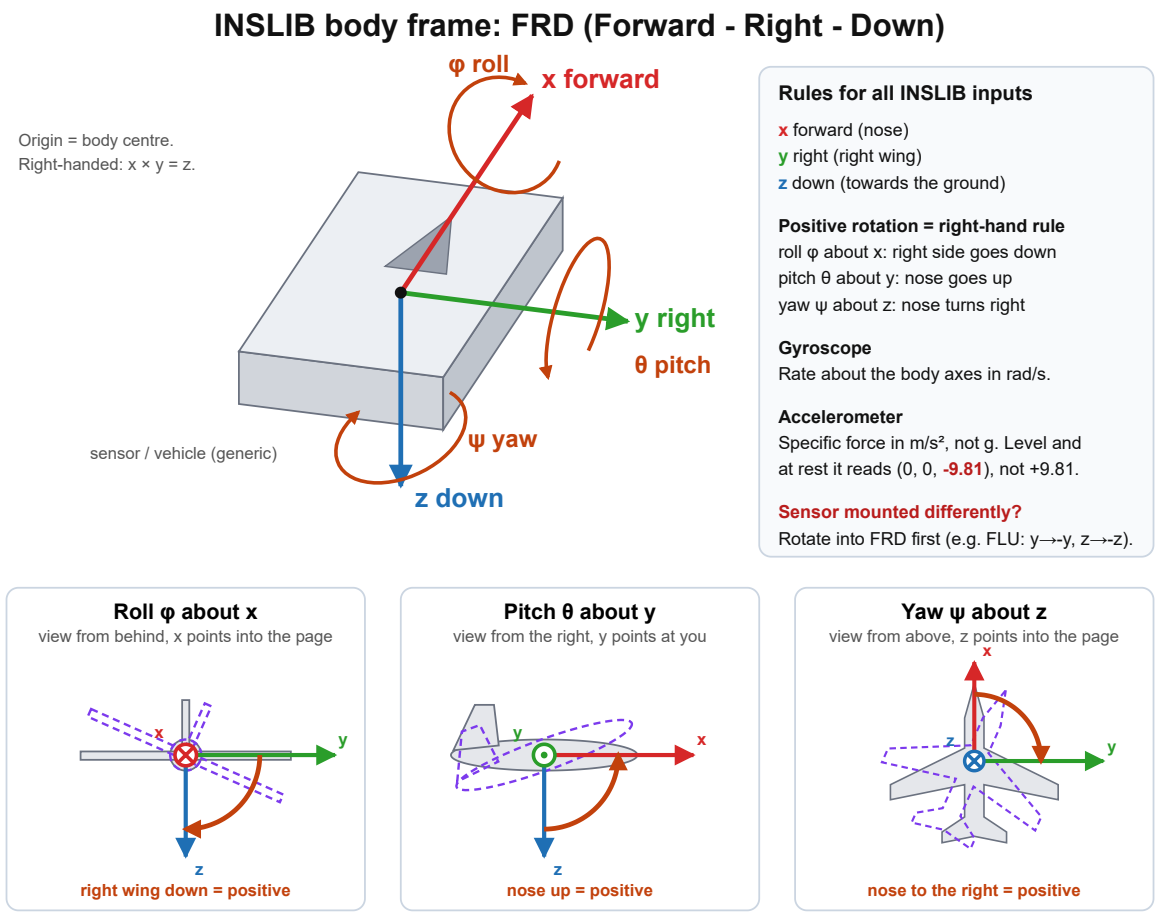



Figure 1: Body frame FRD with the positive rotation directions of roll, pitch and yaw. All sensor inputs must be rotated into this frame before they are passed to the library.

# 3 Software architecture

## 3.1 Structure

Figure 2 shows the dependencies between the modules. The three estimation modules (INS, AHRS, baro_alt) are independent of each other, so each of them can be used standalone. They only share the math layer (geodetic_toolbox, magnetic_model, KFCore) and, in case of INS and AHRS, the IMU noise fallback constants (sensor_defaults.h). The `nav_suite` wrapper is the only module which knows all filters.

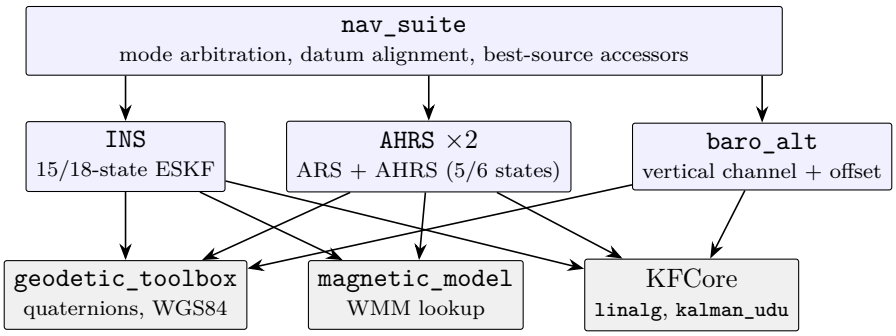



Figure 2: Module dependencies. The grey modules are helper modules: `geodetic_toolbox`, `KFCore` and `magnetic_model` can be reused outside of this repository.

## 3.2 Data flow: one measurement epoch

All measurements of one epoch are collected in a single `ins_measurements_t` bundle (every field is optional and marked with an `is_valid` flag). This bundle is pushed to `nav_suite_update` (or to `ins_update`, if the core filter is used alone). The processing order inside `ins_update` is fixed:

1. Input sanitisation: every consumed field is checked for NaN/Inf before any arithmetic (§3.4).
2. Time-jump handling: backward steps and oversized forward gaps are classified and handled (drop / reset / skip). A backward step that *persists* is interpreted as a restarted timestamp source (§3.4).
3. Strapdown integration of the nominal state with the IMU rate (§5.2).
4. Covariance prediction, which is throttled to the configured Kalman period instead of the IMU rate (§5.4).
5. Measurement fusion in a fixed order: magnetometer, absolute yaw, zero-velocity, zero-rotation, local position, GNSS position/velocity, barometric height, absolute speed, ranges to known anchors, GNSS course-over-ground yaw.
6. History snapshot into a ring buffer (needed for delayed measurements, §5.6) and health check.

In addition the wrapper feeds timestamp, accelerometer and gyroscope (and magnetometer) to the two AHRS instances, timestamp, accelerometer, attitude and pressure to the vertical filter, and baro/GNSS pairs to the offset filter.

### 3.3 Parallel redundant filters and mode arbitration

The suite runs redundant estimators of different complexity:

- INS needs position aiding to stay observable. It provides the full (3D) navigation solution.
- AHRS/ARS only consume IMU (and mag) data and therefore keep producing attitude through arbitrary outages of the position aiding. They serve as cross-check and as fallback for the INS attitude.
- baro_alt plays the same role for the vertical channel: height and climb rate from baro and accelerometer only.

**Height-source selection.** If a barometer was sending measurements when `INS` started *on GNSS*, `INS` fuses the barometric height directly as its own vertical position measurement (§5.5.2), instead of the vertical component of the GNSS position. The GNSS position fusion then only supplies the horizontal $(N, E)$ rows and is graded only on its horizontal accuracy (§5.5.1). The GNSS velocity is not affected on any of the three axes. This selection targets the comparatively poor vertical accuracy of GNSS. A local-position bootstrap keeps its own vertical reference, regardless of a present barometer. Typically a better one. With `baro_height_disable` the selection can be suppressed completely (§5.7).

For the *own* vertical state of `INS` this means that it does not depend on GNSS anymore. It therefore neither steps nor degrades at the `FULL`/`COASTING`/`ATTITUDE_ONLY` boundary, only the horizontal solution does. Please note that this is a statement about the internal state of `ins` and not about what the suite reports: outside of `FULL` mode the height accessors still prefer `baro_alt` (§7.1), which keeps a height available when `ins` itself has none. `baro_alt` runs in parallel in any case, as a really independent barometric estimate with its own state and accelerometer bias (§7.2). Without a barometer `ins` fuses the full 3D GNSS position and the vertical channel follows the same mode machinery as the horizontal channel. The rationale is given in §10.

`nav_suite_get_mode()` arbitrates a solution mode: `FULL` (ins ready, position aiding within the last 2 s) → `COASTING` (ins ready, but dead-reckoning) → `ATTITUDE_ONLY` (ins unavailable, AHRS attitude valid) → `NONE`. Accessors such as `nav_suite_get_rpy()` and `nav_suite_get_height()` always return the best available source. A consumer thus keeps receiving a continuous output through outages. The order of the attitude fallback is ins → magnetometer-AHRS → ARS.

**Heading across an `ins` re-initialisation.** A quality-loss re-arm tears the `ins` instance down and derives its attitude again at the next bootstrap. With a magnetometer this is unproblematic (the AHRS hands over a measured heading). Without one, `ins` has no heading source and starts from the "heading unknown" prior ($\sigma \approx 180°$). The ARS, however, kept running during the outage. Its yaw is not a heading, since it free-integrates and drifts. But the *change* it reports is good over the time scale of an outage, and a change is all that a re-init needs.

Therefore the wrapper latches the pair (`ins` heading, ARS yaw) in every epoch where `ins` is ready and confident about its heading ($\sigma \leq 15°$). The re-initialising filter is then offered the latched heading, advanced by the yaw the ARS has turned since. The uncertainty is the latched $\sigma$ plus the accumulated ARS drift, added in quadrature (its $z$ gyro-bias $\sigma$ integrated over the elapsed time and inflated 10×, like every ARS-derived hint value). Above 30° the carried heading is withheld and the "unknown" prior applies again. If the ARS itself re-bootstraps, the carry-over is dropped, because its yaw restarted from the config seed.

The same hint is offered while an initialised `ins` is not ready because its coasting window expired. Its state is then frozen for the rest of the outage, and the re-acquisition on the next usable fix would otherwise reset the yaw to "heading unknown" (§5.9), with nothing but position fixes to recover it.

`nav_suite_get_rpy()` reports the same reconstruction for yaw whenever it falls back to the ARS. This way the heading of the suite does not jump to the seed-relative yaw of the ARS when `ins` drops out. Roll and pitch are always the own values of the ARS. The per-filter accessor `nav_suite_get_rpy_ars()` keeps reporting the raw ARS yaw, because this is what a cross-check between the two estimators has to read.

**IMU loss.** An IMU gap of `imu_loss_timeout_sec` (0.2 s) breaks every filter of the suite. On the first epoch that finds the IMU stale the wrapper stops the ARS, the AHRS and `baro_alt`, `ins` stops itself (§3.4). The mode becomes `NONE` and the heading carry-over is dropped. When the IMU returns, ARS and AHRS restart from the `ins` leveling window (§5.7), widened under motion and seeded with their gyro bias. `baro_alt` restarts on its old datum, seeded with its accelerometer bias. The first start uses the same window, except under a prescribed initial attitude (`auto_init` off), where the ARS starts from it and the AHRS from the ARS.

Figure 3 shows the mode transitions and Figure 4 shows the resulting behaviour of the parallel filters across a GNSS outage: the attitude and, if barometer data is available, the height remain continuously available while the ins position degrades and recovers.

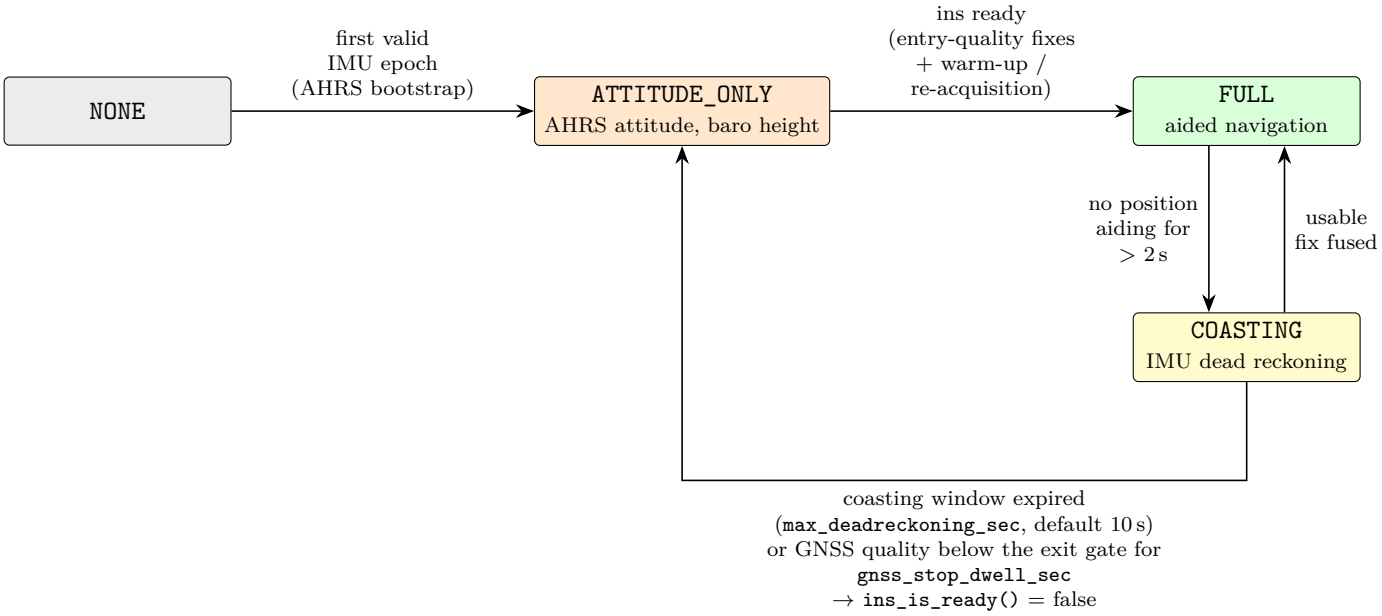



Figure 3: Solution-mode state machine (`nav_suite_get_mode`). The way out of `ATTITUDE_ONLY` depends on the cause that put it there. Both ways require that the fix stream satisfies the entry-quality gate and its dwell (`gnss_start_max_*`, `gnss_init_dwell_sec`). After an expired coasting window the first usable fix *re-acquires*: it re-anchors position and velocity while attitude and IMU biases are kept, and the suite returns directly to `FULL` without a warm-up. After a quality loss the instance was torn down, so the return is a full re-bootstrap: new position and velocity, attitude re-seeded from the ARS/AHRS (heading from the magnetometer-AHRS, or from the pre-outage heading advanced by the ARS, §3.3), IMU biases carried over with inflated uncertainty. The readiness then waits out the usual warm-up. The n-frame origin is carried over as well, so local NED coordinates mean the same thing before and after the outage. Only a fix that the platform could not have reached while unobserved, i. e. further away from the last known position than a hard speed bound allows over the elapsed time, falls back to anchoring a fresh origin. The distance to the origin does not enter this decision, it is just how far the mission has travelled. `allow_unlimited_deadreckoning` disables the coasting-window cause, `gnss_stop_disable` disables the quality-loss cause and `auto_reacquire_disable` disables the tear-down, which falls back to holding `ins_is_ready()` down while the filter runs on.

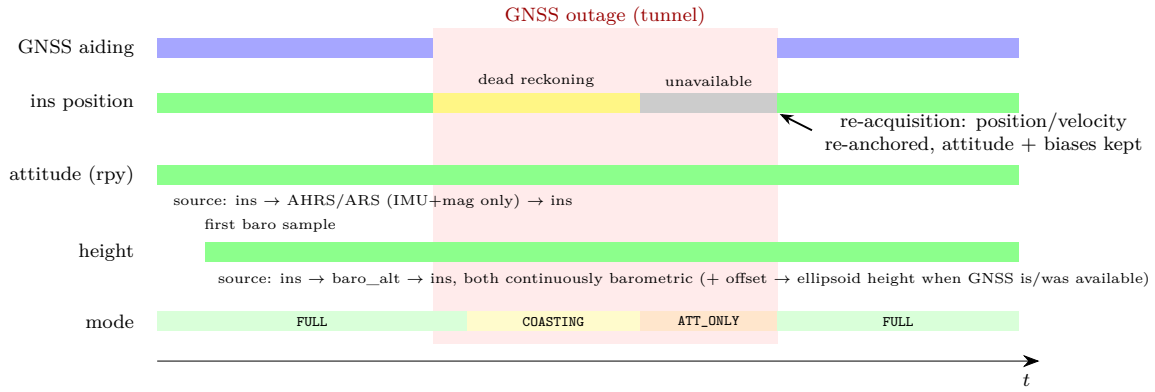



Figure 4: Graceful degradation across a GNSS outage. The AHRS/ARS filters (IMU+mag only) and the baro_alt filter (baro+accel+attitude) keep running unconditionally, so attitude and, with barometer data, height stay available throughout. Both switch sources across the outage (attitude: ins → AHRS/ARS → ins, height: ins → baro_alt → ins, since the accessors prefer ins only in `FULL` mode). The barometric height source does not remove this switch, but it makes it benign: both sides are then the same barometric quantity on the same vertical datum, instead of a GNSS-derived height handing over to a barometric one. The remaining step is thus the disagreement between two independent estimators of one quantity, which the suite cross-checks continuously. The horizontal position of ins coasts on the IMU, degrades after the coasting window and re-acquires with the first fix.

All local heights share one vertical datum (the ins NED origin). Whichever filter initialises first defines it: if ins is ready when the first pressure sample arrives, baro_alt anchors at the current NED height. If baro_alt was running first (auto-init with a late first fix), the ins origin is shifted vertically at bootstrap (§5.10), so that the local heights agree. The absolute solution is untouched by this. The alignment is what keeps the own barometric height of ins (once selected as the height source) and the independently computed height of baro_alt consistent from the first shared epoch onward. The two still carry separate accelerometer-bias states and are not expected to stay bit-identical over a long run, only to track the same physical quantity closely. Since the height accessors hand over between exactly these two estimates at a mode change, the suite does not assume that they stay close. It compares them in every epoch and warns on a persistent gap, and it does the same cross-check on the vertical velocity (the two catch different failures: a divergence too slow to trip the velocity bound still integrates into an unbounded height gap).

### 3.4 Robustness policy

Three layers protect the state estimate, uniformly across all filters, following common best practices of navigation filters [8]:

1. NaN/Inf boundary screening: every comparison with NaN is false, so NaN passes through gates written as `if (x <= limit) reject;`. One corrupt sample would poison the state and force the health check to shut the filter down. Therefore all inputs are checked with `isfinite()` at the API boundary. Corrupt blocks are invalidated and counted (`n_invalid_input`) and the processing continues. Finiteness is not the only way an input can be wrong: absolute headings (the yaw measurement and the attitude hint) are accepted in any single-turn convention ($[-\pi, \pi]$, $[0, 2\pi)$) and normalised to $[-\pi, \pi]$. A heading beyond one full turn is a unit or unwrapping error however (degrees written into a radian field, an accumulated course). It is rejected and counted, instead of being wrapped into a plausible-looking but wrong heading.

2. Statistical screening with two policies: each scalar measurement row is tested with a Mahalanobis (chi-square) gate inside the UDU update (§4). *Persistent absolute references* (GNSS, magnetometer, absolute yaw, barometer) are *downweighted*, i.e. their noise is inflated until the gate is met, but they are never skipped. A skip policy would deadlock when a persistent offset exceeds an optimistic prior (the measurement is rejected forever while the covariance never shrinks). *High-rate local references* (lighthouse, UWB, mocap) are *skipped* on outliers. They glitch (occlusion, reflections) and a dropped sample is cheap at their rate.

3. Health check: after each epoch the state and the covariance factors are checked for finiteness and non-negative variances. A tripped check marks the filter as uninitialised instead of publishing corrupt values. In the suite the AHRS and baro filters then re-bootstrap autonomously from the stream.

A fourth case is neither a corrupt value nor an outlier but a broken *timebase*: the sensor source reboots and its timestamp counter starts over, so every following epoch is stamped far in the past. Each of them is correctly dropped as older than `INS_MAX_DELAY_MS`. The drop happens before the prediction step however, so the internal time reference stays in the abandoned timebase, the next epoch measures the same negative delta and the filter stays inert forever, while still being fed good data. The filter therefore counts *consecutive* drops of this kind. Once the run reaches `INS_TIME_RESTART_EPOCHS`, it forces the same defined reset as for an oversized forward jump and re-arms into the collecting state (`n_time_restart_reset`). Any epoch inside the tolerated depth clears the run, which separates a restart from ordinary reordering: reordering arrives in bursts, a restart never ends. Unlike a forward jump, this reset is not suppressed by `allow_unlimited_deadreckoning`, since coasting would wait for a timebase that never returns.

Drivers should not let it get that far. A driver which reconstructs a 64-bit timeline from a narrower counter (as `tools/insrcv.c` does from a 32-bit microsecond source) can tell an overflow from a restart by *where* in its range the counter was, and can stitch a restart onto the timeline it has already emitted. The filter then keeps its converged state. Discriminating on the step size alone does not work: a restart from a high counter value is indistinguishable from an overflow, and one from a low value looks like no anomaly at all.

A gap in the *IMU* stream is a third case. Aiding epochs arriving in the gap keep the prediction running, so no time jump is seen, yet the motion of the gap is lost. The filter therefore checks the age of the last IMU sample on every epoch. Once it reaches `imu_loss_timeout_sec` (0.2 s) the filter stops before fusing anything (`n_imu_loss`), keeps the n-frame origin and the accelerometer, gyroscope and magnetometer biases ($1\sigma \times 3$, clamped to the cold-start prior) and re-bootstraps like a cold start. Without a fix it stays down. A forward time jump is handled the same way, also with `allow_unlimited_deadreckoning`.

### 3.5 Memory and timing design

Everything is statically sized. The dominating consumer in `ins_t` is the history ring buffer (30 entries × UDU factors). With the default compile-time maximum of 18 states the struct is ≈50 KiB, with `INS_UNKNOWNS_MAX=15` it is ≈38 KiB. The loops of the per-epoch path are bounded by compile-time constants or by the runtime state count $n \leq$ `INS_UNKNOWNS_MAX`. Slowly varying Earth quantities (curvature radii, $\boldsymbol{R}_n^e$, gravity) are cached and refreshed only every 100 m of travel. The per-step latitude/longitude/height bookkeeping then costs some double multiply-adds (§5.3).

### 3.6 Sensor noise defaults

Any noise or tuning field that is left at zero is replaced by a conservative consumer-MEMS default (MPU6050/BMI160 class). An uncharacterised sensor thus yields a stable (slightly cautious) filter instead of an overconfident one, which is however not optimally tuned. The *gyro* defaults (ARW density, bias random walk) are shared between ins and ahrs via `sensor_defaults.h`: in the suite both filters observe the same physical gyro, so their fallback must agree. The *accelerometer* defaults are *not* shared, because each filter fuses a different physical quantity: ins integrates the raw specific force (pure VRW density, $\approx 350\,\mu g/\sqrt{\mathrm{Hz}}$), ars/ahrs fuses a discrete leveling correction, whose noise doubles as a manoeuvre-rejection margin ($0.25\,\mathrm{m\,s^{-2}}$), and baro_alt propagates an attitude-projected vertical acceleration, whose noise also absorbs the roll/pitch projection error ($0.025\,\mathrm{m/s^2}/\sqrt{\mathrm{Hz}}$, $\approx 7\times$ the density of ins). Empirically they are not interchangeable.

## 4 Estimation backend: UDU factorisation

All filters[3] keep the covariance factorised as

$$\boldsymbol{P} = \boldsymbol{U}\ \mathrm{diag}(\boldsymbol{d})\,\boldsymbol{U}^{\mathsf{T}}, \tag{1}$$

with $\boldsymbol{U}$ unit upper triangular. In single precision a full $\boldsymbol{P}$ update loses symmetry and positive definiteness quickly. The factorised form is roughly equivalent to doubling the working precision and guarantees $\boldsymbol{P} \succeq 0$ by construction [3, 4, 8].

**Prediction (Thornton).** The time update

$$\boldsymbol{P}^{-} = \boldsymbol{\Phi}\,\boldsymbol{P}^{+}\,\boldsymbol{\Phi}^{\mathsf{T}} + \boldsymbol{G}\ \mathrm{diag}(\boldsymbol{Q})\,\boldsymbol{G}^{\mathsf{T}} \tag{2}$$

is performed directly on the factors with the modified weighted Gram–Schmidt orthogonalisation of Thornton (`kalman_udu_predict`). The process noise enters in *noise-input form*: $\boldsymbol{G}$ maps physical white noise sources (sensor noise, bias random walks, per-state extra noise) onto the error states and $\boldsymbol{Q}$ holds the corresponding variances.

**Measurement update (Bierman).** Every fusion step assumes a linear(ised) measurement model

$$\boldsymbol{z} = \boldsymbol{H}\,\boldsymbol{x} + \boldsymbol{v}, \qquad \boldsymbol{v} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{R}), \tag{3}$$

which relates the state $\boldsymbol{x}$ to the measurement $\boldsymbol{z}$ via the observation matrix $\boldsymbol{H}$ and zero-mean noise $\boldsymbol{v}$ with the covariance $\boldsymbol{R}$. For the error-state filters (ins, ahrs) $\boldsymbol{x}$ in (3) is the error state $\delta\boldsymbol{x}$, see (6) below. Measurements are processed as sequential scalar rows (Bierman rank-one updates, `kalman_udu`). For one scalar row with the measurement vector $\boldsymbol{h}$ (a row of $\boldsymbol{H}$), the residual $z$ and the measurement variance $R$ the classical Kalman fusion equations are

$$\boldsymbol{k} = \frac{\boldsymbol{P}\,\boldsymbol{h}}{\boldsymbol{h}^{\mathsf{T}}\boldsymbol{P}\boldsymbol{h} + R}, \qquad \delta\boldsymbol{x} \leftarrow \delta\boldsymbol{x} + \boldsymbol{k}\,z, \qquad \boldsymbol{P} \leftarrow (\boldsymbol{I} - \boldsymbol{k}\,\boldsymbol{h}^{\mathsf{T}})\,\boldsymbol{P}, \tag{4}$$

with $\boldsymbol{k}$ the Kalman gain. `kalman_udu` never forms (4) explicitly. It produces the same $\boldsymbol{k}$ and $\delta\boldsymbol{x}$ update, but propagates the covariance change directly through the $\boldsymbol{U}$, $\boldsymbol{d}$ factors (Bierman's rank-one algorithm). This avoids the numerically fragile $\boldsymbol{P} \leftarrow (\boldsymbol{I} - \boldsymbol{k}\boldsymbol{h}^{\mathsf{T}})\boldsymbol{P}$ subtraction. In addition, rows with correlated noise are decorrelated first: if the supplied $m \times m$ measurement covariance $\boldsymbol{R}$ has off-diagonal entries, $\boldsymbol{z}$, $\boldsymbol{H}$ and $\boldsymbol{R}$ are whitened with the Cholesky factor of $\boldsymbol{R}$ (`decorrelate()`). The scalar updates then see independent unit-variance measurements.

[3] Except the 1-state local-height/GNSS offset filter, where the covariance is a plain scalar variance and the Kalman equations are written out directly.

**Robust update.** Each scalar row is screened with a chi-square test on its normalised innovation before it is applied [5]: with innovation $z$, innovation variance $S = \boldsymbol{h}^\mathsf{T}\boldsymbol{P}\boldsymbol{h} + R$ and threshold $\chi^2_{\text{th}}$,

$$\frac{z^2}{S} > \chi^2_{\text{th}} \quad \Longrightarrow \quad \text{either skip the row, or inflate } R \text{ such that } \frac{z^2}{\boldsymbol{h}^\mathsf{T}\boldsymbol{P}\boldsymbol{h} + R'} = \chi^2_{\text{th}}. \tag{5}$$

The choice between *skip* and *downweight* (inflate) is the policy per measurement type from §3.4. A downweighted gross outlier receives a near-zero gain, but the measurement keeps contributing information. A persistent offset is therefore eventually followed, instead of deadlocking the filter.

**Error-state pattern.** ins and ahrs are *error-state* (indirect) filters: the nominal state (quaternion, position, ...) is propagated by nonlinear mechanisation equations, while the Kalman filter estimates the small error $\delta\boldsymbol{x}$ with linear(ised) dynamics. The error state is implicitly zero after every update, so the Thornton step only propagates the covariance and each fusion computes a correction $\delta\boldsymbol{x}$, which is absorbed into the nominal state immediately. The sign convention used throughout the library is

$$\delta\boldsymbol{x} = \boldsymbol{x}_{\text{nominal}} - \boldsymbol{x}_{\text{true}}, \qquad \boldsymbol{x}_{\text{nominal}} \leftarrow \boldsymbol{x}_{\text{nominal}} - \delta\boldsymbol{x}. \tag{6}$$

Residuals are formed as $z = h(\boldsymbol{x}_{\text{nominal}}) - z_{\text{measured}}$, which directly gives $z = \boldsymbol{H}\,\delta\boldsymbol{x} + v$.

# 5 INS: the navigation ESKF

## 5.1 State definition

**Nominal (total) state.** Float-only, in the local n-frame:

| Symbol | Field | Frame | Unit |
|---|---|---|---|
| $\boldsymbol{p}$ | `pos_local` | NED, relative to origin | m |
| $\boldsymbol{v}$ | `vel_ned` | NED | $\mathrm{m\,s^{-1}}$ |
| $\boldsymbol{q}_b^n$ | `qbn` | body→NED | – |
| $\boldsymbol{b}_a$ | `acc_bias` | body | $\mathrm{m\,s^{-2}}$ |
| $\boldsymbol{b}_g$ | `gyr_bias` | body | $\mathrm{rad\,s^{-1}}$ |
| $\boldsymbol{b}_m$ | `mag_bias` | body (for 18-state mode) | µT |

The absolute position is captured in double precision, geodetically in both places: the n-frame origin as $(\varphi, \lambda, h)_{\text{origin}}$ and the current latitude/longitude/height $(\varphi, \lambda, h)$, which is tracked incrementally from it (§5.3).

**Error state.** 15 elements, extended to 18 when the estimation of the magnetometer hard-iron bias is enabled per instance at init:

$$\delta\boldsymbol{x} = \begin{bmatrix} \delta\boldsymbol{p}^\mathsf{T} & \delta\boldsymbol{v}^\mathsf{T} & \boldsymbol{\psi}^\mathsf{T} & \delta\boldsymbol{b}_a^\mathsf{T} & \delta\boldsymbol{b}_g^\mathsf{T} & [\,\delta\boldsymbol{b}_m^\mathsf{T}\,] \end{bmatrix}^\mathsf{T}, \tag{7}$$

with $\boldsymbol{\psi}$ the small-angle attitude error in the n-frame (the so called *psi-angle model* [1, Sec. 8.2]):

$$\boldsymbol{R}_b^n\big|_{\text{nominal}} = \Big(\boldsymbol{I} + [\boldsymbol{\psi}]_\times\Big)\,\boldsymbol{R}_b^n\big|_{\text{true}}. \tag{8}$$

A correction is applied as a left (n-frame side) quaternion multiplication $\boldsymbol{q} \leftarrow \boldsymbol{q}(-\boldsymbol{\psi}) \otimes \boldsymbol{q}$ (`ins_quat_small_angle_correction`), which is consistent with (6).

The 15/18 selection follows a *runtime-n* pattern: all arrays are dimensioned to the compile-time maximum `INS_UNKNOWNS_MAX` (default 18) and the active size $n$ is fixed per instance at init. Memory-constrained integrations can override the maximum with 15 and reclaim the storage. `ins_init` then rejects the option instead of corrupting memory.

### 5.2 Strapdown mechanisation

Each IMU epoch propagates the nominal state over $\Delta t$ (`ins_strapdown`). The Earth rotation and the transport rate are compensated [1, Sec. 3.3]:

$$\boldsymbol{\omega}^n_{in} = \boldsymbol{\omega}^n_{ie} + \boldsymbol{\omega}^n_{en}, \qquad \boldsymbol{\omega}^n_{ie} = \omega_E \begin{bmatrix} \cos\varphi \\ 0 \\ -\sin\varphi \end{bmatrix}, \qquad \boldsymbol{\omega}^n_{en} = \begin{bmatrix} v_E/(R_N+h) \\ -v_N/(R_M+h) \\ -v_E\tan\varphi/(R_N+h) \end{bmatrix}, \tag{9}$$

with $\omega_E$ the WGS84 Earth rate and $R_M, R_N$ the meridian and normal curvature radii. The body rate relative to the n-frame is

$$\boldsymbol{\omega}^b_{nb} = \underbrace{\tilde{\boldsymbol{\omega}}^b_{ib} - \boldsymbol{b}_g}_{\text{bias-corrected gyro}} - (\boldsymbol{R}^n_b)^{\mathsf{T}}\,\boldsymbol{\omega}^n_{in}. \tag{10}$$

**Attitude.** The quaternion is advanced by the equivalent rotation vector $\boldsymbol{\omega}^b_{nb}\Delta t$, which is implemented in `ins_quat_rotate` with a Taylor-series expansion for numerical robustness near zero rotation [1].

**Velocity.** With the bias-corrected specific force $\boldsymbol{f}^b = \tilde{\boldsymbol{f}}^b - \boldsymbol{b}_a$, the n-frame correction terms

$$\boldsymbol{a}_{\text{corr}} = \boldsymbol{g}^n - (2\,\boldsymbol{\omega}^n_{ie} + \boldsymbol{\omega}^n_{en}) \times \boldsymbol{v} \tag{11}$$

(gravity including the centrifugal term, plus Coriolis) and a second-order rotation (coning-style) correction of the integrated specific force [2, 1]

$$\Delta\boldsymbol{u}^b = \boldsymbol{f}^b\,\Delta t + \tfrac{1}{2}\,\boldsymbol{\omega}^b_{nb} \times \boldsymbol{f}^b\,\Delta t^2, \tag{12}$$

the velocity update is

$$\boldsymbol{v}_{k+1} = \boldsymbol{v}_k + \boldsymbol{R}^n_b\big|_k\,\Delta\boldsymbol{u}^b + \boldsymbol{a}_{\text{corr}}\,\Delta t. \tag{13}$$

Please note that $\boldsymbol{R}^n_b$ is evaluated at epoch $k$ (before the attitude update). The rotation correction term accounts for the attitude motion across the interval.

**Position.** Trapezoidal integration:

$$\boldsymbol{p}_{k+1} = \boldsymbol{p}_k + \tfrac{1}{2}\,\Delta t\,(\boldsymbol{v}_k + \boldsymbol{v}_{k+1})\,. \tag{14}$$

### 5.3 Incremental absolute-position bookkeeping

The filter states are float in the local n-frame, whereas the absolute $(\varphi, \lambda, h)$ is maintained in double precision from the *applied* n-frame deltas:

$$\Delta\varphi = \frac{\Delta p_N}{R_M + h}, \qquad \Delta\lambda = \frac{\Delta p_E}{(R_N+h)\cos\varphi}, \qquad \Delta h = -\Delta p_D. \tag{15}$$

The curvature terms (and gravity) change only by $\sim d/R_{\text{earth}}$ per metre of travel and are cached. They are refreshed after every 100 m of accumulated travel. $\boldsymbol{R}^e_n$ is not among them: no epoch-path computation uses it, so it is built on demand in `ins_get_velocity_ecef()` instead of being refreshed inside the worst-case epoch.

### 5.4 Covariance prediction

The prediction is throttled: it runs when at least `kalman_update_dt_sec` has elapsed (default order 50 ms to 100 ms) and not at the IMU rate ([1] names 0.1 s as a typical value). The error dynamics are slow and the strapdown already propagates the nominal state per sample. The discrete transition matrix over the elapsed $\Delta t$ is [1, eq. 10.1]

$$\boldsymbol{\Phi} = \begin{bmatrix} \boldsymbol{I} & \boldsymbol{I}\Delta t & \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} & -[\boldsymbol{f}^n]_\times \Delta t & -\boldsymbol{R}_b^n \Delta t & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{I} & \boldsymbol{0} & -\boldsymbol{R}_b^n \Delta t \\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{I} \end{bmatrix}, \qquad \boldsymbol{f}^n = \boldsymbol{R}_b^n \, \boldsymbol{f}^b, \tag{16}$$

where $\boldsymbol{f}^b$ is the latest bias-corrected specific force. The $-[\boldsymbol{f}^n]_\times$ block is the velocity/attitude coupling under the sign convention (6)(8): $\delta\dot{\boldsymbol{v}} = [\boldsymbol{\psi}]_\times \, \boldsymbol{f}^n = -[\boldsymbol{f}^n]_\times \, \boldsymbol{\psi}$. In 18-state mode the magnetometer bias block is appended as identity (random walk).

Process noise enters (2) through

$$\boldsymbol{G} = \left[\begin{array}{cccc|c} \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0} & \\ \boldsymbol{R}_b^n & \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0} & \\ \boldsymbol{0} & -\boldsymbol{R}_b^n & \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{I}_n \\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{I} & \boldsymbol{0} & \\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{I} & \end{array}\right], \qquad \boldsymbol{Q} = \Delta t \cdot \begin{bmatrix} \mathrm{PSD}_{\mathrm{acc}} \\ \mathrm{PSD}_{\mathrm{gyr}} \\ \sigma^2_{b_a\text{-RW}} \\ \sigma^2_{b_g\text{-RW}} \\ \boldsymbol{q}_{xx} \end{bmatrix}, \tag{17}$$

i. e. twelve IMU noise inputs (accelerometer and gyro white noise rotated into the n-frame, both bias random walks) plus $n$ per-state extra-noise inputs $\boldsymbol{q}_{xx}$ (position/velocity/attitude stabilisation noise and, in 18-state mode, the magnetometer-bias random walk). If the measurement supplies per-axis PSDs, they are used. Axes left at zero fall back to the conservative MEMS defaults (§3.6), so that no zero process noise is injected.

### 5.5 Measurement models

All fusions follow the pattern of §4: the residual $\boldsymbol{z} = h(\text{nominal}) - \boldsymbol{z}_{\text{meas}}$ and the Jacobian $\boldsymbol{H}$ are built, the robust Bierman update is run and the estimated $\delta\boldsymbol{x}$ is subtracted from the nominal state.

#### 5.5.1 GNSS position

The measurement arrives as geodetic $(\varphi, \lambda, h)$ in double precision, which is the form receivers report, with a full $3 \times 3$ NED covariance. No coordinate conversion runs on the epoch path: the fix is differenced directly against the anchor of the filter at the time of validity of the measurement and the difference is mapped to metres with the curvature radii. A source that is natively ECEF (RTK, UBX-NAV-HPPOSECEF) converts once in the caller, outside of the worst case of the filter. With the GNSS antenna lever arm $\boldsymbol{l}^b$:

$$\boldsymbol{z}_p = \underbrace{\mathcal{M}((\varphi,\lambda,h)_{\text{tov}} - (\varphi,\lambda,h)_{\text{meas}})}_{\text{NED difference}} + \boldsymbol{R}_b^n\big|_{\text{tov}} \, \boldsymbol{l}^b, \tag{18}$$

$$\boldsymbol{H}_p = \begin{bmatrix} \boldsymbol{I}_3 & \boldsymbol{0} & -[\boldsymbol{l}^n]_\times & \boldsymbol{0} & \boldsymbol{0} \end{bmatrix}, \qquad \boldsymbol{l}^n = \boldsymbol{R}_b^n\big|_{\text{tov}} \, \boldsymbol{l}^b, \tag{19}$$

following [1, eq. 8.62] (the lever-arm equations for distributed sensors are also collected in [9]). The $-[\boldsymbol{l}^n]_\times$ block makes a lever-arm-induced residual observable in the attitude states: without it the filter would attribute the residual purely to a position error and could not extract heading/tilt information from an offset antenna.

Quality gates reject measurements whose reported standard deviations exceed the configured horizontal/vertical limits before the fusion (counted in the diagnostics). When `ins` uses the barometric height source (§5.5.2), only the horizontal $(N, E)$ rows of (19) are fused. The vertical row is dropped and not downweighted: the two height sources are not simultaneous alternatives for the same epoch, the selection is fixed once at bootstrap. The position gates then grade the fix on its horizontal accuracy alone, here as well as at the 3D exit gate (§5.9). A fix whose vertical row is discarded before the fusion must not lose the horizontal solution its aiding, nor drop the whole 3D solution, over an accuracy it does not contribute anymore (a good HDOP with a poor VDOP is a typical satellite geometry). The velocity gates are unaffected on all three axes. The same holds for the vertical position limit of the *entry* gate: the entry gate runs during a cold bootstrap, before any height source is latched, and this one fix defines the origin. Its vertical accuracy is thus the accuracy of the absolute vertical datum, whatever runs on top of it afterwards.

### 5.5.2 Barometric height (ins-internal)

If barometer data was present when `INS` started (§5.7), the barometric altitude is fused directly as a scalar measurement of the vertical position state, on the same datum anchor $h_0$ established at bootstrap (§5.10, §7.1) and with the same ISA conversion as `baro_alt` (47):

$$z_h = -p_z - (h_{\mathrm{ISA}}(p) - h_0), \qquad \boldsymbol{H}_h = \begin{bmatrix} -\boldsymbol{e}_3^{\mathsf{T}} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}, \tag{20}$$

with $\boldsymbol{e}_3 = (0, 0, 1)^{\mathsf{T}}$ picking the down component of $\boldsymbol{p}$ ($h = -p_z$, §7.1). It is fused through the same robust UDU update as every other measurement (§4). Outliers are downweighted, like in the barometer fusion of `baro_alt` (§7.2), since this is a persistent absolute reference and not a high-rate local one (§3.4). This replaces the vertical row of the GNSS position measurement (19) for the lifetime of the filter instance, it does not supplement it: the choice of the height source is made once at bootstrap, depending on whether a barometer was observed (§5.7), and is not re-evaluated per epoch. The GNSS velocity (§5.5.3), including its vertical component, is fused unconditionally either way (§10).

Furthermore, like every other fusion this runs on live epochs only: while the coasting window is expired the filter is inert (§5.9). The barometer keeps being cached across such an outage and the height is picked up from it again at the re-acquisition. This is what keeps the vertical solution continuous across a passage the frozen state knew nothing about.

### 5.5.3 GNSS velocity

NED velocity with full covariance. The lever-arm velocity is compensated and its attitude coupling is included [1, eq. 8.74]:

$$\boldsymbol{z}_v = \boldsymbol{v}_{\mathrm{tov}} + \boldsymbol{R}_b^n\big|_{\mathrm{tov}} \left(\boldsymbol{\omega}_{nb}^b \times \boldsymbol{l}^b\right) - \boldsymbol{v}_{\mathrm{meas}}, \qquad \boldsymbol{H}_v = \begin{bmatrix} \mathbf{0} & \boldsymbol{I}_3 & -\left[\boldsymbol{R}_b^n(\boldsymbol{\omega}^b \times \boldsymbol{l}^b)\right]_\times & \mathbf{0} & \mathbf{0} \end{bmatrix}. \tag{21}$$

(Using $\boldsymbol{\omega}_{ib}^b$ instead of $\boldsymbol{\omega}_{nb}^b$ neglects the Earth/transport rate crossed with the GNSS antenna lever arm, which is $\sim 0.1\,\mathrm{mm\,s^{-1}}$ for a 1 m arm and negligible against the GNSS velocity noise.) When position and velocity are fused together, an optional position/velocity cross-covariance block completes the full $6 \times 6$ measurement covariance. Correlated blocks are whitened before the scalar updates (§4).

### 5.5.4 GNSS covariance conditioning

The covariance reported by a receiver is not fused as it stands. It passes a fixed pipeline, whose parameters live under `gnss:` in `config.yaml` (§11) and which is applied per epoch to the position and velocity blocks before either of them reaches (19):

$$\text{envelope} \rightarrow \text{scale} \rightarrow \text{height scale} \rightarrow \text{floor} \rightarrow \text{cap} \rightarrow \text{manoeuvre term}. \tag{22}$$

**Envelope.** A rise of the reported $1\sigma$ takes effect at once, a fall only with a first-order decay of `acc_envelope_tau_sec`: $e \leftarrow \max(\sigma_{\text{rep}},\, e\,(1-\alpha))$, $\alpha = \text{clamp}(\Delta t/\tau, 0, 1)$, kept per block and axis group. The accuracy figure of a receiver returns to nominal well before a re-acquired fix has settled.

**Scale and height scale.** $\boldsymbol{Q} \leftarrow s^2\boldsymbol{Q}$, plus an extra downweight of the position Down axis, applied as the correlation-preserving congruence $\text{diag}(1,1,s_h)\,\boldsymbol{Q}\,\text{diag}(1,1,s_h)$.

**Floor and cap.** Per-axis clamps of the diagonal (horizontal and vertical separately), so a floored or capped axis picks up (or loses) independent variance on that axis alone. The cap is what makes the loose fusion gate safe: `gnss_max_*` defaults to the caps, so a degraded fix is downweighted instead of being discarded, while the unbounded reported accuracy stays out of the 32-bit hot path.

**Manoeuvre term.** Extra independent variance on the velocity diagonal, $\sigma_{\text{extra}} = k\|\boldsymbol{a}\|$, taken from the windowed n-frame acceleration of the *antenna* (the acceleration of the body plus the centripetal term $\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{l})$ of the lever arm). A Doppler velocity averages over the measurement interval $T$ of the receiver, while the state is instantaneous. Under acceleration the two thus differ by about $aT/2$. The reported accuracy cannot describe this error, because it belongs to the platform and not to the signal.

All of it weights the *fusion* only. Every fix-quality gate (§5.9) keeps grading the stddev the receiver reported, so downweighting a noisy receiver can never turn into refusing to use it.

### 5.5.5 Absolute speed (odometry, OBD-II)

A scalar ground speed (an OBD-II vehicle speed, a wheel-odometry pulse rate, a Doppler log) is the magnitude of the velocity with no direction attached:

$$h(\boldsymbol{x}) = \|\boldsymbol{v}^n\|\,, \qquad z_s = \|\boldsymbol{v}_{\text{tov}}\| - z_{\text{meas}}, \qquad \boldsymbol{H}_s = \begin{bmatrix}\mathbf{0} & \hat{\boldsymbol{v}}^{\mathsf{T}} & \mathbf{0} & \mathbf{0} & \mathbf{0}\end{bmatrix}, \quad \hat{\boldsymbol{v}} = \frac{\boldsymbol{v}^n}{\|\boldsymbol{v}^n\|}. \tag{23}$$

The nonlinearity costs nothing in an error-state filter: the residual is formed from the nonlinear $h(\cdot)$ evaluated on the *nominal* state and $\boldsymbol{H}$ is only its Jacobian with respect to the error state, exactly as for every other reference here.

$\hat{\boldsymbol{v}}^{\mathsf{T}}$ picks out the component of the velocity error *along the current direction of travel* and nothing else. The cross-track error is invisible to a quantity that carries no direction. The along-track component is nevertheless the one that runs away first when GNSS stops, since attitude and accelerometer-bias errors leak into the velocity through the strapdown and it is the integrated velocity error that becomes position error. One scalar per second therefore buys most of what a GNSS-denied stretch needs, without the state cloning that a delta-position (wheel-tick) formulation would require.

Two properties of this sensor class shape the implementation. First, $\hat{\boldsymbol{v}}$ is undefined at $\boldsymbol{v}^n = \mathbf{0}$ and numerically meaningless just above it, so the fusion is gated on a minimum *filtered* speed. What has to be well conditioned is the Jacobian, and this is built from the state and not from the reading. A really stopped platform is the job of the automatic ZUPT (§5.8).

Second, the error budget has two parts that scale differently, and lumping them together would bias the filter. An OBD-II `PID 0x0D` value is quantised to $1\,\text{km}\,\text{h}^{-1}$, which is a $0.080\,\text{m}\,\text{s}^{-1}$ $1\sigma$ that does not grow with speed. The scale error of the speed signal of the vehicle does grow: rolling radius, tyre wear and the scaling the ECU applies are a few percent, which at the 3 % default of the library is $0.83\,\text{m}\,\text{s}^{-1}$ at $100\,\text{km}\,\text{h}^{-1}$, ten times the quantisation noise and *systematic*. Its sign is not predictable from the type-approval margin, which keeps an indicated speed from ever falling below the true one. That margin constrains the dashboard, whereas `PID 0x0D` is the own value of the ECU and a different number from the one the cluster shows. A

constant scale factor removes the calibrated part and a relative variance prices what is left:

$$z_{\text{meas}} = s_{\text{scale}}\, z_{\text{raw}}, \qquad R = \sigma^2_{\text{sample}} + (k_{\text{rel}}\, z_{\text{meas}})^2 . \tag{24}$$

No scale-factor state is estimated: this stays a measurement-boundary feature and leaves the runtime-selected state size untouched. `tools/inslib_speed_scale.py` fits $s_{\text{scale}}$ against the GNSS speed and prints the `config.yaml` value.

By contract the reading is unsigned (an OBD-II speed PID cannot express reverse), so a negative value is treated as a unit or sign error and is dropped instead of being taken as its magnitude. Latency is the norm on this input (a Bluetooth OBD-II round trip takes tens of milliseconds) and is handled by the same history anchoring that GNSS uses (§5.6).

### 5.5.6 Ranges to known anchors

A range measurement is the distance between the ranging antenna of the platform and an anchor at a known position. Up to `INS_RANGE_MAX` entries per epoch go into `ins_measurements_t.range[]`, each with the anchor position in ECEF (`double`), the measured range, its $1\sigma$, its own delay and an anchor id for diagnostics. One lever arm (`range_leverarm_b`) applies to all entries. Each entry is one scalar row:

$$h(\boldsymbol{x}) = \|\boldsymbol{a}^e - \boldsymbol{p}^e_{\text{ant}}\|\,, \qquad z_r = h(\boldsymbol{x}_{\text{tov}}) - z_{\text{meas}}, \qquad \boldsymbol{H}_r = \begin{bmatrix} -\boldsymbol{u}^{\mathsf{T}} & \mathbf{0} & \boldsymbol{u}^{\mathsf{T}}\,[\boldsymbol{l}^n]_\times & \mathbf{0} & \mathbf{0} \end{bmatrix}, \tag{25}$$

with $\boldsymbol{u}$ the unit line-of-sight vector from the antenna to the anchor in the n-frame and $\boldsymbol{l}^n$ the lever arm. $\boldsymbol{H}_r$ is the local-position Jacobian projected onto $-\boldsymbol{u}$, so the lever arm couples the range into the attitude.

The fusion is split into two layers. A geometry layer turns an entry into $\boldsymbol{u}$, $z_r$ and a variance and a fusion kernel only sees those three.

**Geometry without per-epoch trigonometry.** The absolute position of the filter is geodetic, the anchors are ECEF. Converting in every epoch would cost double precision trigonometry, which is slow on single-precision embedded targets. The antenna position is therefore formed from a cached reference point (geodetic position, its exact ECEF position, its NED-to-ECEF rotation) plus the n-frame offset from it, i. e. with double additions only. The reference is recomputed exactly, once the position moves farther than `INS_RANGE_REF_RADIUS_M` from it. The error of the approximation is about $s^2/R_\oplus$ at a distance $s$ from the reference, which is millimetres inside the radius. The range is rotation invariant, so using the rotation of the reference instead of the current one only tilts $\boldsymbol{u}$ by $s/R_\oplus$. The anchor minus antenna difference is formed in double and only the result goes to single precision, which resolves terrestrial anchors to millimetres. Satellite distances will need the norm in double as well, together with the pseudorange geometry layer.

**Barometric height.** While the filter runs on the barometric height source (§5.5.2), ranges correct North/East only by default: the Down component of $\boldsymbol{u}$ is removed from $\boldsymbol{H}_r$ and the height uncertainty is considered instead of corrected:

$$R = \sigma^2_{\text{meas}} + u_D^2\, P_{DD}. \tag{26}$$

Zeroing the Down column alone would leave a height error in $z_r$ and leak it into North/East. The barometric datum and the ellipsoid heights of the anchors disagree, and a multipath-lengthened range over a steep line of sight would push a height, which the barometer determines better. For the same reason the GNSS position fusion is restricted to the horizontal under barometric height. `opt.range_height_with_baro` switches the height correction back on, for installations whose anchors sit well above and below the platform. Without the barometric height source ranges always correct all three axes.

**Scope and coasting.** Ranges are fused on a live filter only. They neither initialize nor re-acquire it. Whether the ranges of an epoch count as position aiding for the coasting window (Figure 3), like a GNSS fix, is decided by the covariance after the fusion: they count if the horizontal position $1\sigma$ in its worst direction, the square root of the larger eigenvalue of the North/East block, stays at or below `opt.range_aiding_max_hpos_stddev_m`. A single range bounds the position along its line of sight only, so one anchor keeps the filter alive only until the across-track uncertainty has grown past the limit. Anchors with a usable geometry keep it ready through a GNSS outage and also keep it from the GNSS quality-loss exit. `nav_suite` stays in FULL mode as long as they count. The criterion is only as good as the stated range $1\sigma$: multipath is a positive bias and a too small $1\sigma$ keeps the filter ready on a biased solution. The anchors are absolute, so a `nav_suite` local-frame offset does not apply to them.

### 5.5.7 Position decimation against an unknown correlation

That optional cross-covariance block is the exception, not the rule. A receiver computes position and velocity in one internally coupled solution, but no common output message carries the correlation between them: `gnss_Qll_pos_vel_ned` is filled only by the few receivers that report it and is zero otherwise. Fusing both blocks in every epoch as if they were independent therefore counts one solution twice. The covariance drops below what the measurements support, which in turn shrinks the Kalman gain of every other aiding source and pushes the difference into the accelerometer-bias and attitude states, which are free to absorb it.

`gnss.pos_decimation` answers this by spending only one block per epoch. With the factor $N$, every $N$-th epoch that offers a usable position *and* a usable velocity fuses the position alone, the other $N-1$ fuse the velocity alone. For $N \leq 1$ the combined fuse is restored. An epoch that offers only one of the two is never touched, since the decimation removes information and never adds any.

Decimation is a rather blunt instrument, but that is intended: it assumes nothing about a correlation that cannot be observed. This is unlike an assumed correlation coefficient, which is unverifiable against a receiver that reports none and drives $\boldsymbol{R}$ towards singularity when guessed too high. It also attacks the second, larger correlation in the same data: the position error of a receiver is dominated by multipath, residual ionospheric delay and its own smoothing, all correlated over minutes, while the Doppler-derived velocity is comparatively white. The textbook cure for this time-correlated part is a Gauss-Markov position-bias state per axis, which would allow the full rate for both. That is however a change to the state vector and not what this option does.

Two cases suspend the decimation. While the coasting window is expired (§5.9) the position is the entire point of the fix, so it is always fused and re-anchors the filter. And a position-aiding gap longer than the fix-stream continuity bound restarts the cycle at a position, because after a gap the dead-reckoned position is the stale quantity. An epoch whose position is withheld still counts as position aiding for the coasting window: the fix carried a usable position and the filter declined to spend it, which is not the same as not having one.

### 5.5.8 Local NED position

Indoor references (e. g. Lighthouse/SteamVR, UWB, motion capture, total station) measure `pos_local` directly in the n-frame of the filter. The residual and the Jacobian have the same structure as in (19), only without any coordinate conversion. The frame alignment (origin offset and rotation into NED) is the responsibility of the caller. Outliers are *skipped* and not downweighted (§3.4).

### 5.5.9 Absolute yaw

A heading reference (dual-antenna GNSS, mocap pose, gyro compass) is fused as a scalar residual on the yaw component of the attitude error:

$$z_\psi = \mathrm{wrap}_{\pm\pi}\left(\hat{\psi}_{\mathrm{tov}} - \psi_{\mathrm{meas}}\right), \qquad \boldsymbol{H}_\psi = \begin{bmatrix}\mathbf{0} & \mathbf{0} & \boldsymbol{e}_3^\mathsf{T} & \mathbf{0} & \mathbf{0}\end{bmatrix}, \tag{27}$$

since to first order a heading residual equals the yaw component of the n-frame attitude error. The fusion is skipped near gimbal lock ($|\sin\theta| > 0.99$), where yaw is ill-defined.

**Dual-antenna heading.** A moving base receiver pair (u-blox NAV-RELPOSNED `relPosHeading`) measures the azimuth $h$ of the vector between its two antennas and not the vehicle yaw. With that baseline $\boldsymbol{b}^b$ fixed in the body frame, the level-frame baseline $\boldsymbol{c} = \boldsymbol{R}_y(\theta)\,\boldsymbol{R}_x(\phi)\,\boldsymbol{b}^b$ has azimuth $\psi + \mathrm{atan2}(c_y, c_x)$, so

$$\psi_{\mathrm{meas}} = \mathrm{wrap}_{\pm\pi}(h - \mathrm{atan2}(c_y, c_x)) \tag{28}$$

is exact for any mounting and tilt (`ins_yaw_from_baseline_heading()`, REQ-NAV-087). For a baseline along the body $x$ axis $c_y = 0$ and the azimuth is the yaw at any roll and pitch. Any other mounting couples the attitude in: a baseline across the vehicle reads 3.6 deg off at 20 deg roll and 10 deg pitch, if it is fed as yaw directly. The function refuses a baseline whose horizontal projection is shorter than a tenth of its length, because antenna phase noise dominates the azimuth there. The firmware glue and both replay harnesses call it with the attitude the suite holds at the time (level while it holds none) and pass the heading delay of the receiver as `yaw_delay_ms`. With no magnetometer and no motion this is also what gives auto-init its yaw.

**Automotive mode (course-over-ground yaw).** This is opt-in (`automotive_mode`) and reuses the same scalar update with $\psi_{\mathrm{meas}} = \mathrm{atan2}(v_E, v_N)$ from the GNSS velocity vector, above a minimum ground speed (`automotive_min_speed_mps`, default $2\,\mathrm{m\,s^{-1}}$). Below it the course over ground is not well defined. The mode assumes a non-holonomic vehicle that travels in the direction it points. This is reasonable for a car, but wrong for a glider or a boat drifting sideways: in coordinated flight the heading of the airframe aligns with the *airspeed* vector and not with the *ground-track* vector that GNSS measures, so a crosswind adds a wind-drift (crab) angle independent of speed. Raising the minimum speed does not fix this. The measurement standard deviation is the heading uncertainty from velocity noise over speed, floored (`automotive_min_yaw_stddev`, default 5°), so it never claims more precision than this assumption warrants. Raising this floor (e. g. to a conservative crab-angle estimate such as 15°) lets the fusion still act as a weak long-term anchor against free-yaw drift on a platform, where the assumption itself is looser than for a car, without asserting automotive-grade heading precision. Outliers (e. g. reversing, ∼180° residual) are chi-square-downweighted like any absolute yaw reference and never dropped. It is an alternative to the magnetometer heading aiding, if none is available.

### 5.5.10 Magnetometer (yaw-only vector fusion)

The magnetometer is fused as a 3D vector measurement of the body-frame field with the model $\boldsymbol{m}^b = (\boldsymbol{R}_b^n)^\mathsf{T}\boldsymbol{m}^n + \boldsymbol{b}_m$:

$$\boldsymbol{z}_m = (\boldsymbol{R}_b^n)^\mathsf{T}\,\boldsymbol{m}^n + \boldsymbol{b}_m - \tilde{\boldsymbol{m}}^b. \tag{29}$$

The full attitude Jacobian of this model would be $(\boldsymbol{R}_b^n)^\mathsf{T}\,[\boldsymbol{m}^n]_\times$ on the $\boldsymbol{\psi}$ block. *Deliberately, only the yaw column is kept:*

$$\boldsymbol{H}_m[:,\psi_z] = (\boldsymbol{R}_b^n)^\mathsf{T}\,(\boldsymbol{m}^n \times \boldsymbol{e}_D), \qquad \boldsymbol{H}_m[:,\psi_x] = \boldsymbol{H}_m[:,\psi_y] = \mathbf{0}, \qquad \frac{\partial \boldsymbol{z}_m}{\partial \delta\boldsymbol{b}_m} = \boldsymbol{I}_3. \tag{30}$$

The zeroed roll/pitch columns ensure that magnetic dip and hard-iron / deviation disturbances can never tilt the attitude estimate. The leveling information comes exclusively from the accelerometer. The vector measurement can only rotate the estimate about the vertical. Residual components that a yaw rotation cannot explain (magnitude or dip mismatch) meet zero entries and produce no state update [1, eq. 10.7].

**Hard-iron bias observability (18-state mode).** The bias term is constant in the *body* frame while the field term $(\boldsymbol{R}_b^n)^\mathsf{T}\boldsymbol{m}^n$ rotates with the attitude. This makes $\boldsymbol{b}_m$ separable from the yaw error, but only under attitude changes. Convergence is to be expected during and after rotations. This is why the bias is estimated *inside* the filter, where the yaw/bias cross-covariance is maintained, instead of in an external calibration estimator.

**Tuning the hard-iron states.** Two parameters control this: `mag_bias_init_stddev_ut` (10 µT by default), the prior on how much hard iron the filter should be prepared to find, and `mag_bias_pred_stddev_ut_sqrts` ($0.129\,\mu\mathrm{T}/\sqrt{\mathrm{s}}$ by default, which is 1 µT per minute), the random walk. Neither is right everywhere. The initial sigma is the stronger lever: a recording whose magnetometer was calibrated where it sits wants a tight prior, while a unit that was just installed in an unknown vehicle does not, since 10 µT to 20 µT of vehicle hard iron is ordinary. Both are exposed to the replay harness as `mag.bias_init_ut` and `mag.bias_rw_ut_sqrts`.

**Field-strength disturbance gate.** Once a position is known (§9), the measured magnitude is compared against the WMM total field $B_{\mathrm{WMM}}$. With the relative deviation $\epsilon = \big|\,\|\tilde{\boldsymbol{m}}\| - B_{\mathrm{WMM}}\big|/B_{\mathrm{WMM}}$ and the tolerance $\tau$ (default 0.30), a sample outside the band has its per-axis noise inflated by $(\epsilon/\tau)^2$. It is downweighted and not dropped, according to the persistent-reference policy. The gate is defense-in-depth over the vector chi-square test (it bites when the measurement noise is set loose) and can be disabled for uncalibrated magnetometers. The fusion requires per-axis variances (no default) and is rate-limited. It is skipped when the horizontal reference field is unusable, either because the reference vector itself is degenerate or because the position lies inside a dip pole exclusion zone (§9.2).

**Weight and rate limit: a long-term anchor, not a yaw sensor.** Since only the yaw column survives, the length of that column is the horizontal field $\|\boldsymbol{m}_h^n\|$, so a per-axis noise $\sigma_m$ acts as a heading measurement of roughly

$$\sigma_\psi \approx \frac{\sigma_m}{\|\boldsymbol{m}_h^n\|}. \tag{31}$$

At mid-latitudes $\|\boldsymbol{m}_h^n\| \approx 20\,\mu\mathrm{T}$, so even a sensor-grade $\sigma_m = 3\,\mu\mathrm{T}$ is already about 8.6° of heading per sample. At a typical 10 Hz output rate the filter accumulates that down to under a degree within seconds. A MEMS gyro of $0.02\,^\circ/\mathrm{s}/\sqrt{\mathrm{Hz}}$ drifts far less than that over the same seconds, so fusing at the sensor rate buys no heading accuracy. What it does buy is exposure: motor currents and ferrous structure bend the field by far more than $\sigma_m$, and those disturbances are strongly time-correlated while the update treats every sample as independent evidence. A high rate therefore systematically over-counts a biased heading.

Both knobs default weak, and both defaults live in the library and not in each harness: a sample that arrives with no per-axis variance is fused with `INS_DEFAULT_MAG_STDDEV_UT` (`sensor_defaults.h`), a noise of the same order as the field itself, i. e. $\sigma_\psi \approx 1$ rad per sample, and the rate limit defaults to 1 Hz (`ins_options_t.magnetometer_min_delay_ms`, 0 → default, negative → off). That combination still binds the heading over minutes, which is all the gyro needs, without letting a disturbed field pull it within seconds. Please note that this is a defence against *noise and transients only*: a standing hard-iron offset is a *bias*, and where the magnetometer is the only absolute heading reference, the steady-state error it causes is

independent of $\sigma_m$. Lowering the weight only stretches the time the filter takes to reach the same wrong answer. (It does shift the balance once a second heading source is present, e. g. GNSS course in automotive mode.) The bias is the job of the calibration (`mag: fixed_bias` / `estimate_bias`) and not of the weighting.

### 5.5.11 Zero-velocity and zero-rotation updates (ZUPT/ZARU)

On trigger (external flag or the automatic detector, §5.8), see [9] for both updates:

$$\text{ZUPT:} \quad \boldsymbol{z} = \hat{\boldsymbol{v}}, \qquad \boldsymbol{H} = \begin{bmatrix} \mathbf{0} & \boldsymbol{I}_3 & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}, \tag{32}$$

$$\text{ZARU:} \quad \boldsymbol{z} = \hat{\boldsymbol{b}}_g + (\boldsymbol{R}_b^n)^\mathsf{T} \boldsymbol{\omega}_{in}^n - \bar{\boldsymbol{\omega}}_{\text{meas}}, \qquad \boldsymbol{H} = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \boldsymbol{I}_3 \end{bmatrix}, \tag{33}$$

i. e. the zero-rotation update is a direct gyro-bias measurement, with the Earth and transport rate compensated as seen in the body frame. Both are rate-limited to one fusion per Kalman period. $\bar{\boldsymbol{\omega}}_{\text{meas}}$ is the gyro *averaged over the current stillness run* instead of a single sample: a momentary sample of a vibrating-but-still platform (idling engine) would inject the vibration directly into the gyro bias states. With a tight $\sigma_{\text{ZARU}}$ this corrupted the heading by up to $0.6\,^\circ\,\text{s}^{-1}$ at every stop on real automotive data. The average is a running mean and not a median: bounded memory and $O(1)$ per sample (WCET rule).

## 5.6 Delayed measurements: the history buffer

GNSS (and optionally local position / yaw) measurements arrive with a latency of up to 500 ms (`INS_MAX_DELAY_MS`). The filter keeps a ring buffer of recent states (nominal state, UDU factors, $(\varphi, \lambda, h)$, $\boldsymbol{R}_b^n$, $\boldsymbol{\omega}$; one entry every $\geq$20 ms, 30 entries). The residual of a delayed measurement is formed against the state at its time of validity, while the covariance update is applied to the *current* $\boldsymbol{U}, \boldsymbol{d}$. This is the standard approximation for short delays [1, 9]: the error state is nearly constant over a few hundred milliseconds ($\boldsymbol{\Phi} \approx \boldsymbol{I}$ for the error dynamics), whereas the residual would be grossly wrong if it were formed against the already moved on current state. A measurement that cannot be anchored (no history entry within 50 ms of the requested time) is skipped and counted.

The age is the sum of a whole-millisecond part (`gnss_delay_ms`, `local_pos_delay_ms`) and a microsecond part (`gnss_delay_us`, `local_pos_delay_us`). A fix is handed over with the first IMU epoch at or after its time of validity, so even a receiver without latency delivers it up to one IMU period old, far below the 20 ms history grid. Yaw and speed take the same two parts (`yaw_delay_us`, `speed_delay_us`). The state at the time of validity $t_v$ is interpolated between the two states around it at $t_a \leq t_v < t_b$, two history entries or the newest entry and the current state, with $w = (t_v - t_a)/(t_b - t_a)$: position, velocity and body rate linearly, the attitude by normalized linear quaternion interpolation (shorter arc). A fix 24 ms old is thus placed 4 ms past the 20 ms entry, not on it. Only if no pair closer than 100 ms brackets $t_v$ (a gap after a time jump) does the nearer state stand in, its position carried over the remaining $\Delta t = t_v - t_a$ at first order:

$$\hat{\boldsymbol{p}}(t_v) \approx \hat{\boldsymbol{p}}(t_a) + \hat{\boldsymbol{v}}(t_a)\,\Delta t, \tag{34}$$

with $\Delta t\,\boldsymbol{I}_3$ in the velocity block of the position Jacobian. Fused as current instead, a fix leaves a bias of speed times age along the track: 2.5 ms at $10\,\text{m}\,\text{s}^{-1}$ is 2.5 cm, the size of an RTK error budget.

## 5.7 Auto-initialisation

When enabled, `ins_init` leaves the filter in a collecting state and `ins_update` buffers bias-corrected IMU samples (bounded buffer) until the first usable position fix (GNSS or local). At that point:

- Roll/pitch from accelerometer leveling over a static window, with per-axis *medians* over the window (allowed here, since this is the init path and not the per-epoch path), gated on the gyro magnitude and $|\|\boldsymbol{f}\| - g|$:

$$\phi = \operatorname{atan2}(-f_y, -f_z), \qquad \theta = \operatorname{atan2}\Big(f_x, \sqrt{f_y^2 + f_z^2}\Big)\,. \tag{35}$$

- Yaw by priority: external yaw measurement (with its variance), else tilt-compensated magnetometer heading against the reference field (with the configured initial yaw variance), else unknown. In the last case yaw stays 0 with a $\sigma_\psi = \pi$ prior and the running fusion is left to pull it in (e. g. from lever-arm or velocity coupling).

- Position/velocity from the fix. With a local-position fix the geodetic anchor of the caller is kept as origin and the fix sets `pos_local`, so subsequent local measurements stay consistent. A GNSS fix locates the antenna: the origin is the fix, and the IMU starts at `pos_local` $= -\mathbf{R}_{b\to n}\,\mathbf{l}_b$, the lever arm rotated with the attitude that was just determined. The initial velocity is the velocity of the fix minus $\mathbf{R}_{b\to n}(\boldsymbol{\omega} \times \mathbf{l}_b)$, the own motion of the antenna on a turning platform. With the yaw still unknown only the vertical components are applied, since a horizontal arm rotated by an arbitrary yaw can land further off than not moving it at all.

- The height source is latched once here, for a GNSS bootstrap only, from whether a barometer is *streaming* at this instant: the most recent plausible sample cached during the collecting window must exist *and* still be current (`INS_BARO_ANCHOR_MAX_AGE_SEC`). Unlike the GNSS entry gate (§5.9) the sample is not quality-graded, since a pressure reading carries no reported covariance. But bare presence is not enough either, since the decision is never revisited and that same sample becomes the datum anchor. Streaming → barometric height (§5.5.2) with the anchor $h_0$ taken as the own $h_{\mathrm{ISA}}$ of `ins` for that sample at $h_{\mathrm{init}} = 0$. It is matched to the independently running height of `baro_alt` by the vertical-datum alignment right after this call returns (§3.3) and not read from `baro_alt`. Absent or stale → the vertical row of the GNSS position (19) as before, with a warning for the stale case: a barometer that stopped during the collecting window is a fault and not a configuration.

  There are two ways out of the barometric source. A local-position bootstrap never selects it regardless of barometer presence, since its own vertical accuracy is typically better than that of a barometer. And `baro_height_disable` suppresses it outright, for the same situation reached through the other interface: a GNSS-labelled input that is not satellite GNSS and already reports a better-than-barometric vertical accuracy, e. g. a precise indoor tracking system fed through `gnss_pos`/`gnss_vel` for convenience.

  It is not re-evaluated afterwards: a barometer that later fails is a plain aiding gap on that one measurement (§10) and not a source change. Since that gap leaves the vertical channel without an absolute reference while every other health signal stays green, it is reported explicitly (`ins`: "no barometric height aiding for …", alongside the `n_baro_height_used` counter).

The initial yaw variance is independently configurable from roll/pitch (`init_stddev: yaw_init_stddev_rad_deg`, `ins_init_t.rpy_init_stddev_rad[2]`), because leveling and heading come from physically different accuracy sources.

## 5.8 Automatic ZUPT/ZARU detection

A stillness detector (enabled by default) arms the ZUPT/ZARU fusions without an external trigger. It tests on three independent criteria that must all hold:

**Variance (primary statement).** The per-axis sample *variance* of the raw (not bias-corrected) gyro and accelerometer over a short window, evaluated when the window covers a minimum duration and sample count, with the result latched until the next window completes. Variance instead of magnitude, because a constant sensor bias shifts the sample mean but not its spread: a magnitude-only gate reads a biased but perfectly stationary IMU as moving, and this failure is self-locking (this detector is what would estimate the bias in the first place). It is configurable (`auto_zupt_static_gyr_stddev_deg` / `auto_zupt_static_acc_stddev_mps2`, $0 \rightarrow$ built-in default). Raising them out of range effectively disables the criterion, which a platform whose standstill is inherently noisy needs (a station-keeping multicopter puts more spread on the IMU from its rotors than handheld motion does).

**Magnitude bounds.** $\|\boldsymbol{\omega}\| \leq \omega_{\max}$, $\left|\|\boldsymbol{f}\| - g\right| \leq a_{\max}$, applied to the bias-*corrected* sample. It is retained alongside the variance criterion, because variance alone cannot distinguish a constant rotation rate or centripetal acceleration (steady turntable or circular drive: near-zero spread, plainly not stationary) from a standstill. The two criteria cover different failure modes and neither subsumes the other.

**Velocity gate.** $\|\boldsymbol{v}_{\mathrm{GNSS}}\| \leq v_{\max}$, using *exclusively* a recent, sufficiently precise external (GNSS) velocity observation. The observation must additionally report its own 1-sigma (per axis) at or below `auto_zupt_max_vel_stddev_mps`: a low reported speed with a large uncertainty is not evidence of standstill. Without a GNSS velocity that meets both the recency and the precision requirement, this gate does not apply. It passes unconditionally and only the IMU criteria above decide. This is a design decision, so that the ZUPT updates are not hardwired to the GNSS availability.

Triggering additionally requires a dwell time and is rate-limited. During the stillness run the raw gyro is accumulated for the ZARU average. All thresholds above, plus $v_{\max}$ (`auto_zupt_max_vel_mps`), are reachable from the `config.yaml` of both replay harnesses (`imu.auto_zupt_*`, §11), each $0 \rightarrow$ the built-in default.

Furthermore the detector can be switched on and off at runtime (`ins_set_auto_zupt_disable`), independent of the `auto_zupt_disable` option resolved at init. E.g. a multicopter that briefly holds position mid-flight can guarantee that no zero-velocity/zero-rotation update fires for exactly that window and resume the automatic detection once grounded. Disabling clears the in-progress dwell timer and the latched variance verdict, so re-enabling always requires a fresh stillness run instead of firing off a stale timer. Airborne vehicles should make use of this.

Moreover the ARS and magnetometer-AHRS (§3.3) have no velocity state, so they run their own, independent fallback stillness detector against the same physical IMU: the variance and magnitude criteria above, minus the velocity gate that ins can apply. It therefore also arms during genuine constant-velocity cruise. This is an accepted cost, since it is armed by default and is the only way to correct the otherwise permanently unobservable ARS/AHRS z-axis gyro bias when ins has no trigger to propagate (e. g. no aiding source at all).

These are separate implementations, but *one* configuration: inside `nav_suite` the thresholds, dwell and pseudo-measurement stddevs of that fallback are not set on `ahrs_config_t` at all. `nav_suite_init()` fills them from the same `ins_options_t.auto_zupt_*` set described above, and the vertical zero-velocity update of `baro_alt` inherits `zero_vel_stddev_mps` the same way. A caller therefore states what standstill means once, for the whole suite. Tuning them separately was a defect source and not a feature: tightening the gates of ins used to leave the ARS/AHRS (and with them the vertical ZUPT) on untouched defaults, invisibly.

Consequently there are two opt-outs and not one: `auto_zupt_disable` removes every stillness source in the suite, while `auto_zupt_velocity_blind_disable` removes only this velocity-blind fallback and leaves ins deciding for everyone. The latter is for a profile that the

IMU alone cannot tell from a standstill (in particular a noiseless synthetic constant-rate climb or constant-velocity leg: zero sample variance and a gravity-only specific force flatten exactly the trajectory under test). It has the same runtime switch as the own detector of ins (`ahrs_set_auto_zaru_disable`), and `nav_suite_set_auto_zupt_zaru_disable` throws both with one call (plus, transitively, the vertical zero-velocity update of `baro_alt`, which has no detector of its own and is driven entirely by these two).

### 5.8.1 Tuning the detector for a platform

The shared defaults are sized for a low-cost consumer MEMS IMU (`sensor_defaults.h`). A different sensor class or a different carrier usually needs the thresholds moved, and the two ways to get it wrong leave *different* fingerprints in the replay scoring output:

**Arms too rarely.** `n auto-zupt` near zero. Biases that the detector exists to observe (notably the ARS/AHRS z-gyro bias, which nothing else can see) drift away unchecked. Typical cause: a platform whose standstill is inherently noisy (a station-keeping multicopter puts more spread on the IMU from its rotors than handheld motion does), so *raise* the variance thresholds.

**Arms while moving.** The error *scatter* grows while the *bias* stays put. This combination is the typical indicator: each spurious update yanks the velocity to zero and the next fix pulls it back, which adds noise without shifting the solution, whereas a genuine accuracy loss normally moves both. Typical cause: a sensor that is quieter than the defaults assume, so *lower* them.

Both are directly readable from a replay run: the summary line reports `n auto-zupt` and the resolved thresholds are logged at `INFO` on startup (`ins: auto-zupt/zaru: window stddev ...`), so there is no need to infer what the filter actually used.

The knobs should be turned in this order. The **variance** thresholds (`auto_zupt_static_gyr_stddev_deg`, `auto_zupt_static_acc_stddev_mps2`) come first, as this is the criterion that is meant to do the actual rejecting. The **magnitude** bounds (`auto_zupt_static_gyr_deg`, `auto_zupt_static_acc_mps2`) are sized for bias *tolerance* and not for motion rejection, so tightening them to reject motion re-opens the self-locking failure that the variance criterion exists to avoid. The two are tuned as a pair. The **velocity** gate (`auto_zupt_max_vel_mps`, `auto_zupt_max_vel_stddev_mps`) only does anything if the receiver actually reports a velocity *with* its covariance. On a position-only stream it never applies, whatever it is set to.

**Worked example: `datasets/kfgins`.** An automotive fibre-optic-gyro dataset that trips the second failure at the built-in defaults. Its GNSS stream is position-only (no velocity), so the velocity gate never applies and the two IMU criteria decide alone. A navigation-grade FOG in a smoothly cruising car is also far quieter than the consumer MEMS that the defaults are sized against. The detector armed mid-drive: 357 triggers instead of 328, attitude-error scatter up thirtyfold, biases unmoved. Sweeping only `auto_zupt_static_gyr_stddev_deg`:

| threshold [$^\circ\,s^{-1}$] | `n auto-zupt` | roll err. $\sigma$ [$^\circ$] | yaw err. $\sigma$ [$^\circ$] | ARS yaw drift [$^\circ\,min^{-1}$] |
|---:|---:|---:|---:|---:|
| 0.25 (default) | 357 | 0.120 | 0.148 | 0.775 |
| 0.15 | 335 | 0.053 | 0.049 | – |
| 0.12 | 329 | 0.004 | 0.026 | 0.253 |
| **0.10** | **315** | **0.004** | **0.026** | **0.011** |
| 0.08 | 245 | 0.004 | 0.026 | 0.120 |
| 0.03 | 92 | 0.004 | 0.026 | 0.725 |
| $\leq 0.02$ | 0 | 0.004 | 0.104 | – |

Two things generalise from the shape of that table. The `ins` metrics have a broad *plateau* (here 0.03 to 0.12), so the value is not picked at the first setting that passes: a threshold chosen on an edge is tuned to the noise of one dataset and not to the platform. And over-tightening is not harmless: in the bottom row, with no triggers at all, the yaw error grows again, because the genuine standstill updates were doing real work.

Where the `ins` plateau leaves the choice open, the *ARS* resolves it. The same key configures every stillness detector in the suite (§5.8), and the ARS runs free-yaw here (no magnetometer), so its $z$-gyro bias is observable only through ZARU. This turns its yaw drift into a smooth optimum in the last column: too high and spurious mid-drive updates corrupt the bias, too low and there are too few updates to determine it. That optimum is visible only on a fine sweep, sampled coarsely the column reads as noise.

### 5.9 Dead reckoning, re-acquisition, readiness

The filter tracks the time since the last absolute position aiding. Within the coasting window (`max_deadreckoning_sec`, default 10 s, which is roughly how long a consumer-MEMS strapdown position stays usable), incoming fixes fuse normally and the filter keeps dead-reckoning on the IMU. Once the window expires, `ins_is_ready()` degrades to false *and the strapdown stops*: attitude, IMU biases, position and velocity all freeze at their last value instead of dead-reckoning through an outage of unknown length.

The Kalman time update and the measurement fusion stop with it: an expired window makes the whole filter instance *inactive*. Nothing is propagated, nothing is fused and neither the state nor the covariance changes on such an epoch. `ins_is_ready()` has been false since the window expired, so no consumer is entitled to a solution meanwhile. Inside `nav_suite` the vertical channel is served by `baro_alt` and the attitude by the ARS/AHRS, which both run the same sensors and neither of them is frozen.

The *first usable fix re-acquires* instead of fusing. The re-acquisition:

- Re-anchors position (and velocity, if measured) directly from the measurement. The exception is the vertical position under the barometric height source (§5.5.2), which is re-anchored on the last cached barometer sample instead, through the same datum-referenced conversion the fusion uses. The barometer knows where the platform is. The raw sample carries no filter lag of its own and its single-sample accuracy becomes the vertical variance. A GNSS fix is moved from the antenna to the IMU first: position by $\mathbf{R}_{b\to n}\,\mathbf{l}_b$, velocity by $\mathbf{R}_{b\to n}(\boldsymbol{\omega}\times\mathbf{l}_b)$ with the gyro rate of this epoch (the frozen filter computed none during the outage), rotated with the attitude that the filter will hold after an attitude hint is applied. Without a heading in that hint the yaw is unknown (next item) and only the vertical components are applied.

- Without a measured velocity (a GNSS fix without one, any local position fix) keeps the frozen velocity only as a value: it dates from the start of the outage, so its variance is reset to an unknown-velocity prior and the following fixes determine it.

- *Keeps attitude and IMU biases*: frozen, unchanged since the window expired. Re-leveling would need a static phase, which a vehicle does not have at a tunnel exit

- *Inflates the variances of everything it keeps* by the process-noise PSD of that state times the frozen duration. This is the same random-walk growth that the time update would have accumulated, computed once from the known elapsed time and capped at the configured initial priors, so a re-acquisition after an arbitrarily long outage is at worst as uncertain as a cold start, never worse.

- *Resets the yaw* prior to "heading unknown" ($\sigma \approx 180°$, the same prior the bootstrap uses with no heading source) unless an attitude hint carries an absolute yaw.

- Rebuilds the covariance as a diagonal: fresh position/velocity variances from the measurement noise, inflated attitude/bias variances kept (lost cross-correlations rebuild within a few aided epochs).
- Invalidates the stale history buffer.

The inflation runs *before* the attitude hint and the yaw prior, so an externally supplied variance always wins over this estimate. With `allow_unlimited_deadreckoning` the window and the re-acquisition are disabled (for applications that prefer a drifting solution over an unavailable one). The readiness additionally requires a minimum number of Kalman epochs and runtime after (re-)initialisation.

"Absolute position aiding" is measured on the fusion checks and not on what was actually fused: a position withheld by the decimation of §5.5.7 arrived and was usable, and keeps the window open like any other. Only a position that fails the gates or never arrives opens a gap. This is also why a GNSS *velocity*-only fix, which is the usual receiver behaviour at a tunnel exit where Doppler re-converges first, must not refresh the timestamp on its own: the position fix arriving a beat later would find the window un-expired and would be run through an ordinary residual fuse it has no chance of surviving, instead of re-anchoring. For the same reason the decimation is suspended while the window is expired.

**Aiding *quality* versus aiding *presence*.** The coasting window above answers "is aiding arriving at all". A separate question is whether the aiding that *is* arriving is good enough to carry a 3D solution. The filter answers it with three independent threshold sets on the same reported GNSS $1\sigma$ values:

- `gnss_max_*`: the *fusion* gate, i. e. may this individual fix be fused (the loosest set, per measurement);
- `gnss_start_max_*`: the *entry* gate, i. e. is the fix stream good enough to enter the 3D solution. It is the strictest set and it must hold continuously for `gnss_init_dwell_sec` at $\geq$1 Hz. The velocity threshold carries the decision here (default $0.25\,\mathrm{m\,s^{-1}}$ $1\sigma$ per axis), because the vertical channel, the accelerometer biases and the yaw observability of a moving platform all lean on the GNSS velocity;
- `gnss_stop_max_*`: the *exit* gate. Once every fix over `gnss_stop_dwell_sec` (default 10 s) fails it (default $0.4\,\mathrm{m\,s^{-1}}$ on velocity), the 3D solution is left and the suite falls back to attitude-only. The filter does not merely stop claiming a solution, it stops: it re-arms into the collecting state, so integration and fusion cease and every accessor reports "unavailable" instead of a value drifting off a fix stream that was already judged too poor. Returning means a full re-bootstrap through the entry gate and dwell, never a single recovered fix. Position, velocity and attitude are re-derived there. The IMU biases carry across with an inflated uncertainty clamped to the cold-start prior, and so does the n-frame origin, so local NED coordinates issued before the outage still mean the same places afterwards.

Only epochs that actually offer aiding are judged, so a plain outage stays the business of the coasting window. The asymmetry (strict and short to enter, loose and long to leave) is the hysteresis of the transition: a premature entry starts the solution from a poor state, a premature exit throws away a converged one that is still the best information available, and a single threshold would let a marginal stream flap the mode. `ins_init` clamps the entry set to at most the fusion set and the exit set to at least the entry set, so a mis-ordered configuration cannot invert it.

### 5.10 Vertical origin relocation

`ins_shift_origin_down(`$\Delta z$`)` moves the n-frame origin along the local down axis without changing the absolute solution. Down is the ellipsoid normal, which is the axis the height is measured along, so the shift touches the origin height alone: $h_{\text{origin}}$ decreases by $\Delta z$ while $\varphi_{\text{origin}}$ and $\lambda_{\text{origin}}$ stay bit-for-bit unchanged, `pos_local[2]` (state and history) shifts by $-\Delta z$, and $(\varphi, \lambda, h)$, covariance and Earth caches are untouched. The shift is deterministic, so no uncertainty is added. The suite uses this to align the ins vertical datum with an already running barometric filter at bootstrap (§3.3).

## 6 AHRS: standalone attitude filters

### 6.1 Model

The AHRS estimates a quaternion $\boldsymbol{q}_b^n$ and a gyro bias $\boldsymbol{b}_g$ with an error state of 5 (ARS: $[\delta\phi\ \delta\theta\ \delta\boldsymbol{b}_g]$) or 6 (AHRS: $[\delta\phi\ \delta\theta\ \delta\psi\ \delta\boldsymbol{b}_g]$) elements, i.e. without position, velocity and accelerometer bias, which are not observable here [1]. It uses the same n-frame psi-angle model, sign convention and UDU backend as INS. In ARS ("directional gyro") mode the yaw is integrated from the gyro but never corrected, in AHRS mode it is stabilised by magnetometer heading measurements. The mode is fixed per instance. The runtime-$n$ pattern matches the 15/18 handling of ins. The Earth rotation ($15\,^\circ\,\mathrm{h}^{-1}$) is *not* compensated here.

**Propagation.** At IMU rate the quaternion is advanced with the bias-corrected rate $\boldsymbol{\omega} = \tilde{\boldsymbol{\omega}} - \boldsymbol{b}_g$. The covariance prediction is throttled to 20 Hz with [1, eq. 10.9]

$$\boldsymbol{\Phi} = \boldsymbol{I} + \boldsymbol{F}\Delta t, \qquad \boldsymbol{F} = \begin{bmatrix} \boldsymbol{0} & -\boldsymbol{R}_{[0:c,:]} \\ \boldsymbol{0} & \boldsymbol{0} \end{bmatrix}, \qquad \boldsymbol{G} = \begin{bmatrix} -\boldsymbol{R}_{[0:c,:]} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I}_3 \end{bmatrix}, \qquad \boldsymbol{Q} = \Delta t \begin{bmatrix} \sigma^2_{\text{gyr}}\,\boldsymbol{1}_3 \\ \sigma^2_{b\text{-RW}}\,\boldsymbol{1}_3 \end{bmatrix}, \tag{36}$$

where $\boldsymbol{R} = \boldsymbol{R}_b^n$ and $c = 2$ (ARS) or 3 (AHRS) selects the attitude rows.

### 6.2 Accelerometer leveling

The accelerometer is low-pass filtered (first-order RC, $\alpha = \Delta t/(\tau + \Delta t)$, $\tau = 1/(2\pi f_c)$, default cut-off 10 Hz) and fused at a throttled rate (default 5 Hz) against the gravity direction:

$$\boldsymbol{z}_a = \boldsymbol{f}^b_{\text{LP}} + (\boldsymbol{R}_b^n)^{\mathsf{T}}\,\boldsymbol{g}^n \;=\; (\boldsymbol{R}_b^n)^{\mathsf{T}}\,[\boldsymbol{g}^n]_\times\,\boldsymbol{\psi} + \boldsymbol{v}, \tag{37}$$

so with $\boldsymbol{g}^n = [0\ 0\ g]^{\mathsf{T}}$ only the roll and pitch columns of $\boldsymbol{H}$ are non-zero (the yaw column of $[\boldsymbol{g}^n]_\times$ vanishes). Horizontal accelerations can therefore never feed yaw. The measurement noise is inflated during dynamics:

$$\sigma_{\text{eff}} = \sigma_{\text{acc}} + k \cdot \big|\,\|\boldsymbol{f}_{\text{LP}}\| - g\,\big|, \tag{38}$$

with the *gravity-diff penalty* $k$ (default 8 per $\mathrm{m\,s^{-2}}$). The default $\sigma_{\text{acc}} = 0.25\,\mathrm{m\,s^{-2}}$ includes a margin, so a moderate disturbance still passes the robust chi-square gate with downweighting, instead of being rejected outright.

Beyond that graded inflation there is a hard gate (`acc_reject_gravity_mps2`): once the low-passed $\big|\,\|\boldsymbol{f}\| - g\big|$ exceeds it, the leveling update is dropped outright instead of being downweighted. A specific force that far from gravity is a manoeuvre or a shock and not a leveling reference. Unlike a persistent reference offset the deviation is inherently transient, so a hard drop cannot deadlock the filter the way it would for GNSS or the barometer (§3.4). A negative value disables the gate.

**IMU loss.** A sample arriving `imu_loss_timeout_sec` (0.2 s) or more after the previous one stops the filter: the rotation of the gap is unknown. The gyro bias ($1\sigma \times 3$, clamped to the cold-start prior) is kept for the restart (`ahrs_get_bias_carry`). `ahrs_stop_imu_loss` stops it the same way for a caller that detects the loss itself, as `nav_suite` does (§3.3).

**Attitude-precision restart watchdog.** Past a warm-up after init (`restart_warmup_sec`), the own reported $1\sigma$ of each tracked attitude axis is checked against `restart_att_stddev_rad`. Exceeding it marks the filter uninitialised through the same fail-safe as the health check, e. g. after a long outage of the correcting sensor has let the covariance grow past anything trustworthy. Inside `nav_suite` the instance then re-bootstraps from the live stream on the next epoch, a standalone caller must call `ahrs_init()` again. The warm-up keeps the check from firing while the filter is still converging from a deliberately loose initial covariance. Per-axis thresholds can be disabled individually (negative value), the whole watchdog with `precision_restart_disable`. `baro_alt` carries the same mechanism on its height and vertical velocity (§7.2).

### 6.3 Magnetometer heading (scalar fusion)

In AHRS mode the body field is de-tilted with the current roll/pitch [1, Sec. 10.1]

$$\begin{bmatrix} h_x \\ h_y \end{bmatrix} = \left[ \boldsymbol{R}_y(\theta)\, \boldsymbol{R}_x(\phi)\, \tilde{\boldsymbol{m}}^b \right]_{1:2}, \qquad \psi_{\text{mag}} = \operatorname{atan2}(-h_y,\, h_x) + \delta_{\text{decl}}, \tag{39}$$

and fused as a scalar heading residual

$$z = \operatorname{wrap}_{\pm\pi}(\psi_{\text{mag}} - \hat{\psi}) = -\delta\psi + v, \qquad \boldsymbol{H} = \begin{bmatrix} 0 & 0 & -1 & 0 & 0 & 0 \end{bmatrix}. \tag{40}$$

Only yaw is observed, magnetic disturbances cannot tilt roll/pitch. The fusion is skipped near gimbal lock, for near-zero fields and when the de-tilted horizontal component is unusably small relative to the total field. The latter is a measurement-side guard and only bites within about a degree of a dip pole, so it is backed by the position-side exclusion zone (§9.2), which withdraws the heading reference over the surrounding region where the declination is already unusable.

In contrast to the vector fusion of ins (§5.5.10), the scalar heading uses only the field *direction* and is therefore unit-agnostic. This is a deliberate difference: the standalone AHRS must work with uncalibrated, direction-only magnetometers. Consequently its field-strength gate (same $(\epsilon/\tau)^2$ inflation as in ins) is *opt-in* and off by default: enabling it requires the magnetometer in µT. Armed unconditionally it would trip on every sample of a direction-only sensor and silently kill the heading aiding. ins keeps its gate *opt-out*, because µT is contractual for the vector fusion anyway.

### 6.4 True-north referencing (declination step)

Until a position is supplied the AHRS heading references magnetic north. `ahrs_set_position()` obtains the WMM declination $\delta_{\text{decl}}$ and applies the *change* in declination as a deterministic left rotation about the n-frame down axis:

$$\boldsymbol{q} \leftarrow \boldsymbol{q}_z(\Delta\delta_{\text{decl}}) \otimes \boldsymbol{q}, \tag{41}$$

so the estimated yaw *steps* to true north instantly instead of slewing there over subsequent magnetometer updates. Covariance and gyro bias are untouched, since a known rotation adds no uncertainty. The $+\delta_{\text{decl}}$ term stays in the measurement model and holds the steady state at true north.

The step is only sound while the yaw is *magnetically anchored*, which is what makes the rotation a known one. Inside a dip pole exclusion zone (§9.2) the AHRS suspends the magnetometer yaw fusion and keeps its previous declination, so the yaw is carried by the gyro instead. On the position that leaves the zone the new declination is therefore adopted *without* re-framing the attitude and the fusion pulls the accumulated drift out through the ordinary covariance weighting. Re-framing there would be actively harmful: the declination on the far side of a dip pole differs by tens of degrees (about 80° for a typical crossing) and applying that as a rotation would corrupt an otherwise sound estimate instead of correcting it. Roll and pitch are unaffected throughout, only the heading reference is withdrawn.

The one case with no previous declination to keep is a first position that already lies inside a zone, after the yaw has been fused against magnetic north. That yaw is a magnetic heading and no re-framing will ever make it a true one, so the AHRS widens its yaw variance to "heading unknown" there, instead of letting the covariance of that fusion stand. While the attitude-precision restart watchdog is armed, the widened value stays just below its yaw threshold, since a restart would discard the zone state together with the position. Otherwise the nav_suite attitude hint would hand the magnetic heading to ins as a converged true heading. A yaw that was never fused against the magnetometer keeps its variance.

# 7 Vertical channel: height systems, baro_alt and the offset filter

## 7.1 Height systems

Three vertical reference systems are in play, and mixing them up is the most common source of confusion in the vertical channel (Figure 5):

**Local height $h$.** Height above the NED origin, positive up. It is zero at the datum origin *by construction*, whichever sensor started first. This is what a controller consumes (`nav_suite_get_height`), it is available as soon as any vertical source runs and it is what all parallel filters express themselves in (`ins` as $-\text{pos_local}_z$, `baro_alt` as its state $h$). With a barometer present, the own $h$ of `ins` is itself barometrically fused (§5.5.2) and not merely aligned to the one of `baro_alt`. The two remain independent estimates of the same physical quantity. They share the same raw pressure stream and the same datum, but are not tied together by construction, which is why the suite cross-checks them (REQ-SUITE-019). `nav_suite_get_height` reports the $h$ of `ins` only in `FULL` mode and the one of `baro_alt` otherwise: the vertical state of `ins` stays live through an outage, but `ins` as a whole does not (its horizontal solution is what `FULL` is about), and `baro_alt` is also the only source before `ins` has bootstrapped at all.

**Barometric (ISA) altitude $h_{\mathbf{ISA}}$.** The own height system of the barometer, Equation (47) (see `baro_alt_get_isa_altitude`). It is *not* absolute: it carries the weather-dependent pressure offset and the ISA model error, and it refers to the ISA sea level and not to the ellipsoid. It is exposed for diagnostics and for callers that want the raw barometric height.

**Ellipsoid height $h_{\mathbf{ell}}$.** Absolute, WGS84 (`nav_suite_get_height_ellipsoid`). This is *not always available*, hence the boolean return.

**Who defines the datum.** The governing rule is: *the datum is fixed once by the first source and never moves. A later source conforms to it at its own initialisation.* So the local height starts at zero in every startup order and, which is important for a height controller, never jumps when a source arrives or drops. The first local solution of the 3D filter is therefore always $(0, 0, h)$: horizontal zero (there is no horizontal reference before the first fix), vertical $h$ equal to

the current barometric height (0 if no barometer has run yet, non-zero if the vehicle already climbed on the baro before the fix).

Concretely, when the barometer is already running as `ins` bootstraps, `ins` is the one that must join, and *how* depends on who owns the $n$-frame:

- *Nobody external* (GNSS auto-init, or IMU only): the origin is wherever `ins` happened to bootstrap, i. e. arbitrary. The suite shifts it (`ins_shift_origin_down`), so that the local height matches the barometric one. The datum is the mission start and the absolute solution is untouched. This is safe, because no one else references that frame.

- *An external local-position system* (lighthouse, via `ins_meas_local_pos_t`): this system owns the $n$-frame and the contract of `ins_shift_origin_down` requires the external frame to move along, which the suite cannot do. Shifting anyway would put a constant offset on every `local_pos` innovation. Since local references are *skipped* and not downweighted on outliers, every one would be rejected and the aiding would die silently, while the mode still read `FULL`. The suite therefore reconciles *at the source*: it subtracts a constant offset from every `local_pos` before `ins` sees it, so `ins` bootstraps at $(0, 0, h)$ and stays in lock-step with the external stream. This offset, i. e. the coordinates of the datum origin in the external frame, is exposed through `nav_suite_get_local_pos_offset()` (§8).

When `ins` is instead already initialised as the first pressure sample arrives, the barometer simply anchors at the current NED height and not at 0. This is gated on *initialisation* and not on `ins_is_ready()`: a local height may already have moved from the origin during the readiness warm-up, and anchoring at 0 would bake that difference in forever.

The alignment is held per $n$-frame *origin* and not per `ins` instance. A re-bootstrap that anchors a new origin has to re-join the datum, one that keeps the origin carried across a quality-loss re-arm (Fig. 3) must be left alone, because it never left the datum. The difference matters mid-mission: an alignment treats the barometric height as the datum-relative height, which holds when the barometer has just anchored, but not after it has drifted against the fix (weather, ISA model error, the own pressure of a tunnel). Shifting a carried origin onto that reading would push the accumulated disagreement into the origin and move the datum and every local coordinate the consumer already holds. This is what carrying the origin exists to prevent. The carried origin brings the barometric anchor of the exiting instance along with it, so the re-bootstrap takes its initial height from the barometer on that anchor and not from the bootstrap fix. A drone that loses GNSS, holds its altitude on the barometer and then re-enters the 3D solution therefore does not see a step in its height: the vertical error that the fix happens to carry at that moment (several metres right after an indoor or urban outage) never reaches the height channel, which the vertical row of the fix does not aid under the barometric source anyway.

**Bridging local to absolute.** The bridge is the offset $o$ of §7.3, i. e. *the ellipsoid height of the datum origin*:

$$h_{\text{ell}} = h + o. \tag{42}$$

An absolute height therefore requires GNSS *now* (then `ins` reports it directly), or GNSS for long enough *earlier* to have estimated $o$, after which the local height source alone carries the absolute height through the outage. With no GNSS ever, $o$ is unknown and `nav_suite_get_height_ellipsoid` returns false while the local height keeps working. This is the whole point of the split: the local height is a *guarantee*, the absolute height is an *offer*.

## 7.2 baro_alt: 3-state baro/accelerometer filter

Unlike `ins/ahrs` this is a plain (full-state) *linear* Kalman filter. The model is linear, so no error-state formulation is needed. The barometer/accelerometer vertical channel model is

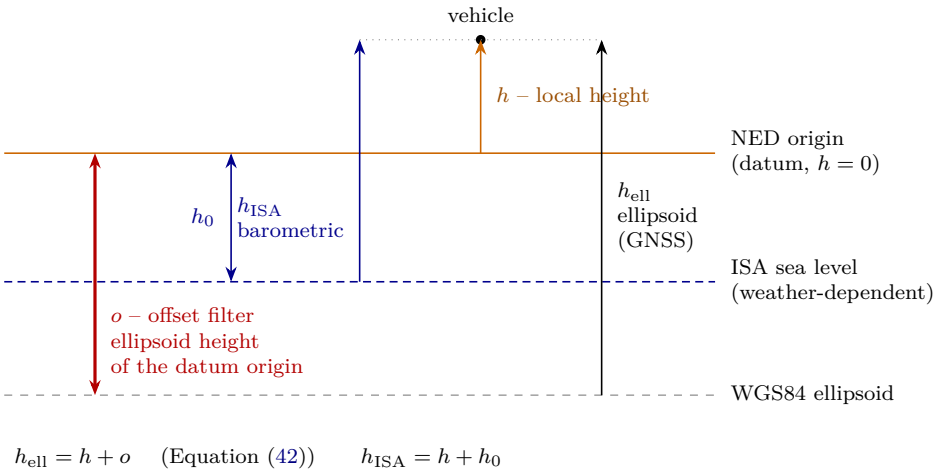



Figure 5: The three height systems. The local height $h$ is measured from the NED origin and is zero at the start point. The offset filter estimates $o$, the ellipsoid height of that origin, which is what turns a local height into an absolute one during a GNSS outage. While GNSS is up and `ins` runs on the barometric height source, it is also what measures how far the own vertical datum of `ins` has drifted away from the ellipsoid, since nothing else re-ties it to GNSS (§7.3). The barometric ISA altitude is a third, non-absolute system, offset from the datum by the anchor constant $h_0$. Note that the ISA sea level drifts against the ellipsoid with the weather, which is exactly why $o$ is modelled as a random walk and not as a constant.

described in detail in [10]. State vector:

$$\boldsymbol{x} = \begin{bmatrix} h & v & a_b \end{bmatrix}^{\mathsf{T}}, \tag{43}$$

height above datum $h$ (m, positive up), vertical velocity $v$ and a slowly varying additive correction $a_b$ to the measured vertical acceleration (absorbing accelerometer bias, attitude error and gravity model error projected onto the vertical axis).

**Prediction: accelerometer as control input.** In each IMU epoch the body specific force is rotated to NED with the *supplied* attitude quaternion (from `ins` or an AHRS), gravity is removed and the sign is flipped to positive-up:

$$a_{\mathrm{up}} = -\left(\left[\boldsymbol{R}_b^n \boldsymbol{f}^b\right]_z + g\right). \tag{44}$$

The state propagates with constant-acceleration kinematics:

$$\boldsymbol{x}_{k+1} = \boldsymbol{\Phi}\,\boldsymbol{x}_k + \boldsymbol{B}\,a_{\mathrm{up}}, \qquad \boldsymbol{\Phi} = \begin{bmatrix} 1 & \Delta t & \tfrac{1}{2}\Delta t^2 \\ 0 & 1 & \Delta t \\ 0 & 0 & 1 \end{bmatrix}, \qquad \boldsymbol{B} = \begin{bmatrix} \tfrac{1}{2}\Delta t^2 \\ \Delta t \\ 0 \end{bmatrix}, \tag{45}$$

(note that $\boldsymbol{\Phi}$ couples $a_b$ into $h, v$, so the effective acceleration is $a_{\mathrm{up}} + a_b$). Process noise in noise-input form, $\boldsymbol{G} = [\boldsymbol{e}_1 \;\; \boldsymbol{e}_2 \;\; \boldsymbol{e}_3]$, $\boldsymbol{Q} = [\sigma_h^2\Delta t,\ \sigma_a^2\Delta t,\ \sigma_b^2\Delta t]$, i.e.

$$\boldsymbol{Q}_d = \mathrm{diag}(\sigma_h^2\,\Delta t,\ \sigma_a^2\,\Delta t,\ \sigma_b^2\,\Delta t). \tag{46}$$

The accelerometer noise enters the own derivative of $v$ directly (unit weight on $\boldsymbol{e}_2$) and *not* through $\boldsymbol{B}$: without a noise column of its own, $h$ would inherit the injected uncertainty of $v$ only one step later, through the $v \to h$ coupling of $\boldsymbol{\Phi}$. The exact continuous-discrete model for this double integrator additionally places small $Q_{hh} \sim \sigma_a^2\Delta t^3/3$ and $Q_{hv} \sim \sigma_a^2\Delta t^2/2$ terms at

every step, which the $\boldsymbol{G}$ with unit weight on $v$ only leaves at zero. $\sigma_h$ (`h_process_noise_m_sqrthz`) is a small, independently tunable stand-in for that omitted term, a margin against discretization/model mismatch and not a primary noise source.

$\sigma_h$, $\sigma_a$ (`acc_noise_mps2_sqrthz`) and $\sigma_b$ (`acc_bias_drift_mps2_sqrthz`) are continuous-time spectral densities ($\sqrt{\text{Hz}}$) scaled by the actual $\Delta t$, so the injected variance per second is IMU-rate-invariant. A fixed per-sample sigma would over- or under-inject noise at other rates. The accelerometer is a control input and *not* a measurement: between barometer samples the filter dead-reckons on it.

**Barometer measurement.** Pressure is converted with the International Standard Atmosphere:

$$h_{\text{ISA}}(p) = 44330 \left(1 - \left(\frac{p}{p_0}\right)^{1/5.255}\right) \text{ m}, \qquad p_0 = 101\,325\,\text{Pa}, \tag{47}$$

anchored at the init pressure sample: the ISA altitude of the anchor corresponds to the height $h_{\text{init}}$ above the datum of the caller (default 0 = "height above start"), which gives the datum zero point $h_0 = h_{\text{ISA}}(p_{\text{anchor}}) - h_{\text{init}}$. Samples are fused as the scalar $z = h_{\text{ISA}}(p) - h_0$ with $\boldsymbol{H} = [1\ 0\ 0]$ through the same robust UDU update. Outliers are downweighted (persistent absolute reference). A typical source of such outliers are pressure disturbances close to the ground, e.g. by the rotor downwash of a multicopter during take-off and landing [9]. Pressure plausibility is enforced at the boundary (10 kPa to 120 kPa).

**Zero-velocity update.** A caller-triggered pseudo-measurement (`baro_alt_zero_velocity_update`, e. g. driven by the same stillness detector as the own ZUPT/ZARU of `ins`, §5.8) observes $v$ directly:

$$z = 0, \qquad \boldsymbol{H} = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}, \tag{48}$$

fused through the same robust UDU update as the barometer (outliers downweighted, not dropped). It is *not* gated on the own $v$ estimate of the filter. That estimate is exactly what is being corrected, so gating on it would suppress the update precisely when the accelerometer-only dead reckoning has drifted furthest and needs it most. Beyond bounding $v$, the update also pulls on $a_b$ through the $v/a_b$ covariance coupling built up by the prediction, so a standstill corrects the acceleration bias even while the barometer is unavailable.

**Precision restart watchdog.** Past a warm-up period after init (`restart_warmup_sec`), the own reported $h$ and $v$ 1-sigma of the filter (from the UDU factors) are checked against `restart_h_stddev_m` / `restart_v_stddev_mps`. If either is exceeded, the filter is marked uninitialized (the same fail-safe mechanism as the general health check), e. g. after a long barometer outage has left the accelerometer-only dead reckoning too uncertain to trust. `nav_suite` then re-bootstraps the vertical channel from the live stream, a standalone caller must call `baro_alt_init` again. Either threshold can be disabled individually (negative value) or the whole watchdog can be turned off (`precision_restart_disable`).

### 7.3 local_gnss_alt: 1-state offset filter

A scalar random-walk Kalman filter estimates

$$o = h_{\text{GNSS,ellipsoid}} - h_{\text{local}}, \tag{49}$$

the ellipsoid height of the datum origin (§7.1). Both heights refer to the IMU point: the fix reports the antenna, so the suite first moves it by the down component of the GNSS lever arm, rotated with its best available attitude, $h_{\text{GNSS,ellipsoid}} = h_{\text{antenna}} + (\mathbf{R}_{b\to n}\,\mathbf{l}_b)_D$. Without

that, $o$ would absorb the lever arm and the absolute height would jump by it whenever `nav_suite_get_height_ellipsoid` switches between ins and $h_{\text{local}} + o$. The filter is *agnostic* to where the local height comes from: with the barometric source $o$ lumps the weather-dependent pressure offset, the geoid undulation and the ISA model error, with an external local height (e. g. a lighthouse system) it lumps the geoid undulation and the own height of the datum. Propagation $\sigma_o^2 \mathrel{+}= \sigma_{\text{RW}}^2 \, \Delta t$ (default $0.3\,\text{m}/\sqrt{\text{s}}$: slow enough to average pair noise, fast enough for the ISA-model error, confidence decays during outages). It is that term and not the weather that sizes the default. Weather drift of metres per hour would be satisfied by an order of magnitude less. The ISA error is instead a fraction of the height *excursion* (a tenth of it is ordinary, since the model assumes a sea-level temperature that the real atmosphere rarely has), so a few hundred metres of terrain climbed in a few minutes move $o$ by tens of metres in those minutes. A random walk in *time* can only bound a drift whose driver is altitude, never model it, and the bound has to stay wide enough that the chi-square screening below still reads such a drift as drift: too slow a walk turns real, persistent drift into a stream of apparent outliers and the downweighting then holds $o$ back precisely when it has the most catching up to do. Update with pair variance $R = \sigma_{\text{local}}^2 + \sigma_{\text{GNSS}}^2$ and the same chi-square downweighting, written out directly ($n$=1, no UDU needed):

$$K = \frac{\sigma_o^2}{\sigma_o^2 + R}, \qquad o \leftarrow o + K\,\delta z, \qquad \sigma_o^2 \leftarrow (1 - K)\,\sigma_o^2. \tag{50}$$

Furthermore both stddevs are derated by a configurable inflation factor and the fusion is decimated to a configurable minimum interval, which together keep a single pair from nudging the offset around.

**Which local height to feed it with.** The suite feeds the filter from the source that *survives a GNSS outage* (the barometer if it is running, otherwise the `ins` local height once an external local position source drives it) and *not* from the momentarily most accurate one. While GNSS is up the `ins` local height is by far the better estimate, but pairing it against GNSS would yield a drift-free $o$ that is useless the moment it is needed, because the height it would then be added to (the one of the barometer) carries a weather drift that $o$ never learned to absorb. The same reference evaluates Equation (42), so offset and height always refer to one source. Under GNSS alone, without barometer and local position aiding, there is no such source: the `ins` height is then itself a GNSS height, and $o$ would only re-estimate the ellipsoid height of the datum origin, which `ins` knows exactly, at the metre-grade $\sigma_o$ of the random walk tuned for the barometer. The filter therefore stays idle and a coasting `ins` reports the absolute height.

**Correcting the datum drift.** Under the barometric height source the absolute height of `ins` is only as good as its vertical *datum*. That datum is anchored once at bootstrap and never re-tied to GNSS (REQ-NAV-055), so the whole absolute error of the barometer accumulates in it for the rest of the run. Its height *relative* to that datum stays excellent, so the absolute accessor keeps `ins` and subtracts only how far $o$ has moved since the datum was fixed. Two details make that safe. The anchor *tracks* $o$ while the filter is still converging and freezes when it settles: a snapshot taken mid-convergence books the remaining convergence as drift that never happened and, being a constant, that error becomes a permanent bias. And only the part of the drift exceeding one $\sigma$ of its own estimate is applied, less that margin: `ins` is the better estimate whenever the datum has *not* moved, which is most missions most of the time. Replacing its decimetre-grade height with $h + o$ outright would buy back tens of metres on a long drive, at the price of the metre-grade noise of the offset filter on every epoch everywhere else. When the fix is delayed, the local height is extrapolated to the GNSS time of validity with the estimated climb rate.

**Datum coupling.** The offset is tied to the datum and would have to move with any datum relocation. Under the "the datum never moves" rule (§7.1) that never arises: when the suite relocates the *ins* origin onto the (stationary) baro datum at a GNSS bootstrap, the datum the offset refers to has *not* moved, so the offset is left untouched. Shifting it there would put a permanent $\Delta z$ error on every absolute height.

# 8 Frame offsets and coordinate conversion

A vehicle rarely lives purely in the local $n$-frame: an indoor drone must relate the frame of the Lighthouse system to the datum of the suite, and an outdoor drone must turn WGS84 waypoints into the local frame its controller flies in. The suite exposes both bridges as accessor pairs, one per side.

**Local-position frame (§7.1).** An external local-positioning system reports in its own frame, whose origin is arbitrary. The suite reconciles it by subtracting a constant offset $\boldsymbol{o}_{\text{lp}}$ from every `local_pos` before `ins` consumes it, so `ins` always solves at $(0, 0, h)$ relative to the datum. $\boldsymbol{o}_{\text{lp}}$ is the coordinates of the datum origin in the external frame, exposed by `nav_suite_get_local_pos_offset()`:

$$\boldsymbol{p}_{\text{external}} = \boldsymbol{p}_{\text{local}} + \boldsymbol{o}_{\text{lp}}, \qquad \boldsymbol{p}_{\text{local}} = \boldsymbol{p}_{\text{external}} - \boldsymbol{o}_{\text{lp}}. \tag{51}$$

Its vertical component aligns to the barometric height that was already established (which is what removes the height jump when the local system comes online late), the horizontal component is the report of the system at the first sample. The offset is tracked until `ins` locks the datum at bootstrap, then frozen, and it survives `ins` resets.

**WGS84 (the GNSS side).** `nav_suite_local_to_wgs84()` and its inverse `nav_suite_wgs84_to_local()` convert between a local NED position (relative to the datum origin) and geodetic lat/lon/ellipsoid-height. Both directions use the same curvature-radii mapping $\mathcal{M}$ that the filter moves its own anchor with (§5.3), applied around the geodetic origin $(\varphi, \lambda, h)_{\text{origin}}$:

$$(\varphi, \lambda, h) = (\varphi, \lambda, h)_{\text{origin}} + \mathcal{M}^{-1}(\boldsymbol{p}^n_{\text{local}})\,, \qquad \boldsymbol{p}^n_{\text{local}} = \mathcal{M}((\varphi, \lambda, h) - (\varphi, \lambda, h)_{\text{origin}})\,. \tag{52}$$

There is no detour through ECEF here: routing it that way would answer the same question with a different approximation than the position bookkeeping of the filter uses, so the local frame and the absolute solution would disagree at the metre level far from the origin. This is the GNSS-side counterpart of the local-position offset: it turns the local solution into WGS84 for telemetry, or WGS84 waypoints into the frame a controller flies in.

Both WGS84 converters, like `nav_suite_get_height_ellipsoid` (§7.1), are available only once a real GNSS fix has anchored the origin to WGS84, which is established by any usable fix, at bootstrap or by fusion. Without it the origin is only the prescribed init position (a rough indoor guess, or a lighthouse frame), so the converters return false instead of emitting a fictitious absolute coordinate.

# 9 World Magnetic Model lookup

`magnetic_model` provides declination $D$, inclination $I$ and total field strength $F$ for a position and decimal year, from generated lookup tables (`wmm_lut.h`, produced offline by pygeomag-based generators) instead of an online spherical-harmonics evaluation. This means a fixed number of table lookups per call, no heap and bounded time. The declination is stored on a fine grid and interpolated bilinearly in space and linearly in time between the two model epochs.

Inclination and field strength use a coarser grid, bilinear in space only (they change slowly over a 5-year epoch). The NED reference vector is assembled self-consistently as

$$\boldsymbol{m}^n = F \begin{bmatrix} \cos I \cos D \\ \cos I \sin D \\ \sin I \end{bmatrix}. \tag{53}$$

Consumers: `ins_set_magnetic_model_from_position()` builds the full reference vector (so the vector fusion yaw of ins is true-north) and arms the field-strength gate, `ahrs_set_position()` uses $D$ for the deterministic declination step and $F$ for the opt-in gate. Like geodetic_toolbox, the module carries no filter policy: "is this measurement trustworthy" is decided in the filters.

### 9.1 Declination is a wrapped angle

$D$ lives in $(-180°, +180°]$, so two neighbouring grid nodes can sit on opposite sides of that cut while being a few degrees apart (for example $-176.5°$ and $+178.5°$). Interpolating the tabulated numbers directly would then run the long way round and return a value off by up to 180°. Both the spatial and the temporal interpolation therefore put the operands on a common branch first: the secular change per node is taken the short way and the three neighbouring nodes are shifted next to the first before the bilinear blend. The affected cells lie along the agonic line trailing each dip pole, which is where the filters are most in need of a trustworthy heading reference.

### 9.2 Dip pole exclusion zones

At a magnetic dip pole the horizontal field vanishes, so $D$ is not merely inaccurate there but ill-conditioned: it rotates through the full circle around the pole and no regular grid resolves that. The exact model puts the residual interpolation error at roughly 179° within a degree of a pole. Outside the 7° exclusion radius the worst case is about 10°, for 0.75 % of the Earth's surface. A magnetometer is no better off, because the horizontal component it needs has fallen to a few percent of the total field.

`magnetic_dip_pole_distance_deg()` reports the great-circle distance to the nearest dip pole and `magnetic_heading_reference_valid()` is false within `MAGNETIC_DIP_POLE_EXCLUSION_DEG` of one. Inclination and total field strength stay usable inside a zone, only the heading reference is withdrawn. Four properties should be stated explicitly:

**Great circle, not latitude and longitude.** Near the northern pole a degree of longitude covers about a twelfth of the ground that a degree of latitude does (46 km against 555 km at 85° N), so a box in coordinates would be wrong by an order of magnitude in one direction.

**Not a latitude limit.** The southern dip pole sits near 64° S, far from the geographic pole, so no cutoff on latitude separates the zones from ordinary terrain without discarding the whole subantarctic.

**The count is data, not an invariant.** The number of dip poles and their positions are measured per epoch by the generator and tabulated (`WMM_DIP_POLE_COUNT`) and the runtime walks whatever it finds. Today's field has two, both drifting, the northern one by roughly 33 km per year. That is a property of the present dipole dominance and not something the code may assume.

**No interpolation in time.** Positions are tabulated once at mid-epoch and the query takes no year, unlike the declination. Over five years the poles move about 1.5°, so a fixed mid-epoch position is at most $\sim 0.8°$ off at either end. This is smaller than the slack

in the radius, and smaller than the uncertainty in picking that radius in the first place. Interpolating would compute a correction finer than the quantity it feeds. The generator measures the drift on every regeneration and refuses to emit positions that exceed the budgeted margin.

Because the poles move, the exclusion radius wants re-examining whenever the tables are regenerated. `magneticmodel/wmm_error_analysis.py` re-measures the shipped table against the exact model and reports the residual error outside the zones for exactly that purpose. It should be sampled at 0.5° or finer: the worst points sit on a thin ridge at the exclusion boundary, and a coarser grid steps over them and understates the maximum.

# 10 Design decisions

This section collects the important decisions and their rationale in one place. The detailed context is in the sections above.

**UDU factorisation instead of a full covariance.** The single-precision hot path requires it. It gives $\boldsymbol{P} \succeq 0$ by construction and roughly double effective precision (§4).

**Error-state formulation (ins, ahrs).** The nominal state absorbs the nonlinear mechanisation, the Kalman filter sees small, nearly linear errors. Attitude errors stay far from singularities and the quaternion is corrected multiplicatively.

**Downweight vs. skip per measurement class.** Persistent absolute references are downweighted (no deadlock on persistent offsets), high-rate local references are skipped (glitches are cheap to drop), see §3.4.

**Running mean instead of median for the ZARU average.** A median needs a buffer and selection ($O(k)$ memory, data-dependent work in the per-epoch path), the running mean is $O(1)$. WCET outranks statistical elegance here, so do not "improve" this.

**Throttled covariance prediction.** The error dynamics are slow, so predicting at IMU rate buys nothing but cycles (§5.4). The same reasoning throttles the AHRS covariance (20 Hz) and its acc/mag fusion rates.

**Vector vs. scalar magnetometer fusion.** ins fuses the field vector (with yaw-only Jacobian columns). µT is contractual there, which enables the magnitude gate by default and the hard-iron bias estimation. The AHRS fuses a scalar heading, which is unit-agnostic by design for direction-only sensors, so its magnitude gate is opt-in (§6.3).

**Hard-iron bias inside the filter, not outside.** The separability from yaw requires the yaw/bias cross-covariance, which only the filter maintains (§5.5.10). Runtime-$n$ (15/18) was chosen instead of a second code path or a compile-time fork.

**Re-acquisition keeps attitude and biases, but not the yaw *prior*.** After a long outage position and velocity are gone, but attitude and biases are the best-preserved states. Resetting them would require a static re-leveling phase, which the vehicle does not have at a tunnel exit. Yaw is the exception: with the state frozen, the own rotation of the platform over the outage never entered it, so without an absolute heading its *variance* goes back to "unknown". Otherwise the heading correction ends up in the gyro $z$ bias (§5.9).

**History-anchored delayed fusion.** The residual is formed at the time of validity and the covariance update is done on the current factors. This is valid because the error state barely moves over a few hundred milliseconds (§5.6).

**Float states, double anchor.** The local float n-frame keeps the hot path fast on FPUs without double units, the double absolute anchor with incremental bookkeeping avoids the metre-scale quantisation of absolute float coordinates (§5.3).

**Independent accelerometer noise defaults per filter.** The three "accelerometer" quantities are physically different measurements (raw VRW vs. leveling correction vs. attitude-projected vertical acceleration). Sharing one constant regressed the vertical filter empirically (§3.6). The gyro defaults *are* shared, as it is the same physical sensor in the suite.

**Deterministic declination step.** The referencing to true north is a known rotation, so it is applied as an instantaneous re-frame of the nominal quaternion and not learned through the fusion (which would slew the heading over seconds and, transiently, disagree with the covariance). It is skipped on leaving a dip pole exclusion zone, where the yaw was carried by the gyro and not by the field, so the rotation would not be a known one (§9.2).

**Dip pole zones are excluded by distance, not by latitude.** The southern dip pole sits near 64° S, so a latitude cutoff cannot separate the degenerate region from ordinary terrain. The number of poles and their drifting positions are measured per epoch and tabulated instead of being assumed in code, because both are properties of the present field and not invariants.

**Conservative defaults everywhere.** A zero or unset noise field never means "no noise", it means "use a conservative consumer-MEMS placeholder". An overconfident filter fails silently, an overly cautious one is just suboptimal.

**Barometric height fused directly in `ins`, not switched in from `baro_alt` on GNSS loss.** Two designs were considered: (a) keep the height of `ins` GNSS-only and let `nav_suite` switch the reported height to `baro_alt` whenever GNSS aiding drops, or (b) have `ins` itself fuse barometric height (§5.5.2), replacing the GNSS-derived height permanently once selected at bootstrap.

(a) was rejected on two counts. A discrete switch between two independently drifting estimates is a discontinuity by construction: no source stays a fixed distance from another over an arbitrary mission, so the step size is unbounded in general and not just in a corner case (on one real dataset the two disagreed by over ten metres after half an hour, entirely within continuous GNSS coverage). Of course a hysteresis band only changes how rarely the jump happens. More fundamentally, splicing the horizontal state of `ins` onto the independently estimated vertical state of `baro_alt` discards the cross-covariance that a single error-state filter maintains (an attitude error couples into all three position axes at once, §5.2), so the spliced vector is no longer a coherent minimum-variance estimate. (b) has neither problem: the height source is fixed once at bootstrap, so the vertical state of `ins` never switches mid-mission and it stays inside the own state and covariance of `ins`.

(b) does not remove the accessor-level handover: `nav_suite_get_height` still falls back to `baro_alt` outside `FULL` mode, because `ins` as a whole can be unavailable while a height is still wanted. What (b) changes is its *character*: both sides are then the same barometric quantity on one datum, so the step is the disagreement between two estimators of it and not the gap between two height systems. That disagreement is cross-checked in every epoch and warned about (REQ-SUITE-019).

Two prices. The height of `ins` then depends on the absolute accuracy of the barometer (weather/ISA model error) instead of the one of GNSS. That is the right trade for the height *relative* to the own vertical datum of `ins`, where the barometer is the better instrument and the vertical accuracy of GNSS is routinely the worse of the two. It is

the wrong trade for the *absolute* height, and not by a little: the datum is anchored once at bootstrap and never re-tied to GNSS, so the whole absolute error of the barometer accumulates in it without bound. On a 68 min drive over 170 m of terrain that reached 19 m RMS against a receiver whose own vertical spread at standstill measured 0.08 m. The premise that GNSS is the worse vertical source does not hold for a good receiver. The absolute accessor therefore *corrects* the height of `ins` under this source by the estimate of the offset filter of how far the datum has since drifted (§7.3), which is exactly the quantity that filter exists to track. It corrects and does not replace: `ins` remains the better estimate whenever the datum has not moved. The relative accessor is unaffected. The second price, accepted: because the selection is never re-evaluated, a barometer failing mid-mission leaves the vertical channel without an absolute reference while GNSS keeps aiding the horizontal solution. Re-evaluating per epoch would reintroduce exactly the discontinuity that (a) was rejected for, so the case is mitigated and not fixed: the sample must be current at bootstrap and not merely present, and the ensuing aiding gap is reported explicitly (§5.7).

# 11 Dataset configuration: `config.yaml`

A dataset directory carries the neutral CSV streams (`imu.csv` and optionally `ref.csv` (without it nothing is scored; only `aiding: ref` and `init: ref` need it), `gnss.csv`, `mag.csv`, `baro.csv`, `speed.csv`, `heading.csv`, `ranges.csv`, `timesync.csv`, the column contract lives in `datasets/replay_format.py`) plus one `config.yaml` that tells the harness *how* to run the filter: aiding source, initialisation mode, the sensor noise model, sensor calibration and lever arms, aiding-quality gates, the (non-filter) scoring thresholds and optional per-stream CSV filename overrides. Every key is optional except the IMU noise model (and `gnss: pos_stddev_fallback_m/vel_stddev_fallback_mps` under `aiding: ref`, see below). A value of 0 (or an all-zero list) means "use the built-in default / no-op" and never "force zero". An unknown key is a hard error in both harnesses and names the offending key: a silently ignored key is indistinguishable from one that had no effect, so a typo would otherwise produce a full run, a score and a plot computed with a configuration nobody asked for.

"Known" here means the *schema* and not what a given harness happens to consume. A dataset directory is read by more than the two replay harnesses: `check_simulated.py` owns `score: lim_groves_pos_rms_factor`, `crazyflie_reader.py` owns the `origin:` and `crazyflie:` sections. So each harness also accepts, and ignores, the parts it does not consume, from a list the two keep identical. Otherwise a dataset would have to choose which of its tools it validates against, and deleting the offending key would silently disarm the tool that does read it. A typo inside a section the harness *does* own remains an error, and a section earns the exemption only once some tool actually reads it.

The live receiver `tools/insrcv.c` reads the same file and is one step more lenient: it implements a *subset* of the schema, and a key it does not consume is named on the console and then ignored instead of aborting the run. Refusing to start a receiver over a `score:` section would be useless, but a tuning value that silently has no effect misleads there as much as in a replay. Keys describing the replay harness itself (`score:`, `inputs:` and the `name`/`aiding`/`init` labels) therefore pass unremarked, and everything else that the receiver does not implement is reported, with the count repeated in its config summary line.

A representative annotated example (keys not shown fall back to their defaults):

```
name: my-dataset              # label for plots / the summary
aiding: gnss                  # gnss | ref | none
init: auto                    # auto | ref
automotive_mode: 0            # 1: derive yaw from GNSS course-over-ground
```

```yaml
chi2_disable: 0              # 1: disable outlier downweighting (diagnostics)
chi2_reject_alpha: 0.0       # 0 -> outlier gate, e.g. 0.05 -> 95% global
allow_unlimited_deadreckoning: 0  # 1: never expire the IMU-only coasting window
max_deadreckoning_sec: 0.0   # IMU-only coasting budget [s], 0 -> default 10 s
kalman_update_dt_sec: 0.0    # ins covariance-prediction period [s], 0 -> ins
                             #    default (20 Hz). The strapdown runs at the
                             #    IMU rate regardless. The IMU sample period
                             #    predicts at the IMU rate: measurably better
                             #    yaw in a turning car, at a CPU cost that
                             #    matters on a microcontroller (REQ-VER-049)
baro_height_disable: 0       # 1: never select the barometric height source,
                             #    keep GNSS position's vertical row. For a
                             #    gnss_pos stream that is not satellite GNSS
                             #    and is more accurate vertically than a
                             #    barometer (e.g. a Lighthouse/UWB rig), and
                             #    for a barometer whose static port does not
                             #    see ambient pressure: a sensor inside a car
                             #    cabin moves tens of metres with ventilation
                             #    and speed, far past any correction GNSS
                             #    could still apply, since the vertical row
                             #    is dropped for the whole run.
gyro_bias_window_sec: 3.0    # static window to seed the initial gyro bias
                             # (harness-only heuristic, no library default;
                             # 0 -> no seed at all)
auto_init_window_sec: 0.0    # IMU window ins levels over during auto-init [s]
                             # 0 -> default, raise it for a low IMU rate

imu:                         # sensor noise model (REQUIRED) + calibration
  gyr_psd:      3.0e-11      # gyro white noise PSD        [(rad/s)^2/Hz]
  acc_psd:      4.0e-06      # accel white noise PSD       [(m/s^2)^2/Hz]
  gyr_bias_rw: 1.4e-06       # gyro bias random walk       [rad/s/sqrt(s)]
  acc_bias_rw: 3.0e-04       # accel bias random walk      [m/s^2/sqrt(s)]
  # optional fixed calibration, corrected = M*(raw - bias):
  gyr_fixed_bias: [0,0,0]    # removed permanently, never estimated [rad/s]
  acc_fixed_bias: [0,0,0]                                          # [m/s^2]
  gyr_misalignment: [1,0,0, 0,1,0, 0,0,1]   # col-major 3x3 (0s->identity)
  acc_misalignment: [1,0,0, 0,1,0, 0,0,1]
  mount_rpy_deg: [0, 0, 0]   # board attitude in the vehicle frame [deg]
  # extra process-noise margin on top of the acc/gyro-driven pos/vel/rpy
  # noise above, 0 -> default. The rpy default assumes a NON-RIGID mount,
  # where attitude model error dwarfs the gyro's random walk; a rigid mount
  # wants far less and sets it here.
  pos_pred_stddev_m_sqrts:   0.0  # [m/sqrt(s)]
  vel_pred_stddev_mps_sqrts: 0.0  # [m/s/sqrt(s)]
  rpy_pred_stddev_rad_sqrts: 0.0  # [rad/sqrt(s)]
  # Stillness: ONE set for the whole suite. It reaches ins, both AHRS
  # instances and (through them) baro_alt's vertical ZUPT -- the filters
  # implement it separately, they are not tuned separately.
  # Every field 0 -> the library's own default.
  #
  # pseudo-measurement stddev: how hard a detected standstill is trusted
  zero_vel_stddev_mps: 0.0        # [m/s]   (ins + baro_alt vertical ZUPT)
  zero_rot_stddev_deg: 0.0        # [deg/s] (ins + ARS/AHRS ZARU)
  # PRIMARY stillness criterion: per-axis RMS sample stddev of the raw
  # IMU over a short window (bias-invariant, hence primary)
  auto_zupt_static_gyr_stddev_deg:  0.0  # [deg/s]
  auto_zupt_static_acc_stddev_mps2: 0.0  # [m/s^2]
  # loose magnitude bounds beside it
  auto_zupt_static_gyr_deg:  0.0  # max |gyro| considered stationary [deg/s]
  auto_zupt_static_acc_mps2: 0.0  # max ||f| - g|                    [m/s^2]
  # velocity gate: applied only while a recent GNSS velocity exists that
  # is itself precise enough (no such fix -> the IMU decides alone; the
  # filter's own velocity state is never used, that would be circular)
```

```
  auto_zupt_max_vel_mps:         0.0  # max |GNSS velocity| to arm  [m/s]
  auto_zupt_max_vel_stddev_mps: 0.0   # max GNSS vel 1-sigma to trust it [m/s]
  # timing
  auto_zupt_dwell_sec:         0.0  # stillness required before a trigger [s]
  auto_zupt_min_interval_sec: 0.0   # min time between triggers          [s]
  # opt-outs
  auto_zupt_disable: 0  # 1 -> no stillness detection anywhere in the suite
  auto_zupt_velocity_blind_disable: 0  # 1 -> only the ARS/AHRS's own
                        # detector off (it cannot tell constant-velocity
                        # cruise from a standstill); ins keeps deciding

gnss:
  enable: 1                  # 0 -> the fixes are still read and counted,
                             # but none of them aids the filter
  leverarm_frd: [0,0,0]      # IMU->antenna lever arm, body FRD   [m]
  delay_ms: 0                # assumed fixed fix latency          [ms]
  max_horizontal_pos_stddev_m: 0.0    # fusion gate: ignore fixes worse than
  max_vertical_pos_stddev_m:   0.0    # this (per fix, see below), 0 -> ins.c's
  max_horizontal_vel_stddev_mps: 0.0  # own default, which is the accuracy CAP
  max_vertical_vel_stddev_mps:   0.0  # below (120 m / 60 m/s): by default no
                                      # fix is rejected for its reported
                                      # accuracy alone, it is capped and fused.
                                      # Set these below the caps for a hard gate.
  # 3D solution ENTRY gate (strict) + its dwell, 0 -> default:
  start_max_horizontal_pos_stddev_m: 0.0    # [m]
  start_max_vertical_pos_stddev_m:   0.0    # [m]
  start_max_horizontal_vel_stddev_mps: 0.0  # [m/s]
  start_max_vertical_vel_stddev_mps:   0.0  # [m/s]
  init_dwell_sec: 0.0        # how long the stream must stay entry-quality
  init_dwell_disable: 0      # 1: enter on the first good fix
  # 3D solution EXIT gate (loose) + its dwell, 0 -> ins.c's own default:
  stop_max_horizontal_pos_stddev_m: 0.0     # [m]
  stop_max_vertical_pos_stddev_m:   0.0     # [m]
  stop_max_horizontal_vel_stddev_mps: 0.0   # [m/s]
  stop_max_vertical_vel_stddev_mps:   0.0   # [m/s]
  stop_dwell_sec: 0.0        # bad-fix run before the 3D solution is left
  stop_disable: 0            # 1: never leave 3D on GNSS quality alone
  min_delay_ms: 0            # fusion rate limit: minimum time between two
                             # fixes the filter actually spends. A fix
                             # arriving sooner is skipped whole (both blocks)
                             # and counted in n_gnss_rate_limited; suspended
                             # while the coasting window is expired, so a
                             # tunnel exit is never paced. A GNSS error
                             # decorrelates over seconds, not over the fix
                             # interval, so fusing a 50 Hz stream at 50 Hz
                             # presents the same information ten times and
                             # collapses the covariance.
                             # 0 -> default (100 ms, 10 Hz), <0 -> no limit
  pos_decimation: 0          # fuse the position on every Nth epoch that
                             # offers BOTH a usable position and velocity,
                             # the velocity alone on the other N-1;
                             # 1 -> off, 0 -> ins.c's own default, which is
                             # OFF: min_delay_ms above covers the same
                             # over-fusing without withholding a block, and
                             # running both thins the position twice
  pos_stddev_fallback_m: [0,0]        # [hor,ver] used if reported cov is 0;
                                      # REQUIRED (no default) if aiding: ref,
                                      # where it is the ONLY noise the
                                      # truth-synthesized fix gets
  vel_stddev_fallback_mps: 0.0        # same, required if aiding: ref
  # The conditioning knobs below weight the FUSION only. The
  # fix-quality gates (max_*, start_max_*, stop_max_*, and the auto-ZUPT
  # accuracy gate) keep grading the stddev the receiver reported, so
```

```
# downweighting a noisy receiver never silently stops the filter from
# using it. Scale them freely against the gates. The pipeline runs
#   envelope -> scale -> height scale -> floor -> cap -> manoeuvre term
# and is described in REQ-NAV-038/041/071/072/073.
acc_envelope_tau_sec: 0.0 # asymmetric tracking of the REPORTED accuracy:
                          # a rise takes effect at once, a fall only decays
                          # out with this time constant. A receiver's
                          # accuracy figure returns to nominal well before
                          # a re-acquired fix has actually settled.
                          # 0 -> default (5 s), <0 -> off
pos_cov_scale: 0.0        # multiply reported pos stddev (0->1)
pos_cov_scale_height: 0.0 # extra downweight of the pos HEIGHT axis (0->1)
vel_cov_scale: 0.0
pos_stddev_floor_hor_m: 0.0          # floor per-axis stddev (independently,
pos_stddev_floor_ver_m: 0.0          # NOT correlation-preserving like
                                      # pos_cov_scale above: raises N/E/D
                                      # diagonal entries alone, so a
                                      # floored axis picks up the effect of
                                      # extra independent noise).
vel_stddev_floor_hor_mps: 0.0        # Unlike the scales, 0 means "use ins.c's
vel_stddev_floor_ver_mps: 0.0        # default", NOT "no floor". The defaults
                                      # (0.5/1.0 m, 0.3/0.5 m/s) describe a
                                      # single-frequency (L1) standalone
                                      # receiver, which is what an
                                      # unconfigured filter should assume: a
                                      # receiver claiming better than such a
                                      # solution can physically deliver is the
                                      # normal case, not the exception.
                                      #
                                      # A BETTER source must say so, or the
                                      # floors throw its precision away. Set
                                      # each floor to what that source really
                                      # delivers, e.g. for RTK fix:
                                      #   pos_stddev_floor_hor_m: 0.02
                                      #   pos_stddev_floor_ver_m: 0.03
                                      #   vel_stddev_floor_hor_mps: 0.03
                                      #   vel_stddev_floor_ver_mps: 0.05
                                      # A NEGATIVE value switches a floor off
                                      # entirely, i.e. trusts the reported
                                      # covariance unconditionally. Prefer
                                      # real numbers over -1: an RTK receiver
                                      # dropping to float or single still
                                      # reports optimistically, and a floor
                                      # sized for the fix mode is what keeps
                                      # that drop from being believed.
pos_stddev_cap_hor_m: 0.0            # upper clamp, same 0 -> default /
pos_stddev_cap_ver_m: 0.0            # <0 -> off convention (120 m, 60 m/s).
vel_stddev_cap_hor_mps: 0.0          # Bounds what an unboundedly bad reported
vel_stddev_cap_ver_mps: 0.0          # accuracy can do to the 32-bit hot path,
                                      # and is what lets max_* default to
                                      # "fuse everything". A cap below the
                                      # floor of the same axis group is raised
                                      # to that floor at init.
vel_noise_acc_scale_hor: 0.0         # manoeuvre-dependent EXTRA velocity noise
vel_noise_acc_scale_ver: 0.0         # in [m/s] per [m/s^2] of n-frame
                                      # ANTENNA acceleration (horizontal norm
                                      # on N/E, |Down| on D), added after the
                                      # cap. A Doppler velocity averages over
                                      # the receiver's measurement interval
                                      # while the state is instantaneous; under
                                      # acceleration the two differ by about
                                      # a*T/2, which the reported accuracy
                                      # cannot describe. The acceleration is
```

```
                                    # the antenna's, i.e. the body's plus the
                                    # centripetal term omega x (omega x l) of
                                    # the leverarm_b below (REQ-NAV-076), so
                                    # a platform turning about its own IMU
                                    # still pays for the arc its antenna
                                    # swings through. With no lever arm that
                                    # addition is exactly zero.
                                    # 0 -> default, <0 -> off. The defaults
                                    # (0.20 / 0.30) ARE that half-interval
                                    # for T = 400 ms, four times the rate
                                    # min_delay_ms paces fusion to (margin: T
                                    # belongs to the receiver's velocity
                                    # algorithm, not its output rate), plus a
                                    # margin on the vertical axis for its
                                    # one-sided satellite geometry. Fixed
                                    # constants: NOT derived from the
                                    # observed fix rate at runtime. At a
                                    # 45 deg lean that adds roughly 2 m/s,
                                    # more than six times the horizontal
                                    # velocity floor; at rest
                                    # it adds nothing, so the standstill
                                    # regime is untouched. Raise them for a
                                    # receiver whose velocity is
                                    # time-differenced carrier phase over a
                                    # long output interval (T is then the
                                    # full interval). Disable them for a
                                    # source with no measurement interval at
                                    # all, e.g. a pose tracker or a simulator.
  vel_noise_acc_window_sec: 0.0    # window the acceleration above is averaged
                                    # over: the SAME T the scales are half of.
                                    # The error priced is what averaging over
                                    # T does to a velocity, so the
                                    # acceleration over T is what sets it.
                                    # Without it a manoeuvre that just ended
                                    # goes unpriced on the fix that still
                                    # carries its error, and airframe
                                    # vibration is priced as if it were a
                                    # manoeuvre (measured: a zero-mean
                                    # 4 m/s^2 oscillation adds 4 (m/s)^2
                                    # instantaneous vs 0.01 windowed). The
                                    # VECTOR is averaged, so a zero-mean
                                    # swing cancels exactly. The rotation
                                    # half of the antenna acceleration uses
                                    # the same window but averages
                                    # omega*omega', which is quadratic and
                                    # therefore does NOT cancel for a
                                    # platform that turns back and forth.
                                    # 0 -> default (200 ms), <0 -> use the
                                    # instantaneous sample

init_stddev:                       # initial-state 1-sigma (each 0 -> built-in default)
  pos_init_stddev_m: 0.0           # [m]
  vel_init_stddev_mps: 0.0         # [m/s]
  rpy_init_stddev_rad_deg:  0.0    # [deg]
  yaw_init_stddev_rad_deg:  0.0    # [deg] (0 -> rpy_init_stddev_rad_deg)
  acc_bias_init_stddev_mps2: 0.0   # [m/s^2]
  gyr_bias_init_stddev_rps_deg: 0.0  # [deg/s]

init_hint:                         # known initial roll/pitch/yaw for auto-init
                                    # (init: auto only), roll/pitch used together,
                                    # yaw independent, each 0 stddev -> no hint
  roll_deg: 0.0
  pitch_deg: 0.0
```

```
  rpy_stddev_deg: 0.0         # 0 -> no roll/pitch hint (usual: gravity levels it)
  yaw_deg: 0.0                # e.g. a known launch heading
  yaw_stddev_deg: 0.0         # 0 -> no yaw hint (usual: stays unknown until aiding)

mag:
  enable: 0                  # replay default (it opens mag.csv);
                              # tools/insrcv.c defaults to 1
  stddev_ut: 0.0             # heading-measurement noise [uT] (0 -> default)
  min_delay_ms: 0            # min. time between fusions (0 -> default,
                              # negative -> fuse every sample)
  wmm_year: 2025.0           # decimal year for the declination lookup
  estimate_bias: 0           # 18-state hard-iron bias
  bias_init_ut: 0.0          # INITIAL hard-iron stddev [uT], 0 -> default
  bias_rw_ut_sqrts: 0.0      # hard-iron random walk [uT/sqrt(s)], 0 -> default
  fixed_bias: [0,0,0]        # hard-iron [uT],  misalignment: soft-iron 3x3
  misalignment: [1,0,0, 0,1,0, 0,0,1]

baro:
  enable: 0                  # Process barometer measurements
  stddev_m: 0.0              # Barometer measurement noise [m], 0 -> default
  acc_bias_rw: 0.0           # accel-bias drift  [m/s^2/sqrt(Hz)], 0 -> default
  acc_bias_init_mps2: 0.0    # INITIAL accel-bias stddev [m/s^2], 0 -> default
  acc_noise_mps2_sqrthz: 0.0 # accel-noise density [m/s^2/sqrt(Hz)]
  h_process_noise: 0.0       # direct height process noise [m/sqrt(Hz)], 0 -> default
  # local-height/GNSS-ellipsoid offset filter tuning (all 0 -> built-in
  # defaults). Raise local_gnss_rw_stddev_mps further for missions with
  # extreme altitude excursions (e.g. a soaring glider gaining kilometres
  # of altitude): the offset absorbs the ISA-model error, which grows with
  # the excursion and can outrun even the default, leaving the offset
  # permanently lagging.
  local_gnss_rw_stddev_mps: 0.0            # [m/sqrt(s)], 0 -> default 0.3
  local_gnss_chi2_threshold: 0.0           # 0 -> default chi2inv(0.95,1)
  local_gnss_min_update_interval_sec: 0.0  # [s], 0 -> default 10
  local_gnss_stddev_inflation_factor: 0.0  # 0 -> default 3
# Scalar ground speed from speed.csv (odometry, OBD-II). Uncertainty and
# delay are constants here, not per-sample columns.
speed:
  enable: 0                  # Process speed.csv
  scale: 0.0                 # multiplies the reported speed, 0 -> 1.0
  stddev_mps: 0.0            # per-sample 1-sigma [m/s], 0 -> default
  stddev_rel: 0.0            # speed-proportional 1-sigma, 0 -> default
  min_speed_mps: 0.0         # skip below this FILTERED speed, 0 -> default
  delay_ms: 0                # how old a sample is at its timestamp
# Dual-antenna GNSS heading from heading.csv (moving base NAV-RELPOSNED).
# The 1-sigma is a per-row column, these keys only scale and floor it.
heading:
  enable: 0                  # Process heading.csv
  baseline_frd: [1, 0, 0]    # base -> rover antenna, body FRD, any length
  require_fixed: 1           # 1 -> only carrier-phase fixed rows
  stddev_scale: 0.0          # multiplies the row's 1-sigma, 0 -> 1.0
  stddev_min_deg: 0.0        # floor after scaling, 0 -> none
  delay_ms: 0                # how old a row is at its timestamp
# Ranges to anchors at known positions from ranges.csv (REQ-NAV-082).
# The 1-sigma is a per-row column, each row is anchored at its own age.
ranges:
  enable: 0                  # Process ranges.csv
  leverarm_frd: [0, 0, 0]    # IMU -> ranging antenna, body FRD [m]
  stddev_scale: 0.0          # multiplies the row's 1-sigma, 0 -> 1.0
  stddev_min_m: 0.0          # floor after scaling, 0 -> none
  height_with_baro: 0        # 1 -> ranges also correct the height
  aiding_max_hpos_stddev_m: 0.0  # 0 -> default (REQ-NAV-085)

ahrs:                        # SHARED ARS/AHRS noise override (0 -> derived
```

```
                              # from imu: above), both sub-filters consume
                              # the same physical gyro/accelerometer
  gyr_noise_psd: 0.0         # [rad/s/sqrt(Hz)], 0 -> sqrt(imu.gyr_psd)
  gyr_bias_rw: 0.0           # [rad/s^2/sqrt(Hz)], 0 -> imu.gyr_bias_rw
  rpy_pred_stddev_rad_sqrts: 0.0  # [rad/sqrt(s)] extra attitude noise
                              # on top of the gyro noise, 0 -> default
  acc_noise_mps2: 0.0        # [m/s^2], 0 -> default
  gyr_bias_init_stddev_rps_deg: 0.0  # [deg/s], 0 -> unchanged (the
                              # gyro_bias_window_sec seed's own stddev, or
                              # ahrs.c's default if that seed never ran)
  # Leveling and magnetometer tuning, the ahrs_config_t fields of the same
  # names (ahrs.h), applied to the ARS and the AHRS. 0 -> ahrs's default.
  acc_freq_hz: 0.0           # [Hz] max. rate of accelerometer corrections
  acc_cutoff_freq_hz: 0.0    # [Hz] accelerometer low-pass cut-off
  gravity_diff_penalty: 0.0 # extra distrust per m/s^2 of |acc| - g,
                              # < 0 -> off
  acc_reject_gravity_mps2: 0.0  # [m/s^2] hard-reject above this
                              # | |acc| - g |, < 0 -> never reject
  chi2_threshold: 0.0        # chi2 gate of the accelerometer update
  kalman_update_dt_sec: 0.0 # [s] covariance prediction period
  mag_yaw_stddev_deg: 0.0    # [deg] heading measurement noise (AHRS)
  mag_freq_hz: 0.0           # [Hz] max. rate of magnetometer corrections
  mag_chi2_threshold: 0.0    # chi2 gate of the heading update

score:                       # scoring/plots + regression gates ONLY (not filter)
  warmup_sec: 60.0
  ahrs: 1                    # also score the AHRS/ARS attitude filters
  leverarm_frd: [0,0,0]      # map the filter position onto the truth point
  min_epochs: 100
  lim_pos_rms_m: 2.0         # regression guard-rails (make simulated/datasets)
  lim_att_bias_deg: 0.1
  lim_yaw_std_deg: 0.2

inputs:                      # optional per-stream CSV filename overrides
  imu:  imu.csv              # each unset key -> the conventional <stream>.csv
  ref:  ref.csv
  gnss: gnss_f9p.csv         # e.g. swap only the receiver, keep the rest shared
  mag:  mag.csv
  baro: baro.csv
  speed: speed.csv
  heading: heading.csv
  ranges: ranges.csv
```

### Top level.

**name** *string.* Dataset label used in plots and the summary.

**aiding** *default* ***gnss.*** Absolute-position source: **gnss** uses real per-epoch receiver covariance from **gnss.csv** (zero-diagonal entries replaced by the **gnss:** fallbacks), **ref** synthesises a fix from the ground truth with the **fix:** noise (not an independent error profile), **none** gives no position aiding at all. In that case **ins** never initialises and only the ARS and **baro_alt** filters run (**nav_suite** ATTITUDE-ONLY fallback).

**init** *default* ***auto.*** **auto**: **ins** auto-initialisation. **ref**: known initial state from the truth epoch aligned with the first IMU sample (a moving start is fine, e. g. an aircraft in cruise).

**automotive_mode** *0/1, default 0.* Derive yaw from GNSS course-over-ground above a minimum speed, typically for wheeled, non-holonomic platforms without a magnetometer. *Not* valid where course differs from heading at speed (aircraft in wind).

**automotive_min_speed_mps** *m/s* Minimum ground speed for course-derived yaw.

`automotive_min_yaw_stddev_deg` *deg* Floor on the fused yaw stddev, raise (e. g. 15) to keep it a weak long-term anchor under wind drift.

`chi2_disable` *0/1, default 0.* Disable chi$^2$ outlier downweighting library-wide.

`chi2_reject_alpha` *default 0* Global significance level $\alpha$ of the chi$^2$ downweight gate of ins. Each absolute reference is fused scalar-row-wise (1 DOF), so a configured $\alpha$ sets one shared gate threshold $\chi^2_{\text{inv}}(1-\alpha, 1)$ for every channel (GNSS, magnetometer, local-position, yaw), evaluated by interpolating a fixed quantile table. $\alpha = 0$ keeps the own single default of ins.c (10.0, identical on all four channels, no differing per-channel gates), a larger $\alpha$ (e. g. $0.05 \to 3.84$) is stricter, a smaller one more tolerant (e. g. $0.01 \to 6.63$). Ignored when `chi2_disable` is set.

`gyro_bias_window_sec` *s* Length of the initial static window averaged to seed the initial gyro-bias estimate.

`auto_init_window_sec` *s, default 0 → default.* Length of the IMU window that the auto-initialisation of `ins` levels over (median of the specific force, §5.7). Raise it for a low IMU rate, so that the window still contains enough samples.

`allow_unlimited_deadreckoning`, `max_deadreckoning_sec` *0/1 and s, default 0 → default.* IMU-only coasting budget before `is_ready()` degrades (§5.9). The flag makes the window never expire and then overrides the budget.

`baro_height_disable` *0/1, default 0.* Never select the barometric height source at bootstrap, keep the GNSS position's vertical row, even with a barometer present.

**`imu:`, noise model (required) and calibration.**

`gyr_psd`, `acc_psd` $(\text{rad/s})^2/Hz$, $(\text{m/s}^2)^2/Hz$. White-noise PSD (`replay.py` suggests values from a trial). **Required.**

`gyr_bias_rw`, `acc_bias_rw` $rad/s/\sqrt{\text{s}}$, $m/s^2/\sqrt{\text{s}}$. Bias random walk. Estimate from a long static log with `tools/allan_variance.py` (Allan deviation, see [9]). **Required.**

`gyr_fixed_bias`, `acc_fixed_bias` *3-list, rad/s and m/s², default 0.* Removed permanently (`raw` − `bias`), never estimated.

`gyr_misalignment`, `acc_misalignment` *9-list, column-major* 3×3*, all-zero → identity.* Scale/cross-coupling matrix $M$, `corrected` $= M($`raw` − `bias`$)$. Also feeds the parallel ARS/AHRS.

Measured by `tools/inslib_imu_calib.py` (or the calibration window of the reference board) from a session of arbitrary static poses, no fixture involved. The same matrix carries the *housing* alignment when one was captured: the unit is rested on a level surface, any tilt it reports is the IMU sitting crooked in the box and the rotation that removes it is premultiplied onto $M$ for the accelerometer, the gyroscope and the magnetometer alike. One placement gives roll and pitch, a second one on a face that is not parallel to the first also gives the yaw, which gravity on its own can never observe.

`mount_rpy_deg` *3-list, deg, default 0.* Attitude of the axes of the sensor board in the vehicle body frame (ZYX roll, pitch, yaw, FRD), e. g. `[0, 0, -9]` for a board whose x axis points 9 deg left of the forward axis of the vehicle. The harnesses (`tools/replay.c`, `tools/insrcv.c`, `tools/replay.py`, `inspostgui.py`) premultiply the rotation $R(\text{roll}, \text{pitch}, \text{yaw})$, which takes a board-frame vector into the vehicle frame, onto the accelerometer, gyroscope and magnetometer matrices, $M \leftarrow R\,M$, after the per-axis calibration those already hold. The filter then works in vehicle axes: lever arms, `heading:` `baseline_frd` and every reported attitude refer to the vehicle and not to the board. The configured matrices themselves are left as written (REQ-VER-036).

A yaw mounting error can be read off a drive with a dual-antenna heading: run once with `heading: enable: 0`, so that the yaw comes from GNSS velocity and accelerations and therefore follows the x axis of the board, and compare it against the heading of the same epochs while driving straight. The median difference is the mounting yaw.

**`pos_pred_stddev_m_sqrts`, `vel_pred_stddev_mps_sqrts`, `rpy_pred_stddev_rad_sqrts`** $m/\sqrt{s}$, $m/s/\sqrt{s}$, $rad/\sqrt{s}$ Extra process-noise margin added *on top of* the physically derived acc/gyro-driven pos/vel/rpy prediction noise and not instead of it.

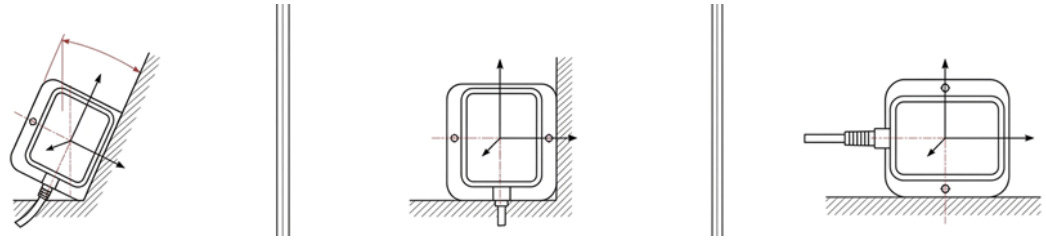


Figure 6: Three of the static poses a calibration session collects: the unit tilted against an edge, resting flat and standing on its connector end, each read against gravity in the world frame ($\boldsymbol{g}$, right of each pose). Held-still orientations that span enough of the sphere are what let the fit separate bias, scale and misalignment without a turntable or fixture (see the coverage sphere below).

The same session also calibrates the magnetometer, when the board streams one (`mag: fixed_bias/misalignment` below): both are fitted from one set of poses and not from two separate sessions. Because of that, *the whole session (not only the parts that look magnetometer-specific) has to happen away from ferrous material and active electronics*: a steel desk, a nearby motor or a metal-legged table all bias the field that the magnetometer fit reads, and nothing in the accelerometer/gyroscope channels can detect that it happened. A genuinely magnetically clean surface is harder to find than it sounds.

**`imu:`, stillness (all optional, $0 \rightarrow$ a default).** One set for the whole suite: it configures `ins`, both AHRS instances and the vertical zero-velocity update of `baro_alt` alike (§5.8). Neither harness configures the ARS/AHRS gates separately any more.

**`zero_vel_stddev_mps`, `zero_rot_stddev_deg`** *m/s, deg/s* Pseudo-measurement stddev: how hard a detected standstill is trusted once flagged. `zero_vel_stddev_mps` also becomes the vertical ZUPT stddev of `baro_alt`, `zero_rot_stddev_deg` the ARS/AHRS zero-rotation stddev. These two are the one exception to "$0 \rightarrow$ the library's own default": both replay harnesses substitute their own 0.05 m/s and 0.1 deg/s, the latter deliberately tighter than the built-in 0.5 deg/s of `ins`, so a config that leaves them at 0 keeps the historical harness tuning instead of silently loosening the rotation channel. Every remaining key below falls back to the own default of the library.

**`auto_zupt_static_gyr_stddev_deg`, `auto_zupt_static_acc_stddev_mps2`** *deg/s, $m/s^2$* The PRIMARY stillness criterion (§5.8): max per-axis RMS sample stddev of the raw gyro/accelerometer over a short window.

**`auto_zupt_static_gyr_deg`, `auto_zupt_static_acc_mps2`** *deg/s, $m/s^2$* The loose magnitude bounds beside it: max $\|\boldsymbol{\omega}\|$ and max $\big|\|\boldsymbol{f}\| - g\big|$ still considered stationary.

**`auto_zupt_max_vel_mps`, `auto_zupt_max_vel_stddev_mps`** *m/s* The velocity gate: max $\|\boldsymbol{v}_{\mathrm{GNSS}}\|$ and the accuracy (1-sigma, per axis) that the GNSS velocity must itself report to count as evidence. Applied only while such a fix is recent enough, with none only the IMU criteria decide. Not propagated to the ARS/AHRS, which have no velocity state.

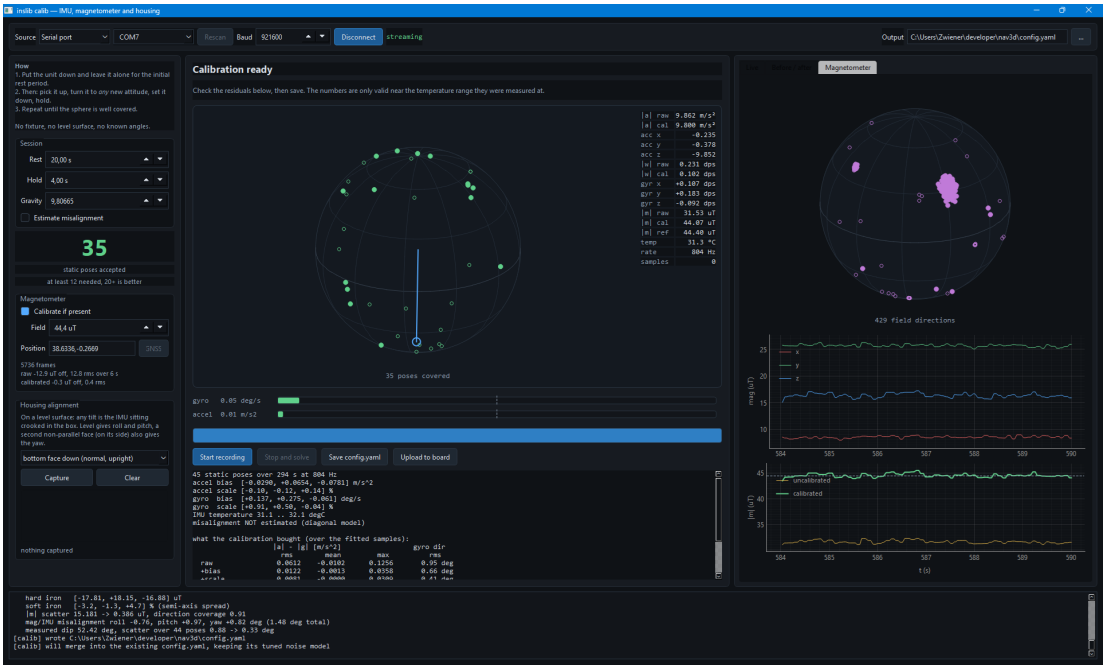



Figure 7: The calibration window of the reference board: the coverage sphere fills in as static poses are accepted, the stillness meter says whether the unit that is currently held down is calm enough to count, and the live traces show the raw and calibrated readings side by side.

**`auto_zupt_dwell_sec`, `auto_zupt_min_interval_sec`** *s* How long the platform must look still before a trigger, and how often triggers may repeat.

**`auto_zupt_disable`** *0/1, default 0.* Turn off every stillness source in the suite: `ins`, both AHRS instances and, transitively, the vertical ZUPT of `baro_alt`.

**`auto_zupt_velocity_blind_disable`** *0/1, default 0.* Turn off only the velocity-blind detector of the ARS/AHRS (§5.8), `ins` keeps deciding for all of them.

**`gnss:`, aiding quality (all optional).**

**`leverarm_frd`** *3-list, m, body FRD, default 0.* IMU-to-antenna lever arm.

**`delay_ms`** *ms, default 0.* Assumed fixed processing/telemetry latency, applied uniformly to every fix, counted from the fix's timestamp. The harness adds the fix's age relative to the IMU epoch it is attached to on top (`gnss_delay_us`, §5.6), and scores the position estimate at each reference row's own timestamp, carried back with the filter velocity.

**`max_{horizontal,vertical}_{pos,vel}_stddev`** Shutdown gates: a fix whose reported std-dev exceeds the limit is ignored.

**`pos_stddev_fallback_m`, `vel_stddev_fallback_mps`** *[hor,ver] m and m/s, default 0.* Substituted for zero-diagonal reported covariance. *REQUIRED (no default) under* ***`aiding: ref`***: the reference-synthesized fix never has a reported covariance to begin with, so this is the only noise it gets, and `load_config()` refuses to run without both.

**`pos_cov_scale`, `vel_cov_scale`** *default 0 → 1.* Multiply the reported stddev, for over-optimistic vendor covariance.

**`pos_cov_scale_height`** *default 0 → 1.* Extra dedicated downweight of the GNSS position *height* (Down) axis only, for receivers whose vertical solution is far worse than their horizontal one. Applied as the correlation-preserving congruence $\boldsymbol{SQS}$ with $\boldsymbol{S} = \mathrm{diag}(1, 1, s)$ (the

N–D and E–D correlation coefficients are kept exactly), on top of `pos_cov_scale`. The horizontal accuracy is untouched.

`{pos,vel}_stddev_floor_...` *default 0.* Per-axis stddev floor, effective `stddev` = max(`scale`·`reported`, `floor`).

**`init_stddev:`, initial-state 1-sigma (REQ-VER-010).** Each field mirrors `ins_init_t`, 0 falls back to the built-in default. Widen these when the "known" start is itself only a GNSS-derived estimate, or if the filter starts overconfident.

`pos_init_stddev_m` m.

`vel_init_stddev_mps` m/s.

`rpy_init_stddev_rad_deg` deg (roll/pitch, and yaw unless the next key is set).

`yaw_init_stddev_rad_deg` deg, 0 → `rpy_init_stddev_rad_deg`.

`acc_bias_init_stddev_mps2` m/s$^2$.

`gyr_bias_init_stddev_rps_deg` deg/s.

**`init_hint:`, known initial attitude for auto-init (optional, `init: auto` only).** A static "I just know it" hint for the auto-init bootstrap of `ins` (e. g. a known launch heading with no magnetometer/GNSS-course yaw aiding to derive it from). Only used while neither the ARS nor the AHRS has initialised yet (a continuously running, converged heading takes priority once available) and only until `ins` itself initialises.

`roll_deg`, `pitch_deg` *deg, default 0.* Known initial roll/pitch, only used together (virtual measurement).

`rpy_stddev_deg` *deg, default 0 → no hint.* Shared roll/pitch 1-sigma for the virtual measurement. Usually left unset, since leveling is already observable from gravity.

`yaw_deg` *deg, default 0.* Known initial yaw, independent of roll/pitch. A set yaw hint also seeds the own initial heading of the yaw-free ARS, so `nav_suite_get_rpy()` already reports it during the ATTITUDE_ONLY window instead of the arbitrary 0 that the ARS would otherwise start from. Roll/pitch are not seeded there (the ARS levels them from gravity on its first epoch).

`yaw_stddev_deg` *deg, default 0 → no hint.* Otherwise the yaw stays unobservable until aiding arrives.

**`ahrs:`, SHARED ARS/AHRS noise override (all optional, default 0).** The standalone ARS/AHRS attitude filters (§3.3) are physically separate filter instances from `ins` and are tuned by default from the same `imu:` noise model above. `ahrs:` overrides that per dataset instead. One call sets the template of both sub-filters, since they consume the same physical gyro/accelerometer. The live receiver `tools/insrcv.c` reads the whole section as well, with the same fallbacks as the replay harnesses.

`gyr_noise_psd` *rad/s/$\sqrt{Hz}$, default 0 → $\sqrt{$`imu: gyr_psd`$}$.* Shared gyro noise density.

`gyr_bias_rw` *rad/s$^2$/$\sqrt{Hz}$, default 0 →* `imu: gyr_bias_rw`. Shared gyro bias random walk density.

`rpy_pred_stddev_rad_sqrts` *rad/$\sqrt{s}$, default 0 → none.* Extra attitude process noise of both sub-filters, added to the gyro noise in variance (REQ-AHRS-027). The gyro terms above describe the sensor, this one is the tuning knob for what the sensor figure leaves out (scale factor and misalignment under rotation, a non-rigid mount), the same split `ins` makes

with `imu: rpy_pred_stddev_rad_sqrts`. Unlike there, the default is none: the generic gyro noise default already carries the margin of a typical MEMS IMU. Set it once `imu:` holds a good sensor's own (Allan) figure, otherwise the ARS/AHRS becomes overconfident in attitude.

`acc_noise_mps2` *$m/s^2$, default 0 → default.* Shared accelerometer noise stddev (leveling).

`gyr_bias_init_stddev_rps_deg` *deg/s, default 0 → unchanged.* Initial gyro-bias uncertainty for both sub-filters.

The remaining keys are the leveling and magnetometer tuning fields of `ahrs_config_t` with the same names (`ahrs.h`), applied to both sub-filters (the `mag_*` keys only matter to the AHRS). 0 keeps `ahrs`'s own default.

`acc_freq_hz` *Hz.* Maximum rate of accelerometer leveling corrections.

`acc_cutoff_freq_hz` *Hz.* Accelerometer low-pass cut-off, tolerance against vibration at the cost of delay.

`gravity_diff_penalty` *stddev per $m/s^2$, negative → off.* Downweights a leveling update by how far $|\mathbf{f}|$ differs from $g$, higher is more resistant to acceleration phases.

`acc_reject_gravity_mps2` *$m/s^2$, negative → never reject.* Hard-rejects the leveling update while the low-passed $\big||\mathbf{f}| - g\big|$ exceeds this.

`chi2_threshold` $\chi^2$ outlier gate of the accelerometer update, per scalar row.

`kalman_update_dt_sec` *s.* Covariance prediction period, the attitude integration still runs at the full IMU rate.

`mag_yaw_stddev_deg` *deg.* Stddev of the magnetometer heading measurement (`ahrs_config_t` holds it in rad).

`mag_freq_hz` *Hz.* Maximum rate of magnetometer heading corrections.

`mag_chi2_threshold` $\chi^2$ outlier gate of the heading update.

**`mag:` and `baro:`, optional sensors.**

`mag: enable` *0/1, default 1 live, 0 offline.* Turn on magnetometer heading fusion. The two defaults differ because the key does two different things. Live (`tools/insrcv.c`) the `0x40/0x06` samples arrive over the wire regardless and the key gates the fusion alone (off, they stay decoded, counted and published), so a receiver with no magnetometer simply never fuses one and the default costs nothing, as with the barometer. Offline (`tools/replay.c`, `tools/replay.py`) the same key gates *opening* `mag.csv` and a missing file is a hard error, so a dataset without a magnetometer stream must not have it on by default. The risk that the live default accepts is an *uncalibrated* sensor: hard iron is a bias and no measurement noise removes the heading error it causes (§5.5.10), only the calibration below does. It is acceptable because the failure is not silent: once a position is known, the field-strength gate downweights samples whose $|B|$ leaves the tolerance band and warns about a persistent deviation, which is what an uncorrected hard iron of any relevant size looks like. `insrcv` additionally says so at startup when the fusion is on and no calibration is configured.

`mag: stddev_ut` *uT, default 0 → default.* Magnetometer measurement noise. The default is of the same order as the horizontal field itself, i. e. roughly a radian of heading noise per sample, see §5.5.10 for why weak is the right starting point. Lower it only for a sensor whose magnetic environment is known to be clean.

`mag:` **`min_delay_ms`** *ms, default 0 → default (1 Hz), negative → no rate limit.* Minimum time between two magnetometer fusions. This is the second half of the weighting decision above: the noise sets how far one sample may pull the heading, this sets how often it may (§5.5.10). A source that is not a real magnetometer exposed to the platform (a synthesized or externally validated heading stream) is the case for switching the limit off.

`mag:` **`wmm_year`** *decimal year.* Epoch for the World Magnetic Model declination lookup. For the live receiver (`insrcv`) it is optional: without it the epoch comes from GPS time in the stream, failing that from the UTC date the receiver reports in NAV-PVT, and failing that from the host clock. The replay harnesses have no date source of their own, so there it is required whenever `mag: enable` is set: without it no reference field is built, the magnetometer is not fused and the harness warns once at start-up. The capture converter of the reference board fills it in from the NAV-PVT date of the capture.

`mag:` **`estimate_bias`** *0/1, default 0.* Run `ins` in its 18-state mode estimating a hard-iron bias online. Observable only under rotation.

`mag:` **`fixed_bias`**, `mag:` **`misalignment`** *3-list uT and 9-list* $3 \times 3$*, defaults 0/identity.* Fixed hard-iron / soft-iron calibration, $\texttt{corrected} = M(\texttt{raw} - \texttt{bias})$.

Measured alongside the IMU by the same tools when the board streams a magnetometer: turned through every attitude the sensor traces an ellipsoid, whose centre is the hard iron and whose shape is the soft iron [9]. Fitting a sphere cannot see a rotation, so the fit is pinned to the symmetric square root and the leftover rotation is solved separately, from the fact that the angle between the local field and gravity cannot change with attitude. `misalignment` therefore also carries the mounting rotation of the magnetometer onto the IMU triad, without which a magnetometer a few degrees off its board contributes those same few degrees to the fused heading.

`baro:` **`enable`** *0/1, default 0 in the replay harnesses, 1 in `tools/insrcv.c`.* Enable the `baro_alt` vertical channel (needs static pressure measurements). The two defaults differ because the key does two different jobs: offline it gates *opening* `baro.csv`, which a dataset without a barometer does not have, while the live receiver is handed pressure samples over the wire whether or not anyone asked for them. A default of 0 there would take the vertical channel away from every session whose config predates the key. Set to 0 live, the samples are still decoded, counted, published to the telemetry stream and shown on the stats line (`baro-off=`), they simply aid nothing, which is the same A/B-comparable exclusion as `gnss: enable: 0`.

`baro:` **`stddev_m`** *m, default 0 → default.* Barometric height noise.

`baro:` **`acc_bias_rw`** $m/s^2/\sqrt{Hz}$*, default 0 → default.* Continuous-time drift density $\sigma_b$ of the vertical-accelerometer bias state of `baro_alt` ($Q = \sigma_b^2\,\Delta t$), larger values let the estimated bias track faster. Note that this is a bias in the local navigation frame and not in the body frame.

`baro:` **`acc_bias_init_mps2`** $m/s^2$*, default 0 → default.* The own INITIAL accel-bias uncertainty of `baro_alt`. This is an initial condition and not a process-noise rate.

`baro:` **`acc_noise_mps2_sqrthz`** $m/s^2/\sqrt{Hz}$*, default 0 → default.* The own direct accelerometer-noise density of `baro_alt`, separate from other accelerometer noise settings, as the measurement here is transformed into the navigation frame and can, for example, include attitude uncertainty.

`baro:` **`h_process_noise`** $m/\sqrt{Hz}$*, default 0 → default.* Small direct process noise density $\sigma_h$ on the height state of `baro_alt` (`h_process_noise_m_sqrthz`, §7.2). It is a safety margin against discretization/model mismatch and not a primary noise source.

**`heading:`, dual-antenna GNSS heading (optional, REQ-VER-035).**

**`heading: enable`** *0/1, default 0.* Open `heading.csv` and fuse its rows as absolute yaw (§5.5.9), the newest row of each IMU interval. A missing file is a hard error, as for `mag/baro/speed`.

**`heading: baseline_frd`** *3-list, default* $[1, 0, 0]$. Direction from the moving base antenna to the rover antenna in the body frame, any length. The file holds the azimuth of that vector as measured, so a remounted antenna pair is a config edit and not a reconversion.

**`heading: require_fixed`** *0/1, default 1.* Skip rows whose carrier-phase solution is not fixed. A float baseline can be decimetres off, which across a metre of antenna separation is degrees of heading while the accuracy of the receiver still claims less.

**`heading: stddev_scale`, `heading: stddev_min_deg`** *default 0 → 1.0 / no floor.* The 1-sigma of the row (the `accHeading` of the receiver) is multiplied by the scale and then floored. The formal accuracy knows nothing about antenna phase-centre variation or a flexing baseline, which is what the floor is for.

**`heading: delay_ms`** *ms, default 0.* Age of a row at its timestamp, anchored in the state history like `gnss: delay_ms`. The converter stamps a heading with the time of the NAV-PVT of the same epoch, so the GNSS delay is the starting point.

Each harness prints how many rows were offered as yaw and how many were skipped as not fixed, for a bad 1-sigma, or by the baseline geometry.

**`ranges:`, ranges to anchors (optional, REQ-VER-038).** `ranges.csv` holds one calibrated range per row: `t_us` (the time the range refers to), the anchor id, the anchor position in ECEF, the range and its 1-sigma. Each producer does its own calibration, the harnesses only weight the rows.

**`ranges: enable`** *0/1, default 0.* Open `ranges.csv` and fuse every row up to each IMU epoch as a range measurement (REQ-NAV-082), anchored in the state history at its own age. Rows beyond the filter's per-epoch maximum move on to the next epoch. A missing file is a hard error.

**`ranges: leverarm_frd`** *3-list, default* $[0, 0, 0]$. From the IMU to the ranging antenna, body frame.

**`ranges: stddev_scale`, `ranges: stddev_min_m`** *default 0 → 1.0 / no floor.* The row's 1-sigma is multiplied by the scale and then floored.

**`ranges: height_with_baro`, `ranges: aiding_max_hpos_stddev_m`** *default 0.* The `ins_options_t` keys of the same names: ranges also correct the height under the barometric height source, and the horizontal 1-sigma below which fused ranges count as position aiding for the coasting window (REQ-NAV-085).

Each harness, `inspostgui.py` included, prints how many rows were offered, fused, rejected by the chi2 gate or skipped, and in how many epochs the ranges counted as position aiding. To score a ranges-only stretch of a recording, cut the GNSS fixes of that stretch with `replay.py --gnss-outage` (§11.4) or the GUI's *GNSS outage* field, or cut the rows from a copy of `gnss.csv` and hand it over with `inputs: gnss`. The reference stays the full track either way.

**Non-holonomic lateral constraint (top level, optional).** `automotive_lateral_constraint` [0/1, default 0] fuses a synthetic measurement stating that a wheeled vehicle does not travel sideways: the body-frame lateral velocity against a truth of zero (REQ-NAV-077). It needs `automotive_mode` and shares its `automotive_min_speed_mps`. The constraint is mainly there

for its attitude row, whose gain is the ground speed: at 20 m/s one degree of roll error is 0.35 m/s of lateral body velocity, and during a GNSS outage that channel is otherwise unobserved. `automotive_lateral_stddev_mps` [m/s] is the measurement noise, `automotive_lateral_max_yaw_rate_deg` [deg/s] skips the fusion in turns where side slip breaks the assumption, and `automotive_lateral_after_sec` [s] holds it off until the filter has been that long without a GNSS fusion.

**`score:`, evaluation only (never touches the filter).** Used by the accuracy summary, the plots and the regression gates. `warmup_sec` [s, default 60] delays scoring after the first fix, `ahrs` [0/1] also scores the attitude filters, `leverarm_frd` [m] maps the filter position onto the ground-truth point (e. g. the antenna, when `ref.csv` is the own solution of the receiver). The position error, the ellipsoid height error and every estimate the plots overlay on the reference (state history, North-East map, both altitude pages, where the curves of the barometer move with it) refer to that point, so a lever arm is never reported as filter error (REQ-VER-037). The replay views of `inspostgui.py` (3D view, altitude profile, North-East map) go the other way: with *Lever arm compensated* checked they move the reference by this lever arm and every GNSS fix by `gnss: leverarm_frd` onto the IMU point the estimate refers to, unchecked they draw both where they were taken. Next to the checkbox the GUI says whether the two lever arms agree, and insdoctor notes when both are set but differ (REQ-VER-039). `min_epochs` sets the minimum scored-epoch count. The `lim_*` keys are the per-dataset guard-rails that `make simulated`/`make datasets` trip on. They are tuned regression thresholds and not accuracy claims, and each one at 0 is simply not gated: `lim_pos_rms_m` on horizontal position, `lim_att_bias_deg`/`lim_att_std_deg` and `lim_yaw_bias_deg`/`lim_yaw_std_deg` on attitude, `lim_baro_{rms,bias,max}_m` on the `baro_alt` *relative* height, `lim_ellipsoid_{rms,bias,max}_m` on the *absolute* ellipsoidal height the suite reports (§7.1), and the `lim_ars_*` attitude-fallback limits (`lim_ars_att_bias_deg`, `lim_ars_roll_std_deg`, `lim_ars_pitch_std_deg`, `lim_ars_yaw_drift_deg_min`), which are only checked with `ahrs: 1`. `attitude` [0/1, default 1] is the opt-out for a dataset whose `ref.csv` carries no attitude truth: with it at 0 the attitude columns are neither scored nor gated.

`ref_delay_ms` [ms, default 0] states that a reference row timestamped $T$ describes the state at $T$ minus that delay, and the harness scores every comparison at that time of validity (REQ-VER-030). Set it to mirror `gnss: delay_ms` whenever `ref.csv` comes out of the same receiver output as the aiding: `gnss: delay_ms` tells the filter how old a fix is, so that it reports where the vehicle is NOW, while the reference row still describes where it was. Comparing both at a shared timestamp therefore penalises the filter by speed times delay, and penalises it more the better it compensates. Leave it at 0 for an independent reference (Lighthouse, synthetic truth) and for a dataset that models the latency in its data and not in its config. A nonzero value with `aiding: ref` is refused, since there the reference is the aiding.

`coast_gap_min_sec` [s, 0 → off] turns on the coasting re-acquisition metric (REQ-VER-029): every gap in the aiding file longer than it is reported with its duration, the straight-line distance across it and the position error of the filter at the first IMU epoch at or after its end, scored against the reference epoch nearest to that IMU epoch (none further than half a second, or half the gap, away). `lim_coast_exit_err_m` [m, 0 → not gated] is the guard-rail on that error. This is the metric for a dataset with a real outage, where the whole-run `lim_pos_rms_m` cannot answer the question: no fixes means no reference inside the gap, and a filter that gives the coasting solution up scores BETTER on the RMS than one that coasts through, because it contributes no epochs where the error is large. The error is sampled before the measurements of that epoch are fused: the returning fix snaps the position onto the truth in the same epoch it arrives, so a sample taken afterwards reports how good the fix was and not how far the coasting drifted. Having no solution when aiding returns counts as a failure of this gate and not as a skipped epoch. The gaps are taken from the aiding file and not from what the filter chose to

fuse, so the number stays comparable across configurations that gate fixes differently.

**`inputs:`, CSV filename overrides (optional).** Per-stream keys `imu`, `ref`, `gnss`, `mag`, `baro`, each a filename resolved relative to the `config.yaml` directory. An unset key keeps the conventional `<stream>.csv` name, so a dataset without an `inputs:` block behaves exactly as before. This lets one stream be swapped (e. g. `gnss: gnss_f9p.csv`) while the remaining CSVs stay shared, so an A/B comparison of two receivers or IMUs needs no directory duplication.

### 11.1 Parameter cross-reference: `config.yaml` ↔ C API

Table 1 lists every `config.yaml` key next to the C struct field or function call that ultimately consumes it, derived from `cfg_set()` in `tools/replay.c` (the canonical, exhaustive mapping, REQ-VER-025). `build_config()` in `tools/replay.py` wires the identical schema onto the same field names via ctypes. A key with no C equivalent is consumed by the harness itself (dataset selection, CSV file names, scoring) and never reaches the library. Fields written per epoch (into `ins_measurements_t`, see §11) are marked *(per epoch)*, everything else is set once, before the replay loop starts.

Table 1: `config.yaml` key → C field / call.

| `config.yaml` key | C field / call (header) |
|---|---|
| **Top level** | |
| `name` | harness-only (dataset label) |
| `aiding` | harness-only (selects `ins` aiding path) |
| `init` | harness-only (selects `ins_options_t.auto_init`) |
| `automotive_mode` | `ins_options_t.automotive_mode` (`ins.h`) |
| `automotive_min_speed_mps` | `ins_options_t.automotive_min_speed_mps` (`ins.h`) |
| `automotive_min_yaw_stddev_deg` | `ins_options_t.automotive_min_yaw_stddev` (`ins.h`) |
| `chi2_disable` | `ins_options_t.chi2_disable` (`ins.h`) |
| `chi2_reject_alpha` | `ins_options_t.chi2_reject_alpha` (`ins.h`) |
| `allow_unlimited_deadreckoning` | `ins_options_t.allow_unlimited_deadreckoning` (`ins.h`) |
| `max_deadreckoning_sec` | `ins_options_t.max_deadreckoning_sec` (`ins.h`) |
| `baro_height_disable` | `ins_options_t.baro_height_disable` (`ins.h`) |
| `gyro_bias_window_sec` | harness-only heuristic (seeds `ins_init_t.gyr_bias_init_rps` / `ahrs_config_t.gyr_bias_init_rps`) |
| `auto_init_window_sec` | `ins_options_t.auto_init_window_sec` (`ins.h`) |
| **`imu:`, noise model and calibration** | |
| `gyr_psd` | `ins_measurements_t.gyr.Qll_diag[]` *(per epoch)* (`ins.h`) |
| `acc_psd` | `ins_measurements_t.acc.Qll_diag[]` *(per epoch)* (`ins.h`) |
| `gyr_bias_rw` | `ins_init_t.gyr_bias_pred_stddev_rps_sqrts` (`ins.h`); also the `ahrs:` default (`ahrs_config_t.gyr_bias_rw`, `ahrs.h`) |
| `acc_bias_rw` | `ins_init_t.acc_bias_pred_stddev_mps2_sqrts` (`ins.h`) |
| `gyr_fixed_bias` | `ins_options_t.imu_gyr_fixed_bias[3]` (`ins.h`) |
| `acc_fixed_bias` | `ins_options_t.imu_acc_fixed_bias[3]` (`ins.h`) |
| `gyr_misalignment` | `ins_options_t.imu_gyr_misalignment[9]` (`ins.h`) |
| `acc_misalignment` | `ins_options_t.imu_acc_misalignment[9]` (`ins.h`) |
| `pos_pred_stddev_m_sqrts` | `ins_init_t.pos_pred_stddev_m_sqrts` (`ins.h`) |
| `vel_pred_stddev_mps_sqrts` | `ins_init_t.vel_pred_stddev_mps_sqrts` (`ins.h`) |
| `rpy_pred_stddev_rad_sqrts` | `ins_init_t.rpy_pred_stddev_rad_sqrts` (`ins.h`) |
| `zero_vel_stddev_mps` | `ins_init_t.zero_vel_stddev_mps` (`ins.h`) |
| `zero_rot_stddev_deg` | `ins_init_t.zero_rot_stddev_rps` (`ins.h`) |
| `auto_zupt_static_gyr_stddev_deg` | `ins_options_t.auto_zupt_static_gyr_stddev_rps` (`ins.h`) |
| `auto_zupt_static_acc_stddev_mps2` | `ins_options_t.auto_zupt_static_acc_stddev_mps2` (`ins.h`) |
| `auto_zupt_static_gyr_deg` | `ins_options_t.auto_zupt_static_gyr_rps` (`ins.h`) |

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| config.yaml key | C field / call (header) |
|---|---|
| auto_zupt_static_acc_mps2 | ins_options_t.auto_zupt_static_acc_mps2 (ins.h) |
| auto_zupt_max_vel_mps | ins_options_t.auto_zupt_max_vel_mps (ins.h) |
| auto_zupt_max_vel_stddev_mps | ins_options_t.auto_zupt_max_vel_stddev_mps (ins.h) |
| auto_zupt_dwell_sec | ins_options_t.auto_zupt_dwell_sec (ins.h) |
| auto_zupt_min_interval_sec | ins_options_t.auto_zupt_min_interval_sec (ins.h) |
| auto_zupt_disable | ins_options_t.auto_zupt_disable (ins.h) |
| auto_zupt_velocity_blind_disable | ins_options_t.auto_zupt_velocity_blind_disable (ins.h) |
| **gnss:, aiding quality** | |
| leverarm_frd | ins_measurements_t.gnss_leverarm_b[3] *(per epoch)* (ins.h) |
| delay_ms | ins_measurements_t.gnss_delay_ms *(per epoch)* (ins.h) |
| pos_stddev_fallback_m | harness-only: fills a zero-diagonal ins_meas_gnss_pos_t.Qll_ned *(per epoch)* |
| vel_stddev_fallback_mps | harness-only: fills a zero-diagonal ins_meas_gnss_vel_t.Qll_ned *(per epoch)* |
| max_horizontal_pos_stddev_m | ins_options_t.gnss_max_horizontal_pos_stddev_m (ins.h) |
| max_vertical_pos_stddev_m | ins_options_t.gnss_max_vertical_pos_stddev_m (ins.h) |
| max_horizontal_vel_stddev_mps | ins_options_t.gnss_max_horizontal_vel_stddev_mps (ins.h) |
| max_vertical_vel_stddev_mps | ins_options_t.gnss_max_vertical_vel_stddev_mps (ins.h) |
| start_max_horizontal_pos_stddev_m | ins_options_t.gnss_start_max_horizontal_pos_stddev_m (ins.h) |
| start_max_vertical_pos_stddev_m | ins_options_t.gnss_start_max_vertical_pos_stddev_m (ins.h) |
| start_max_horizontal_vel_stddev_mps | ins_options_t.gnss_start_max_horizontal_vel_stddev_mps (ins.h) |
| start_max_vertical_vel_stddev_mps | ins_options_t.gnss_start_max_vertical_vel_stddev_mps (ins.h) |
| stop_max_horizontal_pos_stddev_m | ins_options_t.gnss_stop_max_horizontal_pos_stddev_m (ins.h) |
| stop_max_vertical_pos_stddev_m | ins_options_t.gnss_stop_max_vertical_pos_stddev_m (ins.h) |
| stop_max_horizontal_vel_stddev_mps | ins_options_t.gnss_stop_max_horizontal_vel_stddev_mps (ins.h) |
| stop_max_vertical_vel_stddev_mps | ins_options_t.gnss_stop_max_vertical_vel_stddev_mps (ins.h) |
| init_dwell_sec | ins_options_t.gnss_init_dwell_sec (ins.h) |
| init_dwell_disable | ins_options_t.gnss_init_dwell_disable (ins.h) |
| stop_dwell_sec | ins_options_t.gnss_stop_dwell_sec (ins.h) |
| stop_disable | ins_options_t.gnss_stop_disable (ins.h) |
| pos_decimation | ins_options_t.gnss_pos_decimation (ins.h) |
| pos_cov_scale | ins_options_t.gnss_pos_cov_scale (ins.h) |
| pos_cov_scale_height | ins_options_t.gnss_pos_cov_scale_height (ins.h) |
| vel_cov_scale | ins_options_t.gnss_vel_cov_scale (ins.h) |
| pos_stddev_floor_hor_m | ins_options_t.gnss_pos_stddev_floor_hor_m (ins.h) |

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| config.yaml key | C field / call (header) |
|---|---|
| pos_stddev_floor_ver_m | ins_options_t.gnss_pos_stddev_floor_ver_m (ins.h) |
| vel_stddev_floor_hor_mps | ins_options_t.gnss_vel_stddev_floor_hor_mps (ins.h) |
| vel_stddev_floor_ver_mps | ins_options_t.gnss_vel_stddev_floor_ver_mps (ins.h) |
| **init_stddev:, initial-state 1-sigma** | |
| pos_init_stddev_m | ins_init_t.pos_init_stddev_m (ins.h) |
| vel_init_stddev_mps | ins_init_t.vel_init_stddev_mps (ins.h) |
| rpy_init_stddev_rad_deg | ins_init_t.rpy_init_stddev_rad[0..1] (ins.h) |
| yaw_init_stddev_rad_deg | ins_init_t.rpy_init_stddev_rad[2] (ins.h) |
| acc_bias_init_stddev_mps2 | ins_init_t.acc_bias_init_stddev_mps2 (ins.h) |
| gyr_bias_init_stddev_rps_deg | ins_init_t.gyr_bias_init_stddev_rps (ins.h) |
| **init_hint:, auto-init attitude hint** | |
| roll_deg, pitch_deg, rpy_stddev_deg, yaw_deg, yaw_stddev_deg | arguments of nav_suite_set_init_att_hint() (nav_suite.h) |
| **mag:, magnetometer** | |
| enable | harness-only (gates reading mag.csv / setting ins_measurements_t.mag.is_valid) |
| stddev_ut | ins_measurements_t.mag.Qll_diag[] *(per epoch, squared)* (ins.h) |
| min_delay_ms | ins_options_t.magnetometer_min_delay_ms (ins.h) |
| wmm_year | argument of ins_set_magnetic_model_from_position() (ins.h) and ahrs_set_position() (ahrs.h) |
| estimate_bias | ins_options_t.estimate_mag_bias (ins.h) |
| fixed_bias | ins_options_t.mag_fixed_bias[3] (ins.h) |
| misalignment | ins_options_t.mag_misalignment[9] (ins.h) |
| **baro:, barometer** | |
| enable | consumer-only (gates reading baro.csv resp. forwarding a received sample / setting ins_measurements_t.baro.is_valid) |
| stddev_m | ins_measurements_t.baro.stddev_m *(per epoch)* (ins.h, also consumed by baro_alt via nav_suite) |
| acc_bias_rw | baro_alt_config_t.acc_bias_drift_mps2_sqrthz (baro_alt.h) |
| acc_bias_init_mps2 | baro_alt_config_t.acc_bias_init_stddev_mps2 (baro_alt.h) |
| acc_noise_mps2_sqrthz | baro_alt_config_t.acc_noise_mps2_sqrthz (baro_alt.h) |
| h_process_noise | baro_alt_config_t.h_process_noise_m_sqrthz (baro_alt.h) |
| **ahrs:, shared ARS/AHRS noise override** | |
| gyr_noise_psd | ahrs_config_t.gyr_noise_psd (ahrs.h), applies to both nav_suite_t.ars_cfg and .ahrs_cfg |

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| config.yaml key | C field / call (header) |
|---|---|
| gyr_bias_rw | ahrs_config_t.gyr_bias_rw (ahrs.h) |
| acc_noise_mps2 | ahrs_config_t.acc_noise_mps2 (ahrs.h) |
| gyr_bias_init_stddev_rps_deg | ahrs_config_t.gyr_bias_init_stddev_rps[3] (ahrs.h) |
| acc_freq_hz, acc_cutoff_freq_hz, gravity_diff_penalty, acc_reject_gravity_mps2, chi2_threshold, kalman_update_dt_sec | ahrs_config_t fields of the same names (ahrs.h) |
| mag_freq_hz, mag_chi2_threshold | ahrs_config_t fields of the same names (ahrs.h) |
| mag_yaw_stddev_deg | ahrs_config_t.mag_yaw_stddev_rad (ahrs.h), converted from deg |
| **score: and inputs:, evaluation only** | |
| score:* | harness-only (scoring, plots, regression gates, never reaches the filter) |
| inputs:* | harness-only (per-stream CSV filename overrides) |

## 11.2 Free-inertial mode (`free_inertial_start`)

The 3D solution normally begins with a GNSS fix, because the strapdown algorithm needs an absolute position to integrate away from. Free-inertial mode allows the operator to supply the origin manually (instead of a GNSS receiver).

**How it starts.** The declared position is offered to the filter as a position measurement, with the uncertainty the operator states and subject to the ordinary 3D entry gate, and *only until the filter has bootstrapped from it.* Then the source goes silent. Once the bootstrap has happened, the filter runs on the IMU, the auto-ZUPTs (section 5.8) and the barometer, which is selected as the height source at that bootstrap like in any other run (REQ-NAV-053).

**What it switches off on purpose.** Enabling the mode is a statement that coasting is the intent and not a fault, so the two mechanisms that would otherwise end the run are taken out of the loop: the dead-reckoning budget (`max_deadreckoning_sec`) and the 3D exit gate (`gnss.stop_disable`, REQ-NAV-052). Both exist to protect an *aided* solution from quietly degrading, and both would stop the filter mid-experiment for doing exactly what it was asked to do. The 3D entry gate stays in force and every fix is still judged on its own accuracy before it may fuse: GNSS keeps its power to *correct* the solution, it only loses its power to *end* it.

**What it cannot deliver**. Heading is unobservable: without a GNSS course and without a magnetometer nothing measures yaw, and at standstill a MEMS gyroscope cannot find north either, its bias being orders of magnitude above the earth rate. The dead-reckoned track therefore has a shape but no absolute direction, and the yaw variance grows to "unknown", unless the operator states the start heading themselves via `init_hint` (section 11), e. g. a bench test with the unit facing a known direction. In that case the stated uncertainty is what the yaw variance starts from instead. The horizontal position drifts with nothing on board to correct it. The barometer holds the vertical channel and the ZUPTs hold velocity and the IMU biases whenever the platform stands still. That is what makes such a run informative, and it is also all of it.

A typical configuration for an indoor inertial-sensor run, GNSS excluded outright (shipped as `doc/example_conf/config_local_inertial.yaml`, a starting point for a quick test of pure inertial navigation):

```
aiding: none                  # replay only: no gnss.csv needed (insrcv
                              # ignores the key, gnss: enable 0 below is
                              # its counterpart there)

chi2_disable: 1               # outlier rejection off: expected at a standstill
                              # bootstrap and during ZUPT-only coasting

free_inertial_start:          # implemented by insrcv, both replay harnesses
                              # and inspostgui
  enable: 1                   # implies: no DR budget, no GNSS-quality exit
  lat_deg: 49.0               # where the platform starts, as well as known
  lon_deg: 8.4
  height_m: 118.0
  stddev_m: 0.05              # 1-sigma of that statement; must pass the
                              # 3D entry gate (gnss: start_max_horizontal_
                              # pos_stddev_m), so keep it at or below it

init_hint:                    # known start heading, e.g. a bench test with
                              # the unit facing a defined/known direction
  yaw_deg: 0.0
  yaw_stddev_deg: 0.5         # 0 -> no hint (usual: stays unknown, see
                              # above); >0 -> use it

init_stddev:
```

```
  pos_init_stddev_m: 5.00
  vel_init_stddev_mps: 0.05
  yaw_init_stddev_rad_deg: 0.1  # unused while init_hint.yaw_stddev_deg > 0

gnss:
  enable: 0                  # decode and publish, but aid nothing with it

baro:
  enable: 0                  # off here: this example is pure IMU-only
  acc_bias_init_mps2: 0.05
  stddev_m: 2.0

mag:
  enable: 0

imu:                         # the measured noise model of THIS sensor -
  gyr_psd: 5.0e-07           # in this mode nothing else constrains the
  acc_psd: 1.0e-05           # solution, so it is what the run is about
  gyr_bias_rw: 1.5e-05
  acc_bias_rw: 3.0e-04
  zero_vel_stddev_mps: 0.05 # the ZUPT is the only velocity measurement
  auto_zupt_dwell_sec: 0.1
  auto_zupt_static_gyr_stddev_deg: 0.35   # stillness thresholds against
  auto_zupt_static_acc_stddev_mps2: 0.45  # the sensor's own noise
  zero_rot_stddev_deg: 0.03 # left, so its stddev matters more than usual
```

**This example also excludes the barometer.** With `baro: enable: 0` the vertical channel runs on the accelerometer and the ZUPTs alone, drifting like the horizontal solution. This is intended, to demonstrate the pure-IMU extreme (no GNSS, no barometer, no magnetometer). A barometer, where fused, would hold the vertical channel through the same outage that leaves the horizontal solution to drift, exactly as described above.

**Excluding GNSS outright.** `gnss: enable: 0` is a key of its own, usable with or without the free-inertial start and implemented by all three consumers. The fixes are still decoded, counted and (live) published to the telemetry stream, they simply aid nothing. That is different on purpose from deleting `gnss.csv` or setting every gate to reject: the run stays directly comparable against the same recording processed *with* GNSS, one key apart, which is what makes it an A/B and not two different experiments. On a real dataset the difference reads straight off the summary: the same 3073 GNSS fixes are reported either way, while the filter goes from 2950 GNSS fusions to none.

With GNSS excluded the magnetic model takes its anchor from the declared position and not from a fix, since declination and the reference field vary on the scale of degrees of latitude and a hand-entered position is ample for them. Its epoch comes from `mag: wmm_year` if the config states one, from GPS time when the stream carries a pulse, from the UTC date of the own NAV-PVT of the receiver when it does not, and otherwise from the host clock. A date is a clock reading and not a position measurement, so excluding GNSS as an aiding source does not exclude it as a calendar, and no epoch at all would mean no model at all, i. e. an AHRS referencing magnetic north without saying so. The live receiver names the source it used in its log line, so a wrong date is traceable and not merely suspected.

**The same run offline.** All three consumers implement the section: the live receiver `tools/insrcv.c` and both replay harnesses (`tools/replay.c`, `tools/replay.py`). The same `config.yaml` therefore produces the same start, whether the data arrives over UDP or out of `imu.csv`. The offer mechanism is identical in all three: the declared position at 2 Hz until `ins` has bootstrapped from it, never in an epoch that already carries a real fix, and the same two forced options. This is the point of a shared schema: a key that did something live and was quietly ignored on replay would be exactly the silently ineffective setting that this chapter warns about.

Replayed this way a dataset needs no `gnss.csv` at all (`aiding: none`), the declaration does the job that the first fix would otherwise do. For a stationary run the reference is worth having on top: if the platform does not move, the truth is that it does not move, so a `ref.csv` holding the constant start position turns the scored position error of the harness into a direct readout of the free-inertial drift.

```
name: indoor-free-inertial
aiding: none                # no gnss.csv, the declaration starts the filter
init: auto                  # bootstrap from it, exactly as the live path does
```

The same trick works for a run that moves in between, as long as it returns to where (and, if the mount does not turn freely, how) it started: `datasets/inertial_roundtrip` carries a hand-carried, 160.7 s pure-inertial recording whose `ref.csv` holds only the start and end pose, identical because the unit was placed back on the same physical mark. `score: warmup_sec` is set to fall shortly before the timestamp of the end epoch, so the start row (the auto-init anchor) is excluded and only the end row is ever scored. This turns the round-trip position AND yaw error into a tight regression gate with no ground truth needed for the 160 s in between (REQ-VER-034).

### 11.3 Dataset CSV streams

Alongside `config.yaml`, a dataset directory carries the sensor and ground-truth streams as plain comma-separated files in one dataset-neutral format (REQ-VER-003), written by the shared helpers in `datasets/replay_format.py`, so the converters cannot drift apart and both harnesses read the same bytes. Only `imu.csv` drives the filter, `ref.csv` is the independent ground truth (used for scoring, for `init: ref` and to synthesise fixes under `aiding: ref`), `gnss.csv`, `mag.csv`, `baro.csv`, `speed.csv` and `heading.csv` are consumed only when the matching `aiding`/sensor option is on. The first line of each file is a `#` header, the column layouts are:

```
# imu.csv    (required, drives the filter)
  t_us, gx, gy, gz [rad/s], ax, ay, az [m/s^2]
  t_us, gx, gy, gz [rad/s], ax, ay, az [m/s^2], temp_c   (optional column)

# ref.csv    (optional -- ground truth, ZYX roll/pitch/yaw; needed for aiding/init: ref)
  t_us, lat_deg, lon_deg, h_m, roll_deg, pitch_deg, yaw_deg, vn, ve, vd [m/s]

# gnss.csv   (aiding: gnss)   -- 20 fields, one fix per row
  t_us, lat_deg, lon_deg, h_m,
        cov_pos_ned nn,ne,nd,ee,ed,dd [m^2],
        vn, ve, vd [m/s],
        cov_vel_ned nn,ne,nd,ee,ed,dd [(m/s)^2],
        vel_ok

# mag.csv    (optional, mag: enable)
  t_us, mx, my, mz [uT]

# baro.csv   (optional, baro: enable)
  t_us, static_pressure [Pa]

# speed.csv  (optional, speed: enable) -- any further columns are the
#            producer's own record, both harnesses read the first two
  t_us, speed_mps

# heading.csv (optional, heading: enable) -- baseline azimuth, NOT yet
#            the yaw, carr_soln 0 none / 1 float / 2 fixed. Same t_us as
#            the gnss.csv fix of the same epoch. Both harnesses read the
#            first four columns, the rest is the producer's record
  t_us, heading_deg, stddev_deg, carr_soln, length_m, itow_ms
```

Conventions shared by all streams:

**Timestamp** `t_us` `int64` microseconds on one common clock, monotonically non-decreasing. The harness feeds each IMU epoch and then the most recent sample of every other stream at or before that time.

**Frames** Body **FRD** (x forward, y right, z down) for IMU/mag, navigation **NED** for velocity and covariances, angles in degrees (`ref` attitude is Tait–Bryan ZYX roll/pitch/yaw). Position is geodetic `lat_deg/lon_deg` with ellipsoid height `h_m`.

**IMU** `gx..gz` are the body angular rate [rad/s], `ax..az` the specific force [m/s$^2$] (gravity included, so a level static unit reads $\approx (0, 0, -g)$).

**GNSS covariance** The 6 unique elements of the symmetric $3 \times 3$ NED covariance in order `nn,ne,nd,ee,ed,dd`. A *zero diagonal* entry means "unknown" and is replaced at replay time by the `gnss:` fallback stddev (§11). `vel_ok` (0/1) flags whether the velocity data is usable.

**GNSS quality and time** Optional trailing columns of `gnss.csv`: `quality`, the fix type in RTKLIB's solution numbering (1 RTK fixed, 2 RTK float, 3 SBAS, 4 DGNSS, 5 single point, 6 PPP, 0 unknown), `gps_week/gps_tow_s`, the receiver's own GPS time of the fix (its time of validity, 0/0 unknown), `num_sv`, the satellites used in the fix (0 unknown), `jamming` and `spoofing`, the receiver's interference monitor (1 none, 2 warning or indicated, 3 critical or confirmed, the numbering of u-blox' `UBX-SEC-SIG`), and `sig_auth/nav_auth`, whether the signals (ranging codes) and the navigation message (Galileo OSNMA, for example) of the fix were authenticated (1 not authenticated, 2 authenticated, 3 authentication failed). 0 always means unknown or not monitored, a producer writes 0 where it has nothing, and the columns may stop after any of them. The row's `t_us` is when the fix reached the recorder, the receiver's latency included. The filter does not read these columns, a `gnss.csv` without them stays valid. The quality and the satellite count are the `Q` and `ns` of a `.pos` solution file (§11.4). The replay reports the share of each quality and, with a `timesync.csv`, the transport latency of the fixes (arrival minus the fix's own GPS time, median and 5/95 percentile) in the data quality summary, as Dr. INS findings, on a page of the `--plot` PDF and on the GUI's *Plots* tab. The configured `gnss: delay_ms` may exceed the measured latency (the receiver's own filter lag adds to it) but not undercut it by more than 20 ms, that is a warning. A median outside $-50\,\text{ms} \,..\, 2\,\text{s}$ is reported as two clocks that do not belong together, typically a capture whose receiver had its time pulse on the UTC grid. The satellite count (median, minimum) and the share of each known jamming, spoofing and authentication state are reported the same way, any jamming or spoofing indication and any failed authentication as a warning.

**Header/comment lines** Any line beginning with `#` is skipped, so the header row is informational. Rows with fewer than the required number of columns are ignored.

### 11.4 `replay.py` analysis options (command line)

The *filter* behaviour is fixed by `config.yaml` (§11). The Python harness `tools/replay.py` adds a few command-line flags that only affect *analysis and output*, never the estimate itself, plus `--gnss-outage`, which takes data away from the filter. `tools/replay.py --help` is authoritative, the flags that change what is evaluated or reported are:

`--gnss-outage START:DURATION` Simulate a GNSS outage: every fix from `START` to `START+DURATION` seconds after the first IMU sample (the origin of `--eval-start`) is cut from the input before the run, so the filter never learns that a fix existed. Repeatable, `aiding: gnss` only. Combine it with `--eval-start`/`--eval-end` to score the drift inside the gap instead

of diluting it over the whole run. `inspostgui.py` offers the same cut as its *GNSS outage* toolbar field (comma-separated windows) and as `--gnss-outage` with `--batch`.

**`--eval-start SEC, --eval-end SEC`** Restrict the ground-truth error evaluation to a trial-time window (seconds from the first IMU sample, the same origin as the printed `t=...` times). The filter still runs over the *whole* trial, so it stays converged, only the scored attitude/position error statistics and the `--summary-json` output are limited to the window. The sensor data-quality checks (standstill accuracy, IMU noise) still use the full trial. Useful to exclude a bad startup segment or focus on one leg. Either bound may be given alone, the active window is echoed in the "`ins vs ground truth`" header.

**`--estimate-gnss-delay`** Force the GNSS-delay cross-correlation estimate on, even outside the auto-detected case (`aiding: gnss` with usable velocity plus a barometer already runs it automatically, REQ-VER-009). Prints the estimated delay, its correlation quality and, when the correlation is trustworthy, an actionable hint to set config `gnss: delay_ms` (§11) to that value. The same number and hint appear on the GNSS-delay page of the `--plot` PDF. The estimate is *relative to* the own (not necessarily zero) group delay of `baro_alt`.

**`--summary-json PATH`** Write the end-of-run accuracy summary (scored-epoch count, position RMS and maximum, roll/pitch/yaw error mean/std/max, warmup and evaluation window) as machine-readable JSON. Consumed by the simulated-dataset regression harness (`datasets/check_simulated.py`, REQ-VER-016). It carries no pass/fail gates of its own (this stays a pure analysis tool).

**`--plot, --plot-out PATH`** Render the multi-page diagnostic figures (state history, error, bias, ARS/AHRS, `baro_alt`, altitude, GNSS delay, sensor rate, outliers) interactively or, with `--plot-out`, as a single PDF (implies `--plot`).

**`--plot-hz HZ, --plot-track-hz HZ`** Sampling rates of the two `--plot` recorders. `--plot-hz` drives the state history behind every time-series page (roughly 60 fields per tick). `--plot-track-hz` drives the North-East map page alone, which stores nothing but the North/East pair and therefore runs an order of magnitude faster: on a track spanning a few metres, every sample the recorder misses shows up as a chord across the real path. The map page then thins its vertices geometrically, to a tolerance relative to the extent of the track and not to a fixed sample count, so the drawn resolution follows the scale of the view (millimetres on a 2 m track, metres on a 10 km one) at nearly constant page size. `0` makes the map reuse the `--plot-hz` samples.

**`--dump-solution FILE, --dump-solution-hz HZ`** Write the filter's solution to `FILE` at `HZ` of sim time (`0`: every IMU epoch), independent of the reference and the warmup. The extension picks the format (`tools/ins_solution.py`): `.pos` is the RTKLIB solution format, so RTKPLOT shows it next to an RTKLIB solution, anything else is a CSV in the `ref.csv` layout plus the 1-sigma of position, velocity and attitude as trailing columns. Both harnesses read only the first ten columns of a `ref.csv`, so the CSV serves as another run's reference as it is, the smoothed solution of a dataset without ground truth in particular. With `--smooth` the smoothed solution is written next to it as `FILE_rts`. In the `.pos` the RTKLIB fields are used as follows: while the position is aided (at most 1 s since the last absolute position aiding) `Q` is the quality of the last GNSS fix (the `quality` column of `gnss.csv`, RTKLIB's numbering, 1 where it is unknown), `Q` 5 while dead reckoning, `age` the seconds since that aiding, `ns` the satellites of that fix while aided (the `num_sv` column, 0 where it is unknown and while dead reckoning), `ratio` 0, the position and velocity 1-sigma and covariances in ENU as RTKLIB writes them (signed

square roots), velocity in the columns of the RTKLIB demo5 output. The time is GPST from the dataset's `timesync.csv`, the recorder's time sync log: pairs of the `t_us` of a pulse-per-second edge and the GNSS time of that edge, with the flags word of the board's time sync frame. A `TP_SUSPECT` pair is left out. The time has to be GPS time: a capture whose receiver had its time pulse on the UTC grid (`UTC_BASE`, the u-blox default before `tools/ublox_*_config.py` set GPS) is off by the leap seconds, which the `.pos` header only warns about. Offset and rate of the recording clock are fitted to the pairs (a crystal is tens of ppm off, tens of milliseconds over a quarter of an hour), one pair gives the offset alone. Without the file the time is a guess: the first epoch is put at the system clock of the export. `inspostgui.py` writes the same files with its *Export solution* button.

`--smooth` After the replay, run the offline Rauch–Tung–Striebel smoother (`tools/ins_rts.py`, see below) and print its accuracy next to the filter's: position RMS and maximum, velocity RMS and maximum, roll/pitch/yaw error std and mean. The filter run and every number it reports stay exactly what they are without the flag. With `--summary-json` both columns are added under the key `rts`. With `--plot` the PDF gets four pages of the smoother after the filter's bias page: the North-East track of filter, smoother and reference next to the accuracy table, the state history, the error against the reference and the IMU biases, each with the smoother's own 1-sigma band and the filter as a thin grey line. With `--kml` the smoothed track is its own path in the KMZ.

`--smooth-att-sd-max DEG` Cut the smoother's chain wherever the filter's own attitude 1-sigma is above `DEG`. Off unless given.

**Offline RTS smoother.** The smoother turns the error-state filter into a fixed-interval smoother [7] without touching the C library. Its forward pass is a second replay of its own, in which every epoch runs as `predict()` and `correct()` instead of `update()`, so the covariance can be read before and after the correction, and `predict()` hands over the transition matrix $\Phi$ the covariance was propagated with. That second pass runs with `allow_unlimited_deadreckoning`: a real-time filter that gives up dead reckoning after the coasting window and re-acquires leaves nothing to smooth across the outage, run through, its drift (however large) is what the backward pass removes by tying the outage to the fixes on both ends. The backward pass is the textbook recursion [7, 6] in the filter's error convention ($\delta x$ = nominal − truth, attitude as $R_{\text{nominal}} = (I + [\psi\times])\, R_{\text{true}}$):

$$C_k = P_{k|k}\,\Phi_{k+1}^{T} P_{k+1|k}^{-1}, \quad \delta x_k^s = C_k\,(\delta x_{k+1}^- + \delta x_{k+1}^s), \quad P_k^s = P_{k|k} + C_k\,(P_{k+1}^s - P_{k+1|k})\,C_k^T,$$

where $\delta x_{k+1}^-$ is the correction the filter applied at epoch $k+1$ (the nominal state before minus after it). The smoothed state is the filtered nominal state with $\delta x^s$ removed, the attitude through the full rotation. Between two epochs that moved the covariance the error is interpolated linearly, so the smoothed solution has the full IMU rate. The chain is cut where the filter is no plain Kalman filter: ins not ready, a new local origin, or a correction that grows a variance by more than 5 % (re-initialization, re-acquisition, covariance inflation, while float32 round-off of a large update stays at a few $10^{-3}$). The epoch before a cut keeps its filtered value.

The backward pass reuses the forward pass's linearization. Where the filter re-acquires with an unknown heading during strong dynamics, attitude corrections of tens of degrees can make the first seconds worse than the filter. On a slow vehicle the same situation recovers the heading back to the re-acquisition epoch, which is why `--smooth-att-sd-max` is not on by default. The recorder keeps three $n \times n$ float32 matrices per epoch that moved the covariance and about 25 doubles per IMU epoch, an hour of 200 Hz IMU data takes about 400 MB.

`inspostgui.py` has the same smoother as its *RTS smoother* toolbar checkbox (off by default, its forward pass records every epoch and takes several times as long as the plain replay): the

smoothed solution appears as its own trail in the 3D view and on the *Map* tab (*RTS* checkbox), next to the filter in the error, map, altitude, speed and bias plots of the *Plots* tab, in the *Last run* panel and in the summary text, and *Export PDF* and *Export KML* carry the same pages and path as `replay.py`. `python/tests/test_ins_rts.py` checks the backward pass against the textbook RTS of a linear filter and the whole chain on a simulated flight with a GNSS outage.

`replay.py` and `inspostgui.py` run the same replay core (`tools/replay_core.py`): input loading, feed order, scoring, recorders and the printed summary are one implementation, so the GUI and the command line give the same result for the same `config.yaml`. `python/tests/test_replay_core.py` checks that on a dataset with ranges and a GNSS outage.

## 12 Verification environment (overview)

For completeness, the process itself is documented in `requirements/README.md`:

- Requirements database (`requirements/req_*.md`): aerospace-style, machine-checked by `make reqs` (unique IDs, status, parent links and resolution of every `Test:` reference against the actual test code, `@satisfies` trace tags in `src/`).
- Synthetic tests (`make test`): plain-C scenario tests for all filters, no framework, non-zero exit on failure.
- Real-data replays (`make datasets`): the dataset processing harness (`tools/replay.c`, mirrored by `tools/replay.py`) replays datasets for verification.
- Static analysis (`make cppcheck`, `make clang-tidy`): complements the compiler warnings (`-Wall -Wextra` plus a wider set) with two independent tools over the production sources.
- Stack usage (`make stack`): compiles the library with GCC's `-fcallgraph-info=su`, which reports the frame size of every function and the call graph after inlining, and computes the worst-case stack of every public API function along its deepest call chain (`scripts/stack_usage.py`). It fails on recursion, variable length arrays, calls through function pointers or into library functions that `scripts/stack_usage.cfg` does not give a figure for, and on a worst case above its budget in that file. The budgets there are for an x86_64 host. For a target, run the script with its cross compiler and build flags and a config holding the figures of its C library: the script reads several config files and, given the stack a program reserves, checks the main task plus nested interrupt handlers and exception frames against it. Frame sizes depend on the architecture and the optimization level, so a host figure is no substitute for the one of the target. `make check-all` runs every gate above (plus `make coverage` and a `make doxygen` pass that treats any undocumented function or parameter as a failure) independently and prints a pass/fail summary, for pre-push or CI use.
- Coverage (`make coverage`): lcov HTML report including branch (and, with gcc $\geq$ 14.2, MC/DC) coverage.

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